



Warning



All installation works must be performed by a qualified installer and must comply with applicable laws and regulations.



Hi7-N Controller Maintenance Manual

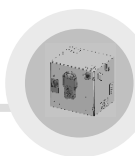
- Hi7-N00(U)-A0
- Hi7-N30(U)-A0
- Hi7-N80(U)-A0
- Hi7-N20(U)-A0



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Printed in Korea – Oct. 2025. 1st Edition
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Overview	1-1
Copyrights	1-1
Safety Cautions	1-2
1. Safety	1-7
1.1. Applicable Standards	1-8
1.2. Safety Performance	1-10
1.3. Safety Trainings	1-10
1.4. Risk Assessment	1-12
1.5. Potential Risks	1-13
1.6. Effectiveness and Responsibility	1-13
1.7. Safety Labels	1-14
1.8. Safety Functions	1-17
1.8.1. Main Safety Functions	1-17
1.8.2. Other Related Functions	1-19
1.9. Stop	1-19
1.10. Safety Measures When Installing	1-20
1.10.1. Installing the Safety Guard	1-20
1.10.2. Placement of the Robot and Peripheral Devices	1-22
1.10.3. Installation of the Robot	1-24
1.11. Safety Works When Operating the Robot	1-26
1.11.1. Safety Measures When Operating the Robot	1-26
1.11.2. Safety Measures When Trial-Operating the Robot	1-28
1.11.3. Safety Measures for Auto Operation	1-29
1.12. Safety Measures When Entering Inside the Safety Fence	1-30
1.13. Safety Measures When Maintaining and Inspecting	1-31
1.13.1. Safety Measures When Maintaining and Inspecting the Controller	1-31
1.13.2. Safety Measures When Maintaining and Inspecting the Robot System and Manipulator	1-31
1.13.3. Actions to Take after Maintenance and Inspection	1-32
1.14. Safety Related to End Effectors	1-33
1.14.1. Gripper	1-33
1.14.2. Tools/Workpieces	1-33
1.14.3. Pneumatic / Hydraulic Systems	1-33
2. Details of Specifications	2-1
2.1. Details of Specifications of Each Controller Model	2-2
2.2. Exterior of the Controller	2-5
3. Installation of the Controller	3-1
3.1. Configuration	3-2
3.1.1. Basic Configuration	3-2

3.1.2. Checking of Various Name Plates	3-3
3.2. Installation and Use Environmets	3-7
3.3. Transport of the Controller	3-8
3.3.1. Packing	3-8
3.3.2. Transport (Modifying the Weight)	3-8
3.3.3. Unpacking	3-10
3.4. Storage of the Controller	3-11
3.5. Discarding of the Controller	3-11
3.6. Connection	3-12
3.6.1. Connection of the Teach Pendant	3-12
3.6.2. Connection of the Manipulator and Controller	3-13
3.6.3. Connectin of the Controller and Primary Power	3-13
3.6.4. The Controller and Grounding	3-16
3.6.5. Other Cautions	3-16
3.6.6. Connection of the Ethernet Port for the User	3-17
4. Basic Configuration of the Controller	4-1
4.1. Configuration	4-2
4.2. Placement of Parts	4-3
4.3. Functions of the Individual Components	4-7
4.3.1. Main Module (H6COM-T)	4-9
4.3.2. Safety Module (BD642)	4-11
4.3.3. Backplane Board (BD604)	4-29
4.3.4. Drive Module	4-31
4.3.5. Power Supply Module (H7PSM)	4-53
4.3.6. Teach Pendant (TP630)	4-59
5. Optional Components of the Controller	5-1
5.1. PCI Communication Card	5-3
5.1.1. Overview	5-3
5.1.2. Configuration of the PCI Communication Card	5-4
5.1.3. Front Part of the PCI Communication Card	5-5
5.2. Brake Release Unit	5-6
5.2.1. Overview	5-6
5.2.2. Brake Release Switch	5-6
5.2.3. Power and Connectors	5-8
5.2.4. Brake Release Unit Status Display LEDs	5-10
5.3. Remote IO	5-11
5.3.1. Overview	5-11
5.3.2. Communication Module (of Crevis)	5-12
5.3.3. IO and Other Modules (of Crevis)	5-12
5.4. Option Safety IO Module(BD680)	5-14
5.4.1. Overview	5-14
5.4.2. Connector	5-14

5.4.3. Device for display	5-15
5.4.4. Device for setting	5-18
5.4.5. Connection for safety output	5-19
5.4.6. Connection for safety input	5-21
5.5. User DIO	5-24
5.5.1. Overview	5-24
5.5.2. Information of hardware	5-25
5.6. Safety communication board(BD671)	5-34
5.6.1. Overview	5-34
5.6.2. Information of hardware	5-34
 6. Periodic Inspections	 6-1
6.1. Inspection Schedule	6-2
6.2. General Cautions for Periodic Inspections	6-3
6.3. General Inspections	6-4
6.4. First Inspection (Inspection When Reaching 750 Hours)	6-5
6.5. Cyclic Inspections	6-6
6.6. Inspection When Going on a Long Vacation	6-8
6.7. Maintenance Parts	6-9
 Appendices	 i
Rules on Occupational Safety and Health Standards, and Notice for Safety Inspection	i
Quality Assurance	vi

Table of Figures

Figure 1.1 Safety Lables	1-14
Figure 1.2 Emergency Stop Buttons on the Controller and Teach Pendant	1-17
Figure 1.3 Connection of the Additional Emergency Stop Device	1-18
Figure 1.4 Connection of the Safety Fence	1-20
Figure 1.5 Beam Type Safety Fence for the LCD Handling Pobot	1-23
Figure 1.6 Cylinder Type Safety Fence for the Industrial Robot	1-24
Figure 2.1 Front Exterior of the Controller	2-5
Figure 3.1 Basic Installation Configuration of the Industrial Robot	3-2
Figure 3.2 Controller's Labels	3-3
Figure 3.3 Controller Wire Connection Position	3-9
Figure 3.4 Transport of the Controller by Using a Forklift	3-10
Figure 3.5 Connection of Hi7-N**(U) Teach Pendant	3-12
Figure 3.6 Connection of the Manipulator and Controller (Hi7-N**(U))	3-13
Figure 3.7 Primary Power Connection Part of Hi7-N**(U) Controller	3-14
Figure 3.8 Power Ground Terminal of the Controller	3-16
Figure 4.1 Hi7-N**(U) Controller	4-2
Figure 4.2 Teach Pendant TP630	4-2
Figure 4.3 Placement of Parts on the Front Exterior of the Hi7-N00(U)/N30(U)/N80(U) Controller	4-4
Figure 4.4 Placement of Parts on the Front Interior of the Hi7-N00(U)/N30(U)/N80(U) Controller	4-5
Figure 4.5 Placement of Parts on the Back of the Hi7-N00(U)/N30(U)/N80(U) Controller ...	4-5
Figure 4.6 Hi7-N00(U),N30(u),N80(U) Controller Transformer Box	4-6
Figure 4.7 Components of the Control module	4-8
Figure 4.8 H6COM-T	4-9
Figure 4.9 Placement of the Connectors and Switches of the BD642 (Safety IO Board) ...	4-11
Figure 4.10 Top Board Display Layout of Servo/Safety Module (BD642)	4-13
Figure 4.11 Front Display Layout of Servo/Safety Module (BD642)	4-16
Figure 4.12 Segment State Display Description	4-19
Figure 4.13 Description of the Setting Devices of the BD642 (Safety IO Board)	4-20
Figure 4.14 Actual Photo of Servo/Safety Module (BD642), and Position of Safety Output Connector (CNSO1)	4-21
Figure 4.15 Safety Output Wiring with Internal Power Supply (NPN Type) for Servo/Safety Module (BD642)	4-22
Figure 4.16 Safety Output Wiring with Internal Power Supply (PNP Type) for Servo/Safety Module (BD642)	4-22
Figure 4.17 Safety Output Wiring with External Power Supply (NPN Type) for Servo/Safety Module (BD642)	4-23
Figure 4.18 Safety Output Wiring with External Power Supply (PNP Type) for Servo/Safety Module (BD642)	4-23
Figure 4.19 Actual Photo of Servo/Safety Module (BD642), and Position of Safety Input Connector (CNSI1)	4-24
Figure 4.20 Safety Input Factory Setting of Servo/Safety Module (BD642)	4-25
Figure 4.21 Safety Input Wiring with Internal Power Supply (NPN Type) for Servo/Safety Module	

(BD642)	4-26
Figure 4.22 Safety Input Wiring with Internal Power Supply (PNP Type) for Servo/Safety Module (BD642)	4-26
Figure 4.23 Safety Input Wiring with External Power Supply (NPN Type) for Servo/Safety Module (BD642)	4-27
Figure 4.24 Safety Input Wiring with External Power Supply (PNP Type) for Servo/Safety Module (BD642)	4-28
Figure 4.25 Backplane Board Structure	4-29
Figure 4.26 Backplane Board Connectors	4-29
Figure 4.27 Connector Configuration of Backplane Board	4-30
Figure 4.28 Component Layout of BD651V60 and BD651V70	4-35
Figure 4.29 Component Layout of BD652V60 and BD652V70	4-38
Figure 4.30 Component Layout of BD653V60	4-43
Figure 4.31 Component Layout of BD654V60	4-45
Figure 4.32 Component Layout of BD658V60 for H6AD1X	4-49
Figure 4.33 Component Layout of BD659V60 for H6AD1Z	4-51
Figure 4.34 Exterior of H6PSM (Hi7-N Controller Power Supply Module)	4-53
Figure 4.35 Power System of the Hi7-N Controller	4-54
Figure 4.36 Connectors of the Electronic Board (BD6C3)	4-55
Figure 4.37 LEDs of BD6C3	4-57
Figure 4.38 Exterior of the Teach Pendant TP630	4-59
Figure 5.1 Exterior of the PCI Communication Card	5-4
Figure 5.2 Front Part of the PCI Communication Card	5-5
Figure 5.3 Switches and Status Checking LEDs of the Brake Release Unit	5-7
Figure 5.4 Switches and Connectors of the Brake Release Unit	5-8
Figure 5.5 Example of Configuration of Commercial Remote IO	5-11
Figure 5.6 Connector Layout of Optional Safety IO Module (BD680)	5-14
Figure 5.7 Display Device Layout of Optional Safety IO Module (BD680)	5-15
Figure 5.8 Front Display Layout of Optional Safety IO Module (BD680)	5-16
Figure 5.9 Setting Device Layout of Optional Safety IO Module (BD680)	5-18
Figure 5.10 Actual Photo of Optional Safety IO Module (BD680), and Position of Safety Output Connector	5-19
Figure 5.11 Safety Output Wiring with Internal Power Supply for Optional Safety IO Module (BD680)	5-20
Figure 5.12 Safety Output Wiring with External Power Supply for Optional Safety IO Module (BD680)	5-20
Figure 5.13 Actual Photo of Optional Safety IO Module (BD680), and Position of Safety Input Connector	5-21
Figure 5.14 Safety Input Wiring with Internal Power Supply (NPN Type) for Optional Safety IO Module (BD680)	5-22
Figure 5.15 Safety Input Wiring with Internal Power Supply (PNP Type) for Optional Safety IO Module (BD680)	5-22
Figure 5.16 Safety Input Wiring with External Power Supply (NPN Type) for Optional Safety IO Module (BD680)	5-23
Figure 5.17 Safety Input Wiring with External Power Supply (PNP Type) for Optional Safety IO Module (BD680)	5-23
Figure 6.1 Inspection Schedule	6-2

List of Tables

Table 1-1 Safety Performance of Safety Modules	1-10
Table 1-2 Safety Labels	1-15
Table 1-3 Installation Standards of the Safety Fence	1-21
Table 1-4 Status of the robot when it stop mode	1-28
Table 2-1 Details of Specification of Each Controller Model	2-2
Table 2-2 Power Requirements	2-4
Table 3-1 Label Type	3-3
Table 3-2 Power Requirements	3-15
Table 3-3 Recommended Minimum Cable Thickness	3-15
Table 3-4 Pin Description (RJ45 Connector Specification; RJ 45P Shield)	3-17
Table 4-1 Names of Individual Parts of the Hi7-N00/N30/N80 Controller	4-3
Table 4-2 Summary of Functions of the Individual Components	4-7
Table 4-3 Types and Usage of the Connectors of Hi7COM-T	4-10
Table 4-4 Types and Usage of the Connectors of the BD642 (Safety IO Board)	4-12
Table 4-5 Top Board Display Description of Servo/Safety Module (BD642)	4-13
Table 4-6 Front Display Description of Servo/Safety Module (BD642)	4-16
Table 4-7 Front LED Description of Servo/Safety Module (BD642)	4-18
Table 4-8 Description of the SW1 setting devices of BD642 (Safety IO Module)	4-20
Table 4-9 Types and Functions of Backplane Board Connectors	4-30
Table 4-10 Configuration of H6D6X (Medium-Sized 6 Axes Integrated Drive Module)	4-31
Table 4-11 Type Symbol of the Medium-Sized 6 Axes Integrated Drive Module	4-32
Table 4-12 Specification of the Medium-Sized 6 Axes Integrated Drive Module	4-32
Table 4-13 Symbols of the IPM of the medium-sized 6 axes drive module	4-33
Table 4-14 Symbols of the Hall Sensors of the medium-sized 6 axes drive module	4-33
Table 4-15 Description of the Connectors of BD651	4-35
Table 4-16 Description of LEDs of BD651	4-36
Table 4-17 Description of the Connectors of BD652	4-39
Table 4-18 Description of the LEDs of BD652	4-39
Table 4-19 Configuration of H6D6A (Small-Sized 6 Axes Integrated Drive Module)	4-40
Table 4-20 Type Symbol of the Small-Sized 6 Axes Integrated Drive Module	4-41
Table 4-21 Specification of the Small-Sized 6 Axes Integrated Drive Module	4-41
Table 4-22 Symbols of the IPM of the small-sized 6 axes drive module	4-42
Table 4-23 Symbols of the Hall Sensors of the small-sized 6 axes drive module	4-42
Table 4-24 Description of the Connectors of BD653	4-44
Table 4-25 Description of the LEDs of BD653	4-44
Table 4-26 Description of the Connectors of BD654	4-46
Table 4-27 Description of the LEDs of BD654	4-46
Table 4-28 Type Symbol of the Optional Drive Module	4-47
Table 4-29 Symbols of the IPM of the Optional Drive Module	4-47
Table 4-30 Symbols of the Hall Sensors of the Optional Drive Module	4-48
Table 4-31 Configuration of H6D1X	4-50
Table 4-32 Description of the Connectors of H6D1X	4-50
Table 4-33 Configuration of H6D1Z	4-52

Table 4-34 Description of the Connectors of H6D1Z	4-52
Table 4-35 Types and Usage of the Fuses of the Electronic Module	4-54
Table 4-36 Type and Function of BD6C3 Connector	4-55
Table 4-37 Types and Functions of BD6C3 LEDs	4-58
Table 5-1 Summary of Optional Configurations	5-2
Table 5-2 Part Names of PCI Communication Cards	5-3
Table 5-3 Description of the Exterior of the PCI Communication Card	5-4
Table 5-4 Configuration of the Front Part of the PCI Communication Card, and the Description of the Functions	5-5
Table 5-5 Usage of the Switches of the Brake Release Unit	5-7
Table 5-6 Types and Usage of the Connectors of the Brake Release Unit	5-9
Table 5-7 Usage and Operation of the LEDs for the Status of the Brake Release Unit	5-10
Table 5-8 Communication Module (of Crevis)	5-12
Table 5-9 IO Module (of Crevis)	5-12
Table 5-10 Relay Module (of Crevis)	5-13
Table 5-11 Analogue IO Module (of Crevis)	5-13
Table 5-12 Pulse Measuring Module (of Crevis)	5-13
Table 5-13 Serial Communication Module (of Crevis)	5-13
Table 5-14 Connector Name, Function, and External Connection Device of Optional Safety IO Module (BD680)	5-15
Table 5-15 Display Device Description of Optional Safety IO Module (BD680)	5-16
Table 5-16 Front Display Description of Optional Safety IO Module (BD680)	5-17
Table 5-17 SW1 Setting Device Description of Optional Safety IO Module (BD680)	5-18
Table 5-18 Board Specifications	5-24
Table 6-1 Daily Inspection	6-4
Table 6-2 First Inspection	6-5
Table 6-3 Cyclic Inspections	6-6
Table 6-4 Inspection of Maintenance Parts of the Category A	6-9
Table 6-5 Maintenance Parts of the Category A-1 (Spare Parts for Standard Parts)	6-9
Table 6-6 Maintenance Parts of the Category A-2 (Important Backup Parts)	6-10
Table 6-7 Maintenance Parts of the Category A-3 (Periodic Replacement Parts)	6-10
Table 6-8 Maintenance Parts of the Category B	6-11
Table 6-9 Maintenance Parts of the Category B-1 (Parts that should be purchased from Hyundai Robotics)	6-11
Table 6-10 Table 6 10 Maintenance Parts of the Category B-2 (Parts that can be purchased directly from the parts maker)	6-11

Overview

The main purpose of this chapter is to describe the matters about the safety for the users of the industrial robot and for the workers who maintain and operate the robot.

Every worker who installs, replaces, adjusts, operate, preserves and maintains the robot system must carefully read and fully understand the operation manual and maintenance manual. Also, keep the manuals nearby so that you can view it whenever necessary.

Our company plans and implements the preservation, maintenance and operation trainings so the person using the robot should ensure that the workers working in the robot system should receive the relevant trainings. It must be ensured that only those who have completed this course can handle the robot.

The users of our industrial robots are responsible for surely understanding and complying with safety-related laws applicable in concerned countries and for properly designing, installing, and operating safety devices to protect workers working in robot systems.

The dangerous areas of the robot system, in other words, the areas where the robot, tools and peripheral devices are operating, should be equipped with safety devices according to ANSI / RIA R15.06-2012 to prevent an object, other than the workers and the workpiece, from entering the dangerous areas. The robot system should be configured in a way that it can be stopped immediately by an emergency stop device when a worker or object needs to enter the dangerous area in spite of possible danger. The workers are responsible for installing, checking and operating these safety devices.

This manual has been created based on standard specifications, so some contents may differ depending on the options and model of the product that you have purchased. In addition, the contents and specifications of this manual are subject to change without prior notice to improve the performance of the product, and Hyundai Robotics is not responsible for situations that could be caused by inaccuracies or typos in the manual. For detailed information on the revision of the manual, you need to visit our internet website (www.hyundai-robotics.com).

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Safety Cautions

Before using the product, you must fully understand the following safety cautions for proper use, user safety, and prevention of property damage

■ Danger



Danger

Imminent danger: Incompliance may cause the death of or serious injuries to the worker.

- Make sure that the safety circuit is never ignored, modified or altered in any way.
- Considering that additional problems may occur due to gravity or brake release, you must take a measure, before carrying out works, such as using a rope and crane that are to be used for transport of individual robots to prevent the brake from running down or additional accidents from taking place during the release of the brake.

■ Warning



Warning

Potential danger: Incompliance may cause injuries to the worker or damage to property, such as significant damage to the product.

- Any act of damaging the safety labels, such as relocating the name plates, warning markings, safety symbols, name markings and wire markings or painting over them or blocking them with a cover is prohibited.
- When the robot is operating, there is a risk of collision between the robot and the worker. Therefore, install a safety fence to prevent the worker from getting close to the robot.
- The robot should be installed and operated according to the guidelines of ISO 10218-2. In addition, it is required to comply with the relevant requirements of international standards and national laws. Our company (or the manufacturer) will not be responsible for any accidents that occur due to not complying with the relevant requirements of international standards and national laws or due to not reviewing the “risk assessment”.
- Must observe the safety work procedures to prevent safety accidents. Do not change or ignore safety devices or circuits under any circumstances, and pay attention to possible electric shock. In auto mode, all normal works should be performed outside the safety guard. Before carrying out works, make sure that there are no people in the operation area of the robot.
- Sufficiently check under your feet when teaching. In particular, you must perform the teaching work outside the safety guard when teaching at a high speed (250mm/s or above).
- When it comes to changing of components or addition of optional equipment (both hardware and software) to the robot both of which may affect safety-related functions, you must check whether the functions are in normal conditions, by paying attention to the items described in “1.11 Safety Works When Operating the Robot”.
- When installing and operating an end effector, you must perform application, maintenance and operation according to ISO/TR 20218-1:2018.

-
- When transporting the product by using lifting equipment, you should comply with the relevant national and local safety regulations and equipment usage guidelines. When moving the product using a crane, you must make sure that no workers are under the product. Also, never work or walk under a crane or the product.
 - If a general safety guard is to be installed and used, the robot should be operated after confirming that the emergency stop operates normally. Also, check if the emergency stop input is disabled. This is an essential measure that must be taken in advance for the safety of workers.
 - If an automatic safety guard is to be installed and used, the robot should be operated after confirming that the emergency stop operates normally. Also, check if the emergency stop input is disabled. This is an essential measure that must be taken in advance for the safety of workers.
 - If a safety input is to be installed and used, the robot should be operated after confirming that the input function operates normally. This is an essential measure that must be taken in advance for the safety of workers.
 - The robot should be installed and operated according to the guidelines of ISO 10218-2. In addition, it is required to comply with the relevant requirements of international standards and national laws.

Our company (or the manufacturer) will not be responsible for any accidents that occur due to not complying with the relevant requirements of international standards and national laws or due to not following the above “caution”.

- If a safety-related input is installed and enabled, you must check whether the function operates normally by referring to “1.11 Safety Measures When Operating the Robot”.
- If an emergency stop output is to be installed and used, the robot should be operated after confirming that the emergency stop output operates normally. This is an essential measure that must be taken in advance for the safety of workers.

■ Caution



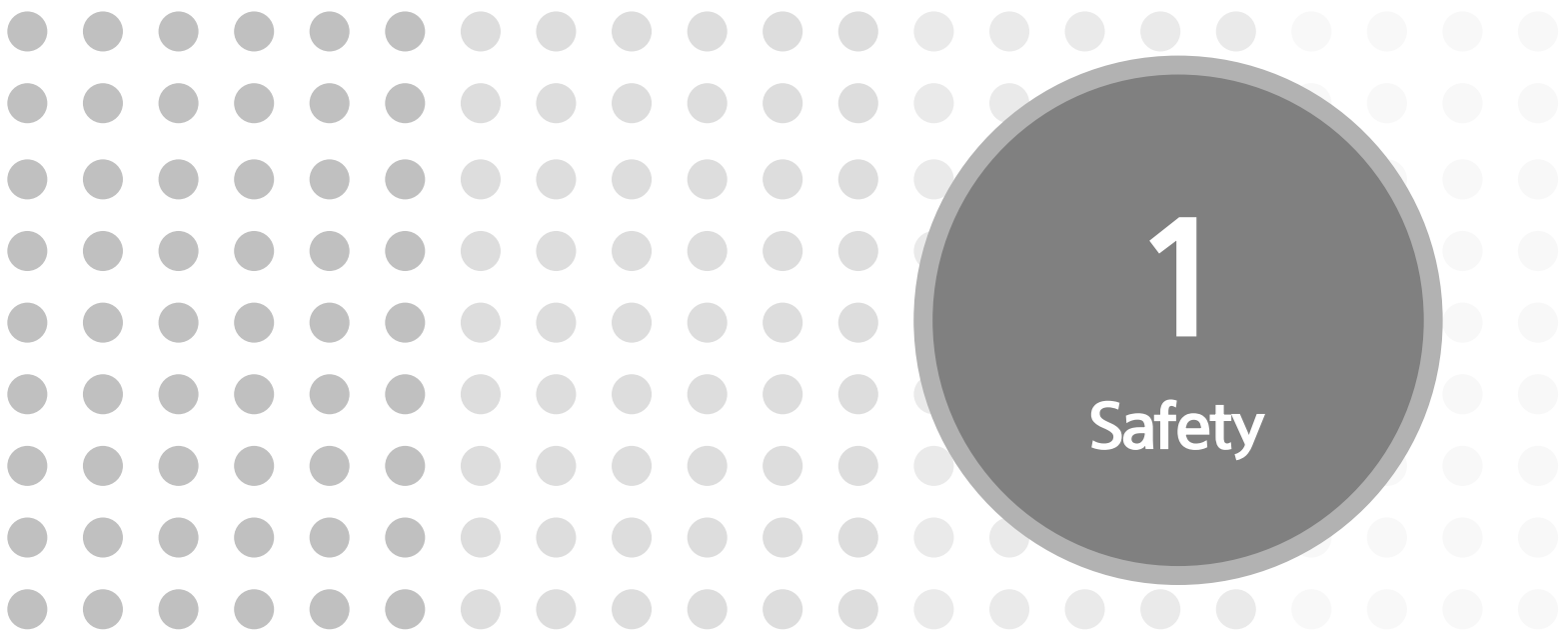
Low-level danger factor: Incompliance may result in minor injury to the worker or damage to property, such as damage to the product.

- Mark the installation and dangerous areas of the robot in a way that they can be clearly differentiated from other facilities and devices in terms of type, color and style.
- As the emergency stop function immediately cuts off the motor power, so reckless use of the function may result in accumulation of fatigue that affects the durability of the robot. The function must be used only in emergency situations.
- Our company is not responsible if the jogging operation does not work due to the failure of the hardware limit switch. You must check it periodically. For the measure to take in case of a failure, please refer to the troubleshooting manual.
- You should not be negligent in paying attention to sudden movements while entry is made possible. Under any circumstances, you must avoid approaching the robot without preparing for possible emergency situation.

- In the case of trial-operation, there may be a design error, teaching error, or defect in manufacturing with regard to the entire system including the teaching program, jigs, and sequencing. For this reason, you should work with elevated safety awareness in trial-operation. Multiple factors can contribute to safety accidents. Observe the following measures considering that safety is very important when trial-operating the robot.
- During maintenance and inspection of the manipulator, the robot arm may fall, or there could be a different type of danger. So, you must proceed with the work according to the instructed procedures
- When moving the axis of the robot that has no driving force applied, there is a risk of the axis dropping due to gravity and also an additional risk due to the release of the brake system. So, you must proceed with the work according to the instructed procedures.
- Before turning on the power, check that there is no worker inside the robot operation area and you are in a safe place.
- Before installing the product, you must perform a sufficient risk assessment and then set the safety functions based on the assessment results. For details on safety functions, refer to “1. Safety” section.
- When installing and repairing the product, contact the customer support team and ask an expert.
- Do not install and use the product in an area that has lots of dust or is dirty. Dust or foreign substances may cause product failure or performance problems.
- Mark the installation and dangerous areas of the robot in a way that they can be clearly differentiated from other facilities and devices in terms of type, color and style.
- If the product is not installed in the recommended locations, the performance and service life of the product may be reduced. Install and use the product according to the recommendations.
- 1. Before connecting the cables, turn “off” the controller's main power switch and then lock it by using a padlock.
 2. The controller has DC400V charged energy. Be careful. Wait at least 5 minutes after turning “off” the power switch, and then wait 5 minutes at least to discharge the charged energy.
 3. When handling the PCB, take precautions not to allow static electricity to damage it
 4. Wiring and connection of wires must be performed by qualified personnel.
- Since the contact part of the connector for each robot may be different from the figure above, you must read the relevant robot maintenance manual carefully before connecting the wire harness.
- 1. When performing the wiring work for the controller and manipulator, separate the signal line and the power line. In addition, use a separate duct for the high power line and the signal line respectively.
 2. Use a protective cover for the wiring, as a measure to prevent the wiring from getting damaged when people are passing nearby.

-
3. Before supplying the primary power, you must check again the relationship with regard to connection, the power specification and power supply specification of the controller.
- The person in charge of maintenance should work after understanding the placement of various devices and parts and their functions inside the controller.
 - The DIP switch is set to OFF mode when shipped from the factory, and the setting should not be changed arbitrarily by the user.
 - The user cannot change the following items arbitrarily, and needs to refer to them only when required to reprogram through FPGA JTAG.
 - The drive module differs depending on the type of the robot, so you must check the type when replacing it.
 - 1. Do not release more than two axes at the same time.
 - 2. Must keep a safe distance from the robot first before using the brake release unit.
 - 3. Use the brake release unit after preparing for the dropping of the robot's axis by using equipment such as a crane.
 - 4. Check the safety matters while working in a group of at least 2 people.
 - When using the brake release unit, follow the procedures below.
 1. Turn off the AC220V power switch and check that the DC24V power switch is turned off.
 2. Connect the AC power cable to the AC power connector.
 3. Turn on the AC220V power switch.
 4. Turn on the DC24V power switch.
 - When the use of the brake release unit is finished, follow the procedures below.
 1. Turn off the DC24V power switch.
 2. Turn off the AC220V power switch.
 3. Disconnect the AC power cable.
 - Do not use AC220V power and DC24V battery power at the same time.
 - Fieldbus communication must be available to use commercial remote IO. Therefore, you should configure the PCI communication card together by referring to the Paragraph 5.1 mentioned above.
 - The drive unit differs depending on the type of the robot, so you must check the type when replacing the drive unit.
 - When transporting the product, you should maintain a proper posture and work in a group of at least two workers. Otherwise, injury to parts of the body such as the back, arms and legs may result.
 - If you transport the product by using lifting equipment, you should comply with relevant national and local safety regulations and equipment usage guidelines.

- When transporting the product, you should fully understand the transport-related contents in the manual and follow the instructions. Our company will not be responsible for any damage to or breaking of the product due to the customer's transport of the product.
- These are important parts to be prepared for daily maintenance and inspection.
- These are maintenance parts to be prepared when multiple units are purchased.





1. Safety

1.1. Applicable Standards

The safety standards applied to the product are as follows.

- ANSI/RIA/ISO 10218-1:2011 Robots and robotic devices - Safety requirements for industrial robots - Part 1: Robots
- ANSI/RIA R15.06-2012 - Industrial Robots and Robot Systems - Safety Requirements
- ISO 10218-2:2011 Robots and robotic devices - Safety requirements for industrial robots - Part 2: Robot systems and integration
- IEC 61508-1:2010 Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 1: General requirements
- IEC 61508-2:2010 Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 2: Requirements for electrical/electronic/programmable electronic safety-related systems
- IEC 61508-3:2010 Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 3: Software requirements
- IEC 61508-4:2010 Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 4: Definitions and abbreviations
- IEC 61508-5:2010 Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 5: Examples of methods for the determination of safety integrity levels
- IEC 61508-6:2010 Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 6: Guidelines on the application of IEC 61508-2 and IEC 61508-3
- IEC 61508-7:2010 Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 7: Overview of techniques and measures
- IEC 61800-5-1:2007/A1:2017 Adjustable speed electrical power drive systems - Part 5-1: Safety requirements - Electrical, thermal and energy
- IEC 61800-5-2:2015 Adjustable speed electrical power drive systems - Part 2: General requirements - Rating specifications for low voltage adjustable speed a.c. power drive systems
- ISO 13849-1:2015 Safety of machinery - Safety-related parts of control systems - Part 1: General principles for design
- ISO 13849-2:2012 Safety of machinery - Safety-related parts of control systems - Part 2: Validation
- IEC 62061:2005/A2:2015 Safety of machinery. Functional safety of safety-related electrical, electronic and programmable electronic control systems
- IEC 61800-3:2017 Adjustable speed electrical power drive systems - Part 3: EMC requirements and specific test methods

- IEC 61000-6-7:2014 Electromagnetic compatibility (EMC) - Part 6-7: Generic standards - Immunity requirements for equipment intended to perform functions in a safety-related system (functional safety) in industrial locations
- IEC 61326-3-1:2017 Electrical equipment for measurement, control and laboratory use. EMC requirements. Part 3-1: Immunity requirements for safety-related systems and for equipment intended to perform safety-related functions (functional safety) - General industrial applications
- IEC 60204-1:2016 Safety of machinery - Electrical equipment of machines - Part 1: General requirements
- ISO 11161:2007 Safety of machinery - Integrated manufacturing systems - Basic requirements

1.2. Safety Performance

The performance of the industrial robot's safety modules is as follows.

Table 1-1 Safety Performance of Safety Modules

The safety performance of SafeSpace2.0's emergency stop and external device interface (basic safety input/output, PROFIsafe) is as follows:

Item	Safety performance	Applicable standards
HFT	1	IEC 61508/62061/61800-5-2
SIL (Safety Integrity Level)	3	IEC 61508/62061/61800-5-2
Category	4	ISO 13849-1
PL (Performance Level)	e	ISO 13849-1
PFH	1.34217E-08	ISO 13849-1

The safety performance of other safety functions is as follows:

Item	Safety performance	Applicable standards
HFT	1	IEC 61508/62061/61800-5-2
SIL (Safety Integrity Level)	2	
Category	3	ISO 13849-1
PL (Performance Level)	d	

1.3. Safety Trainings

In order to use the functions of the product effectively, you must fully understand the contents of the manual and then install, use and maintain the product properly. The users of the product are responsible for fully understanding and complying with robot-related safety laws in the area where the robot is installed and used, and also responsible for properly designing, installing, and operating safety devices to ensure the safety of users working in the robot system.

- All the workers who install, use, and maintain the robot system must read the manual carefully and understand the contents completely. In particular, they must fully understand the safety cautions (⚠).
- Our company plans and implements the trainings for the installation, use and maintenance of the product. So, the product users and the workers must complete relevant training courses before using the product.

- The workers responsible for teaching and inspecting the robot must complete the robot use and safety trainings before using the robot. The contents covered in the safety training courses include the following items.
 - The concept of safety, and the purpose and function of the safety devices
 - Procedures for safe handling of the robot
 - Performance and potential danger of the robot and robotic system
 - Items including the works related to specific applications of the robot

1.4. Risk Assessment

When it comes to configuring an integrated system including a robot, risk assessment is one of the most critical factors that are handled as legal requirements in most countries. A risk assessment of an integrated system cannot be performed only on a single robot because the safety assessment for the installation of the will differ depending on how the robot is integrated into the system.

The system administrator should configure and operate the system according to the guidelines of ISO 12100 and ISO 10218-2 to conduct a risk assessment.

You should conduct the risk assessment by taking into account the entire process of an integrated system including the robot. The main objectives of the risk assessment are as follows.

- Basic settings for the use of the robot, and teaching of the robot
- Trouble diagnosis, and maintenance
- Normal operation of the installed robot

After installing the robot and configuring the system, you must conduct a risk assessment. The risk assessment mainly determines the appropriateness of safety devices in the robot integrated system, as well as the necessity for additional emergency stop devices and other safety devices. It is very important to know the appropriate safety devices and properly configure the robot integrated system. You should configure the integrated system by referring to the relevant information in the manual. For details on configuring the safety functions, refer to “1.8. Safety Functions”. In addition, when installing the robot in a specific location or configuring safety-related functions by using the safety I/O, the important items in the risk assessment of an integrated robot system are as follows.

- Severity
- Frequency of exposure to risks
- Possible occurrence
- Possible avoidance

In the process of configuring an integrated system, if the safety-related functions of the robot do not sufficiently eliminate the risk factors, it is possible to check the necessity for additional protective devices during the risk assessment.

1.5. Potential Risks

If the results of the risk assessment of an integrated system linked to the robot reveal that the risk factors have not been sufficiently removed only with the safety-related functions of the robot, additional protective measures must be taken.

The items to take into consideration when establishing additional protective measures is as follows.

- During installation, fingers can be pinched (caught) between the robot base and the mounting table.
- Injuries (puncture wound, penetrating injury, etc.) from sharp edges or pointed parts of obstacles or tools in the operation area.
- Injuries (bruise, falling over, fracture, etc.) due to collision with the robot
- Injuries caused (puncture wound, penetrating injury, fracture, etc.) due to obstacles around the robot
- Injuries that may occur because the fastening part is not completely fixed
- Injuries (skin damage, shortness of breath, etc.) that may occur when working with toxic and hazardous substances
- Separation of a workpiece from the tool due to sudden power failure
- A mistake caused by confusing with the emergency stop switch of other equipment
- Errors due to arbitrary change of the setting of safety-related functions.

The type of risk that may occur depends on the system configuration, so you must conduct the risk assessment before using the integrated system.

1.6. Effectiveness and Responsibility

It is required to observe the safety requirements according to the safety regulations and laws of the countries and regions where the robot is installed and used. Suppliers and users of the robot integrated systems have a variety of responsibilities, including the following items.

- Risk assessment of the robot integrated system
- Addition and removal of safety devices according to the risk assessment results
- Check whether the integrated system is configured, installed and set up correctly.
- Establishment of the methods and guidelines to use the integrated system, and trainings for the users
- Management of safety devices (Prohibition of user's arbitrary change and manipulation of safety devices)
- Provision of important information, as well as contact information, related to the use and safety of the product
- Provision of all kinds of technical documents including manuals

The safety information in this manual does not cover all risk factors and situations that may arise while using the product.

1.7. Safety Labels

Name plates, warning markings, safety symbols, etc. are attached on the inside and outside of the controller. Any act of damaging the safety labels, such as relocating the name plates, warning markings, safety symbols, name markings and wire markings or painting over them or blocking them with a cover is prohibited. Mark the installation and dangerous areas of the robot in a way that they can be differentiated from other facilities and devices in terms of type, color and style.

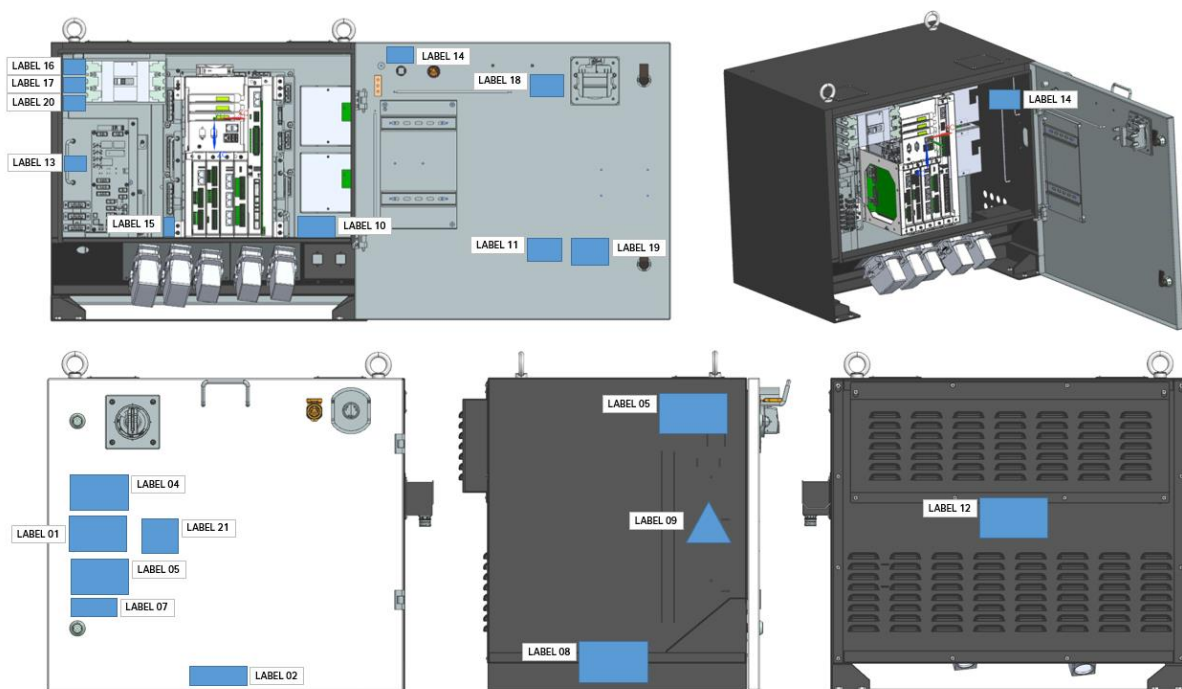
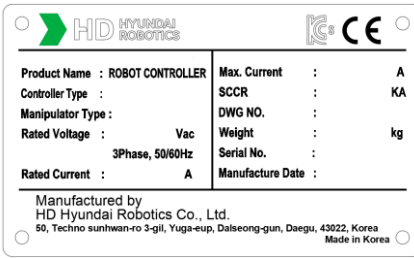
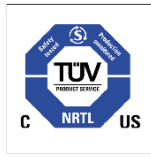
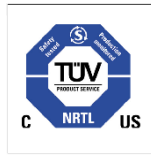


Figure 1.1 Safety Labels

1. Safety

Table 1-2 Safety Labels

No.	Item	Korean	English
1	Name plate	-	
2	NRTL certification mark		
3	Warning for high voltage	경 고 고전압 - 고전압에 의해 중대한 인명사고가 일어날 수 있으므로 다음을 준수해 주십시오. - 제어기 문을 열 때는 반드시 전원을 OFF해 주십시오. - 수리시에는 전원을 OFF한 후 자물쇠를 이용하여 전원스위치를 OFF로 잠가주십시오. 	WARNING High Voltage - High voltage can cause injury or death. - Control cabinet must be turned to "OFF" before opening cabinet door. - The Robot System must be switched off before any maintenance, exchange, repair. - Padlock must be used to lock the power switch to "OFF" . 
4	Caution for installation	주 의 - 설치작업 전에 조작설명서 및 안전지침서를 주의깊게 읽어주십시오. - 조작 중에는 로봇 작업영역 내로 들어가지 마십시오. - 케이블을 연결하기 전에 로봇 본체와 제어기의 일련번호가 동일한지 확인하여 주십시오. 일련번호가 다를 경우, 비정상적인 동작을 일으킬 수 있습니다. 	CAUTION - Carefully read the operation manual and the safety manual before installation and using application. - Do not enter the working range of the Robot system under operation. - Before cables connecting, check that the S/N is identical on the controller and on the manipulator. - If the S/N is different, robot may be operated abnormally. 
5	Marking for high voltage, and the voltage	 AC220 VOLTS Change to 380V	 AC220 VOLTS Change to 380V
6	Caution for air vent	주 의 공기 순환용 흡/배기구를 막지 마십시오. 제어기에 심각한 손상을 입힐 수 있습니다.	CAUTION Ensure no interference for air circulation of ventiduct. Interference may cause controller damage.



Any act of damaging the safety labels, such as relocating the name plates, warning markings, safety symbols, name markings and wire markings or painting over them or blocking them with a cover is prohibited.



Mark the installation and dangerous areas of the robot in a way that they can be differentiated from other facilities and devices in terms of type, color and style.

1.8. Safety Functions

The safety system of the robot is designed in dual configuration (HFT=1) to satisfy the safety performance (PL) = d Cat3 of [ISO13849-1:2015] and the safety integrity level (SIL) 2 of [IEC62061:2005], and continuously monitors the status of safety related devices. When an error is detected by self-diagnosis, or a safety related signal is inputted, the safety functions will stop the robot according to the classification of stop situations determined based on the risk assessment. Also, when any of the dual switches of the safety circuit is activated, the motor drive power and brake drive power will be cut off by the safety functions to secure a safe state. Information on the relevant status can be checked through the Teach Pendant.

! Danger Make sure that the safety circuit is never ignored, modified or altered in any way.

The safety-related main functions of the robot are as follows.

1.8.1. Main Safety Functions

- Emergency stop (IEC 60204-1,10,7)

There is one emergency stop button on the controller and teach pendant respectively. It is possible to connect, if necessary, an additional emergency button to the safety chain circuit of the robot. The emergency stop function is to be applied with higher priority over all other control functions of the robot. The function will immediately cut off the power supply to the motors of individual axes of the robot, stopping the robot and making it impossible to use safety related functions controlled by the robot.

! Caution As the emergency stop function immediately cuts off the motor power, so reckless use of the function may result in accumulation of fatigue that affects the durability of the robot. The function must be used only in emergency situations.



Figure 1.2 Emergency Stop Buttons on the Controller and Teach Pendant

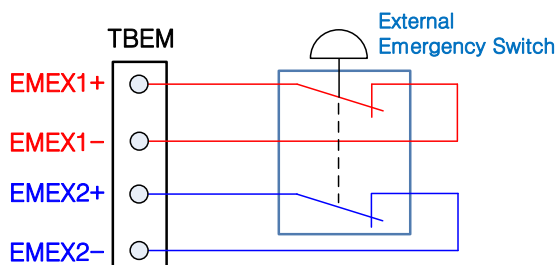


Figure 1.3 Connection of the Additional Emergency Stop Device

- Protective stop (ISO 10218-1:2011)

The robot should have multiple safety inputs so that it can be used in connection with external safety devices such as safety guards, safety pads, and safety lamps. These safety inputs will make the robot stop when there is an input from the robot itself and peripheral facilities, securing a safe state. For details on the connection to the safety inputs, refer to “4.3.2. Safety Module (BD632)”.

- Speed limit (EN ISO 10218-1:2011)

In manual operation mode, the speed of the robot is limited to a maximum of 250 mm / s. The speed limit applies not only to the TCP (Tool Center Point) but also to all other parts of the robot that are to be operated in manual mode. It should be also made possible to monitor the speed of the equipment mounted on the robot.

- Operaiton area limit (ANSI/RIA R15.06-2012)

When applying a robot, in order to secure a sufficient safety area, the operation range of the robot can be limited by using a hardware limit or a stopper. This function can minimize the damage if the robot collides with an external safety device such as a safety guard. Axis 1, 2, and 3 are mainly limited by a stopper or hardware limit. If the operation range is changed due to a mechanical stopper or hardware limit, the operation range limit parameter should be also changed in software as well. Please refer to the operation manual about the change. The operation area limit of each axis can be changed by the user, and at the time of shipment, it is set to the maximum operation range of the robot. The safety system of the Hi7 controller can support up to 4 hardware limit switches as an option. Refer to “4.3.2. Please refer to “Safety Module (BD632)” for the matters related to the connection.

- Operaiton mode selection (ANSI/RIA R15.06-2012)

You can operate the robot in manual, automatic or remote mode. The maximum speed in manual mode is limited to 250 mm/s, and you can perform operation only with the Teach Pendant. In addition, it is possible to mount a mode switch additionally on the control panel by configuring it as an option. For details on the operation, please refer to the operation manual.

1.8.2. Other Related Functions

Required to fully understand the following items and take actions if a person is pinched due to an accident caused by the arm of the robot.

- Release of the manual brake



Considering that additional problems may occur due to gravity or brake release, you must take a measure, before carrying out works, such as using a rope and crane that are to be used for transportation of individual robots to prevent the brake from running down or additional accidents from taking place during the release of the brake.

- After removing power from the controller, connect the brake release unit to the designated connector of the robot or the internal board connector of the controller and then release the manual brake for each axis as needed.
- Refer to the maintenance manual of each robot for the information on individual axes of each robot, and for the designated transport equipment (eg, rope, crane) for each robot.

When the robot is stopped by the limit switch, it is possible to change the position by jogging the robot with the Teach Pendant in the constant setting mode. After designating the soft limit according to the site situation, make sure that it is installed by a trained worker.



Our company is not responsible if the jogging operation does not work due to the failure of the hardware limit switch.
You must check it periodically. For the measure to take in case of a failure, please refer to the troubleshooting manual.

1.9. Stop

The safety system of Hi7 controller can handle the stop operation as shown below. Safety inputs can be classified as follows according to the stop classification criteria specified in IEC 60204-1.

- Stop classification Stopping (uncontrolled stopping) occurs as the power to the machine actuator is immediately removed.
 - ➔ Emergency stop button
- Stop classification 1: Controlled stop in which the machine actuator still has a power that can be used to achieve the stopping. The power will be removed when the stopping is achieved.
 - ➔ Safety inputs except for the emergency stop button

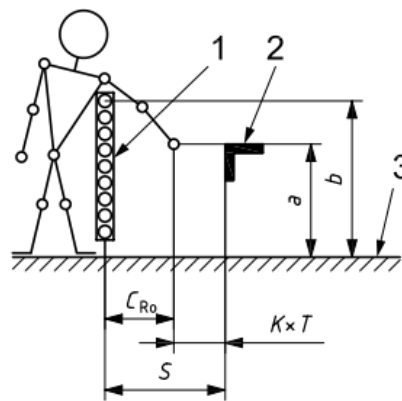
1.10. Safety Measures When Installing

1.10.1. Installing the Safety Guard



Warning When the robot is operating, there is a risk of collision between the robot and the worker. Therefore, install a safety fence to prevent a worker from getting close to the robot.

When the robot is operating, there is a risk of collision between the robot and the worker. Therefore, install a safety fence to prevent a worker from getting close to the robot according to ISO 13855:2010. Configure the system to ensure that the robot stops when a worker opens the door of the safety fence and approaches the facility, during the robot operation, for any reason, such as inspecting the robot or a welding fixture, performing tip dressing or tip changing, etc.



Key

- 1 electro-sensitive protective equipment
- 2 hazard zone
- 3 reference plane
- a height of the hazard zone
- b height of the upper edge of the detection zone of electro-sensitive protective equipment
- C_{RO} additional distance which a part of the body can be moving towards the hazard zone prior to the actuation of the safeguard (see values in Table 1)
- S minimum distance for reaching over

Figure 1.4 Connection of the Safety Fence

Source: ISO 13855:2010 Safety of machinery — Positioning of safeguards with respect to the approach speeds of parts of the human body

Table 1-3 Installation Standards of the Safety Fence

Height of hazard zone <i>a</i>	Height of upper edge of the detection zone of the electro-sensitive protective equipment <i>b</i>											
	900	1 000	1 100	1 200	1 300	1 400	1 600	1 800	2 000	2 200	2 400	2 600
	Additional distance to hazard zone <i>C_{RO}</i>											
2 600 ^a	0	0	0	0	0	0	0	0	0	0	0	0
2 500	400	400	350	300	300	300	300	300	250	150	100	0
2 400	550	550	550	500	450	450	400	400	300	250	100	0
2 200	800	750	750	700	650	650	600	550	400	250	0	0
2 000	950	950	850	850	800	750	700	550	400	0	0	0
1 800	1 100	1 100	950	950	850	800	750	550	0	0	0	0
1 600	1 150	1 150	1 100	1 000	900	850	750	450	0	0	0	0
1 400	1 200	1 200	1 100	1 000	900	850	650	0	0	0	0	0
1 200	1 200	1 200	1 100	1 000	850	800	0	0	0	0	0	0
1 000	1 200	1 150	1 050	950	750	700	0	0	0	0	0	0
800	1 150	1 050	950	800	500	450	0	0	0	0	0	0
600	1 050	950	750	550	0	0	0	0	0	0	0	0
400	900	700	0	0	0	0	0	0	0	0	0	0
200	600	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0
When a value of zero is given, the calculation of the minimum distance, <i>S</i> , should be made in accordance with 6.2 to 6.4.												
NOTE 1 Electro-sensitive protective equipment with a height of the - upper edge of the detection zone below 900 mm is not included since they do not offer sufficient protection against circumventing or stepping over - lower edge of the detection zone above 300 mm in relation to the reference plane does not offer sufficient protection against crawling below. NOTE 2 The data for this table were researched at a study of the German BG, see [22]. NOTE 3 Most values given in Table 1 are lower in relation to the values of ISO 13857:2008, Tables 1 and 2, since parts of the body cannot support themselves on safeguards in case of reaching over.												
^a Approach to the hazard zone by reaching over is impossible.												

Source: ISO 13855:2010 Safety of machinery - Positioning of safeguards with respect to the approach speeds of parts of the human body

- The safety fence should cover the operation area of the robot, and should secure enough space so that there is no interference when the worker carries out works, such as teaching, maintenance, etc. The safety fence should be made solid to prevent it from being moved easily and should be structured in a way not to allow people to enter inside the fence by going over the safety fence.
- In principle, it is required to install and use a fixed type safety fence that does not have dangerous parts such as uneven or sharp parts.
- An entrance door should be installed to allow people to enter inside the safety fence, and a safety plug must be installed on the door in a way that the door should not open unless the plug is removed. In addition, wiring should be configured in a way that allows the motor to be turned off and the brake to be in hold state when the safety plug is removed or the safety fence is opened.

- If you want to operate the robot even when the safety plug is removed, wiring should be configured in a way that allows the robot to play back at low speed.
- Install the emergency stop button of the robot at a location where the operator can quickly press it.
- If a safety fence is not to be installed, safety devices such as photoelectric switches and mat switches should be installed covering the entire area that falls within the specification of the safety guard range of the robot, as substitute devices for the safety fence, making it possible for the robot to stop automatically when a person enters inside the safety fence.
- Make sure that the robot's operation area (danger area) can be identified in some way, such as painting the floor.

1.10.2. Placement of the Robot and Peripheral Devices



The robot should be installed and operated according to the guidelines of ISO 10218-2. In addition, it is required to comply with the relevant requirements of international standards and national laws.

Our company (or the manufacturer) will not be responsible for any accidents that occur due to not complying with the relevant requirements of international standards and national laws or due to not reviewing the “**risk assessment**”.

Installation of the product should be performed by a qualified installer according to relevant national and local regulations and laws.

- When unpacking the product, check it for damage that may occur while transporting or unpacking it.
- Before installing the product after unpacking it, you must check the safety regulations, instructions, information related to the product installation and use environments, and fully understand the installation methods.
- When connecting the primary power of the controller or peripheral device, perform the connection after checking first whether the supply side power is turned off. Since high voltage is used as the primary power source, there is a risk of electric shock.
- Put up a sign “No entry during operation” at the entrance of the safety fence, and inform the workers of the intents.
- Place the controller, interlock panels, and other control panels in a way that they can be operated from outside the safety fence.
- When installing the operation stand, attach an emergency stop button to it also. Wherever you operate the robot, you should be able to stop the robot in an emergency situation.
- Do not allow the wiring or piping of the manipulator, controller, interlock panel, timer, etc. to get caught on by the workers' feet or to get stepped on directly by the forklift. Otherwise, there is a risk of accidents of the worker getting electric shock or the wires getting disconnected.

-
- The diagram shows a circular restricted space. Inside, there is a complex structure with various components. A hatched area represents the restricted space. A legend at the bottom defines the symbols: a hatched area for 'Restricted space', a hatched area with a cross-hatch pattern for 'Maximum space', a hatched area with a cross-hatch pattern and a white rectangle for 'Safeguarded space', a hatched area with a cross-hatch pattern and a white rectangle with a circle for 'Operator', and a white rectangle with a circle for 'Controller'. A label 'Interlocked Barrier Guard' points to a small square on the boundary of the restricted space.

Placement of the Peripheral Devices and Workers for the Industrial Robot

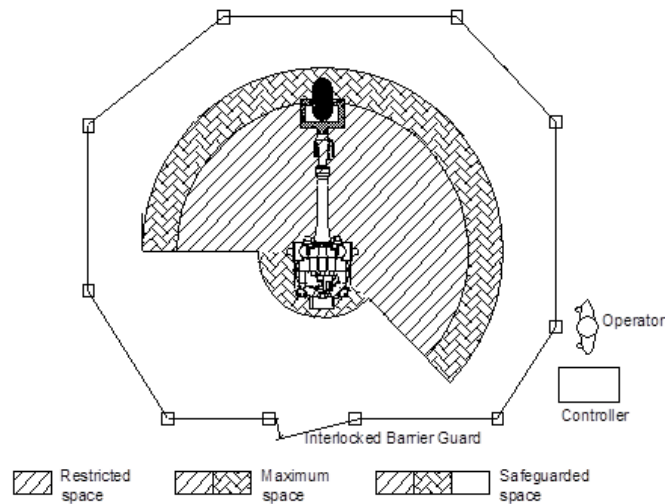


Figure 1.6 Cylinder Type Safety Fence for the Industrial Robot

1.10.3. Installation of the Robot



Warning

The robot should be installed and operated according to the guidelines of ISO 10218-2. In addition, it is required to comply with the relevant requirements of international standards and national laws.

Our company (or the manufacturer) will not be responsible for any accidents that occur due to not complying with the relevant requirements of international standards and national laws or due to not reviewing the “**risk assessment**”.

Installation of the product should be performed by a qualified installer according to relevant national and local regulations and laws.

- When unpacking the product, check it for damage that may occur while transporting or unpacking it.
- Before installing the product after unpacking it, you must check the safety regulations, instructions, information related to the product installation and use environments, and fully understand the installation methods.
- The worker who uses the robot should fully understand the contents described in the application and auxiliary manuals, and operate and handle the industrial robot skillfully.
- The worker who installs the robot should be able to apply the safety instructions during the installation if there is a problem.
- The system supplier should guarantee that all circuits used for safety functions perform their functions surely.

- The main power supply to the robot should be installed in a way that it can be cut off from outside the robot operation area.
- The system supplier should surely guarantee that all circuits used for safety functions perform their functions safely.
- The emergency stop button should be located at a place where the worker can easily approach it when required to stop the robot urgently.
- By taking into consideration the dimensions of the manipulator and the operation range, make sure that there is no interference with peripheral devices.
- Avoid installing the robot in a place that is exposed to direct sunlight, has high level of humidity, has oil or chemicals around and has lots of metal powder or explosive gas in the air.
- Install the robot in an area where the ambient temperature is 0 - 45 °C.
- Secure enough space for disassembling and inspecting the robot easily.
- Install a safety fence and prevent people from entering the robot operation range.
- Make sure there is no obstacle in the robot operation area.
- When installing the robot in an area exposed to direct sunlight or near a heating element, you should take measures in consideration of the thermodynamic state of the controller.
- Take additional measures when installing the robot in an area that has lots of dust such as metal powder in the air.
- Carry out installation in a way that welding current never flows to the robot. In other words, there should be insulation between the spot gun and the wrist of the robot.
- Since grounding is important for preventing a malfunction due to noise and an electric shock, you should install the robot as shown below.
 - Install the dedicated grounding terminal while setting it as Type 3 Grounding higher.
 - Connect the grounding wire to the grounding bus bar inside the control panel.
 - When the manipulator is installed, if it is directly grounded to the floor by an anchor, etc., the controller side and the manipulator side will make a two-point grounding, forming a closed circuit, which conversely may cause malfunctioning due to some reasons such as noise. In this case, connect the grounding wire to the base part of the manipulator, but not to the controller side. In addition, if there is shaking when the robot stops, there is a high possibility that the grounding is incomplete or there is a closed circuit. It is required to check the grounding again.
 - If you use a gun with an embedded transformer, there is a risk of dropping because the primary power cable is connected directly to the spot gun. In this case, in order to protect the control panel and prevent electric shock, connect the grounding wire directly to the base part of the manipulator and do not connect it to the controller.
- Carry out the installation by referring to the robot maintenance manual for each robot.
- After designating the soft limit suitable for the site situation, the positioning and adjustment of the hardware limit must be performed by a trained worker. After installation, you must check whether the functions are working well.

1.11. Safety Works When Operating the Robot



Warning

Must observe safety work procedures to prevent safety accidents. Do not change or ignore safety devices or circuits under any circumstances, and pay attention to possible electric shock.

In auto mode, all normal works should be performed outside the safety guard. Before carrying out works, you must make sure that there are no people in the operation area of the robot.

1.11.1. Safety Measures When Operating the Robot

Observe the following measures considering that safety is very important when operating the robot.

- The worker who operates or may operate the robot, and the supervisor should take certain trainings. Except for those who are recognized to be fully aware of the safety and functions of the robot and designated accordingly, any unqualified person should not operate the robot.
- Before operating the robot, you must check whether the product has been installed by a qualified installer in compliance with the relevant national and regional regulations and laws.
- Before operating the robot, check whether the safety functions are working normally.
- Must wear a safety helmet, protective glasses and safety shoes.
- Two people must work together. One person should be teaching and the other should be monitoring from the operation panel. One of the two should be ready to press the emergency stop switch at any time, and the other should perform the work quickly with sufficient care in the operation area. In addition, check the evacuation route before starting the work.
- Supply the power after confirming that there is no worker within the safety guard.
- Works such as teaching should be performed outside the robot safety guard in principle. However, when required to stop the system and work within the operation range, the worker should bring in the mode switch key (or a switch to change to the auto mode), or the safety plug when going inside. Such action is necessary to make sure that no other worker accidentally changes the robot to the auto operation mode. Also, pay special attention to the direction of the operation of the robot just in preparation for the robot malfunctioning or operating in wrong conditions.
 - ※ The supervisor should observe the followings.
 - The supervisor should be positioned at a location where she or he can see the robot entirely and should be dedicated to the duties of supervision.
 - If there is any problem, press the emergency stop button immediately.
 - Other people except for the persons involved in the work should not be allowed to stay within the operation range.
- In manual operation mode, the speed should be limited to a maximum of 250 mm/sec. At this time, you should progress the work while being prepared, together with the worker outside the guard, to press the emergency stop switch at any time if a problem occurs.

- When operating the robot manually in high-speed mode, you should progress the operation from outside the safety guard.
- When performing a teaching work, you should put up a sign [Teaching in progress].
- When required to enter inside the safety guard, the worker should pull out the safety plug or an equivalent, and then bring it in when going inside.
- Do not use equipment that could be a source of noise near the teaching place or its surroundings.
- Do not operate the robot operation buttons on the teach pendant just by using the feeling at the hands while watching the teaching points. Instead, operate the buttons while checking with bare eyes.



Sufficiently check under your feet when teaching. In particular, you must perform the teaching work outside the safety guard when teaching at a high speed (250mm/s or above).

- Take the following measures when abnormality occurs.
 - If abnormal operation is found, immediately press the emergency stop button.
 - If abnormality is to be checked following an emergency stop, the stop status of the concerned facility must be checked.
 - If the robot stops automatically due to abnormality with the power. Check first that the robot is stopped completely, and then investigate the cause and take measures.
 - If the emergency stop device does not perform functions properly, immediately shut off the main power, and then investigate the cause and take measures.
 - The investigation of the cause of the abnormality should not be performed by any other than the designated person. After the emergency stop, you should restart the system after surely identifying the cause of the abnormality and taking measure accordingly.
- Prepare proper work regulations, considering the installation location and work details, with regard to the robot operation and manipulation methods, and actions to take when abnormality occurs. In addition, work should be carried out according to the work regulations.
- Precautions to take when the robot is stopped
 - Must avoid approaching the robot recklessly while thinking that the robot is stopped. When you approach the robot because you think it is stopped, the robot may suddenly move, causing an accident in many cases. The robot will be in the stopped state in the following cases.

Table 1-4 Status of the robot when it stop mode

No.	State of the robot	Source of driving	Entry
1	In temporary stop mode (Minor abnormality, temporary stop switch)	ON	X
2	In emergency stop mode (Major abnormality, emergency stop switch and safety door)	OFF	O
3	Waiting for input signal from peripheral devices (Start interlock)	ON	X
4	Playback being completed	ON	X
5	Waiting in progress	ON	X



You should not be negligent in paying attention to sudden movements even when entry is made possible. Must avoid approaching without preparing for possible emergency situation under any circumstances.

- If the entrance door needs to be opened to take measures for minor abnormalities (such as nozzle contact, deposition detection and arc abnormality) during a temporary stop, the same measures as taken for opening the door for the teaching should be taken.
- After completing the robot operation, clean the inside of the safety fence to make sure that no tools, oil or foreign substances remain. If the operation area is stained with oil, or if tools are left in the operation area, it could cause an accident such as falling over. Make sure that arranging and organizing are performed always.

1.11.2. Safety Measures When Trial-Operating the Robot



In the case of trial-operation, there may be a design error, a teaching error, or a defect in manufacturing with regard to the entire system including the teaching program, jigs, and sequencing. For this reason, you should work with elevated safety awareness in trial-operation. Multiple factors can contribute to safety accidents. Observe the following measures considering that safety is very important when trial-operating the robot.

- Before operating the robot, check the functions of buttons such as the emergency stop button and the stop button as well as the functions of relevant signals. After that, check the operation related to detection of abnormality. First of all, it is important to check all the signals that stop the robot. When an accident is expected, the most important thing is to stop the robot.
- When performing trial operation of the robot, first set it to manual mode, input a job program with which you can test all axes, and then check the operation by repeating more than 1 cycle for each step. While the robot is moving, open the safety guard or remove the enabling switch (enabling switch on the Teach Pendant) to see whether the robot stop. If a problem is found, press the emergency stop button to check whether the robot stops. If the emergency stop device does not perform its function well, immediately shut off the main power. After that, you should call the responsible aftersales person. If there is no problem, increase the speed in sequence (50% → 75% → 100%), and

check the operation by repeating more than 1 cycle each. Operating at high speed from the beginning may lead to a major accident.

- It is not possible to predict what kind of problem will occur during trial-operation. Never enter inside the safety fence during trial-operation. As the reliability is low, it is very likely that unexpected accidents could occur

1.11.3. Safety Measures for Auto Operation

Observe the following measures considering that safety is very important when operating the robot in auto mode.

- Put up a sign [No entry during operation] on the safety fence entrance and ask the worker to refrain from entering during operation. If the robot is stopped, you may enter inside the safety fence after judging the situation.
- When starting the auto operation, you must check whether there is a worker inside the safety fence. If you work without checking whether there is a worker inside, it may cause an accident involving people.
- When starting the auto operation, check first that the program number, step number, mode, start selection, etc. are in proper state for auto operation. If you start the robot while an irrelevant program or step is selected, the robot may behave unexpectedly causing an accident.
- When starting the auto operation, check in advance that the robot is in the position where the robot can start the auto operation. Check also whether the program number or step number matches with the robot position. Even when the program or step is correct, if the robot is in a different position, an accident may occur due to an operation different from a normal operation
- Be prepared to press the emergency stop button immediately at the start of auto operation. If an unexpected robot operation or unexpected situation occurs, immediately press the emergency stop button.
- Check the operation path, operation status, and operation sound, etc. of the robot to judge whether there is any abnormal state. The robot may suddenly cause an abnormality such as a failure, but it may give some symptoms before the failure occurs. In order to predict this in advance, it is required to well understand the normal operation status of the robot.
- If any abnormality is found, immediately make an emergency stop and take proper measure for it. Using the robot without proper measures could lead to a severe failure that may lead to production interruption and significant accidents involving people.
- While completing measures and checking the operation after an abnormality occurred, do not operate the robot while the worker is still inside the safety fence. Unexpected accidents such as other abnormalities may occur as the reliability is low.
- Before selecting the auto mode, if there is a safety device function that has been stopped, you should progress works after recovering the function back to completely normal state.

1.12. Safety Measures When Entering Inside the Safety Fence

When required to enter the safety door in the robot operation area, a worker and a supervisor who have received certain trainings should perform works in a group of two. Also, they must wear a safety helmet, protective glasses and safety shoes. The supervisor should be prepared to press the emergency stop switch at any time, and the worker must bring in the teach pendant when going inside, making impossible for other people to operate the robot. Must put up a sign on the control panel to indicate that the robot is being operated.

When you enter the robot operation area, you must fully understand the following items.

- No one except for the teaching person should enter the robot operation area.
- The operation setting mode of the controller should be manual mode on the control panel.
- Always wear certified work clothes.
- Do not wear gloves when operating the controller.
- Don't let underwear, shirt, tie etc. come out of the work clothes.
- Do not wear large jewelry such as earrings, rings or necklaces.
- Must wear safety shoes, safety helmet and protective glasses, and, when necessary, should wear safety gear such as safety gloves.
- Before operating the robot, check if the emergency stop circuit is functioning well to turn off the motor when the emergency stop button on the control panel or the teach pendant is pressed
- Work in a position facing the manipulator.
- Follow predetermined work procedures.
- Thinking that the robot may rush towards you unexpectedly, you should prepare a method or place for evacuation.



You should not be negligent in paying attention to sudden movements even when entry is made possible. Must avoid approaching without preparing for possible emergency situation under any circumstances.

1.13. Safety Measures When Maintaining and Inspecting

1.13.1. Safety Measures When Maintaining and Inspecting the Controller

Observe the following safety measures when maintaining and inspecting the robot controller.

- Maintenance and inspection works should be performed only by those who have received special maintenance trainings and fully understand the related contents.
- Progress the work according to the controller maintenance and inspection procedures.
- For the maintenance and inspection works, you must check the surroundings for safety and secure a passage or a place to avoid danger before progressing the work safely.
- Must turn off the power before performing daily inspection or maintenance of the robot or replacement of parts. In addition, in order to prevent other workers from inadvertently turning on the power, put a warning marking such as [Power-On Prohibited] on the primary power supply.
- Always use the designated replacement parts.
- When required to open the controller door, you must turn off the power first and then wait for about 3 minutes before starting the work.
- Use external lighting when sufficient illumination is not secured when performing maintenance and inspection works inside the controller.
- Do not touch the heat sink and regenerative resistor of the servo amplifier because they generate excessive heat. After maintenance, check whether tools, foreign substances, etc. are left behind inside the controller, and then close the door securely.

1.13.2. Safety Measures When Maintaining and Inspecting the Robot System and Manipulator

Observe the following safety measures when maintaining and inspecting the robot system and manipulator.

- Refer to safety measures for the maintenance and inspection of the controller.
- When maintaining and inspecting the robot system and manipulator, proceed with the work according to the instructed procedures.
- Must cut off the main power of the controller. In order to prevent other workers from powering it up again, put a warning marking such as [Power-On Prohibited] on the primary power supply.



During maintenance and inspection of the manipulator, the robot arm may fall, or there could be a different type of danger. So, you must proceed with the work according to the instructed procedures.



When moving the axis of the robot without driving force applied, there is a risk of the axis dropping due to gravity and also an additional risk due to the release of the brake system. So, you must proceed with the work according to the instructed procedures.

1.13.3. Actions to Take after Maintenance and Inspection

Observe the following actions after maintenance and inspection.

- Check if the wires or parts inside the controller are connected normally.
- After maintenance, check whether any tool is left behind inside and around the controller, manipulator or the robot system, and keep them surely arranged and organized. Must close every door.
- If any problem or fatal defect is found, do not turn on the power of the robot.
- Turn on the main breaker in the control panel.
- Check the current position and status of the robot.
- Operate the robot at a low speed.



Before turning on the power, check that there is no worker inside the robot operation area and you are in a safe place



When it comes to the change of components or addition of optional equipment (both hardware and software) to the robot both of which may affect safety-related functions, you must check whether the functions are in normal conditions, by paying attention to the items described in “1.11 Safety Works When Operating the Robot”.

1.14. Safety Related to End Effectors



Warning

When installing and operating end effectors, you must comply with ISO 10218-1:2018 in applying, maintaining and operating them.

Refer to the maintenance manual of each robot for detailed specifications regarding the installation of end effectors.

1.14.1. Gripper

- When a gripper is used to hold a workpiece, there should be a measure to take against abrupt dropping of the workpiece.
- When installing the device onto an end effector or arm, you should use the bolts of specified sizes and the specified number of bolts, and tighten them completely according to regulated torques by using torque wrenches. You should use bolts that are not rusted or stained.
- When manufacturing an end effector, you should take into account that it can be used within the allowable load value of the wrist of the robot. An end effector should have a structure that will not allow a gripped material to be released or dropped even when the power supply or air supply is interrupted, and the corners and protrusions should be surely treated in order to prevent people or objects from suffering any damage.

1.14.2. Tools/Workpieces

- It should be possible to safely change tools such as a milling cutter. Until the cutter stops rotating, the safety devices should surely perform proper functions.
- The tool should be designed in a way that the workpiece will not have any abnormality even when a sudden power failure or control failure occurs. In manual operation, it should be possible to separate the workpiece.

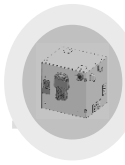
1.14.3. Pneumatic / Hydraulic Systems

- Special safety laws will be applied to the pneumatic and hydraulic systems.
- In this type of systems, as residual energy may remain even after the system is shut down, you should pay attention in particular. Before repairing the pneumatic or hydraulic systems, you must remove the pressure inside the devices.



2

Details of
Specifications



2. Details of Specifications

Hi7-N Controller Maintenance Manual

2.1. Details of Specifications of Each Controller Model

Table 2-1 Details of Specification of Each Controller Model

Model		Hi7-N00-A0 Hi7-N30-A0 Hi7-N80-A0	Hi7-N00U-A0 Hi7-N30U-A0 Hi7-N80U-A0
CPU		2.7GHz Dual core	
Program execution method		Teaching and playback	
Opertion method		Menu-based	
Interpolation type		PTP, linear and circular	
Memory backup method		Batter backup IC memory	
Encoder type		Absolue encoder	
Servo drive unit		6 axes integrated, digial servo	
Maximum number of units		Maximum 32 axes simultaneously	
Step		10,000,000 steps	
Program selection		255 (binary)/8 (discret)	
Display on the teach pendant		7-inch colorful TFT-LCD (800x480)	
Fieldbus interface (optional)		DeviceNet, ProfiNET, Modbus TCP/UDP, EtherCAT	
Digital I/O (optional)		Input: 36 points (maximum 44 points) / Output: 36 points (maximum 44 points)	
Analogue I/O (optional)		Input: 4 points / Output: 4 points	
Conveyor pulse counter (optional)		Line driver / Open collector	
Communication interface		3Ethernet port/ 2 USB 2.0 ports/ 2 RS232 ports	
Circuit board	Main module	H6COM-T	

2. Details of Specifications

Model		Hi7-N00-A0 Hi7-N30-A0 Hi7-N80-A0	Hi7-N00U-A0 Hi7-N30U-A0 Hi7-N80U-A0
	Servo-Safety board	BD642	
	Back plane board	BD604	
	Power supply module	H7PSM(BD6C3)	
Drive module	For medium-sized 6 axes	N00(U)-A0 : H7D6X N80(U)-A0 : H7D6X N30(U)-A0 : H7D6A	
	For one additional axis	H7D1X, H7D1Z	
Wire harness		CMC1, CMC2, CEC1	
Teach pendant		TP630	
Rated supply voltage		3-phase 220V(50/60 Hz) $\pm 10\%$ Option: 3-phase 380V, 400V, 415V and 440V	3-phase 220V(50/60 Hz) $\pm 10\%$ Option: 3-phase 460V and 480V
Maximum power consumption		Hi7-N80(U)-A0: 10.5KVA Hi7-N00(U)-A0: 7.8KVA Hi7-N30(U)-A0: 4.4KVA	
Operation temperature		0 ~ 45 °C	
Operation humidity		75 %	
Protection grade		IP54	
Noise level		Maximum 68 dB	
Exterior diemson*1 (WxHxD)		W680xD520xH550 (mm) Caster 100mm Exclude	
Weight		Standard Specification(Exclude TR) Hi7-N30(U): 80kg Hi7-N00(U): 80kg Hi7-N80(U): 90kg Optional Specification(Include TR) Hi7-N30(U): 140kg Hi7-N00(U): 160kg Hi7-N80(U): 170kg	

Table 2-2 Power Requirements

Controller type	Capacity ^{**1)} [KVA]	Input voltage ^{**2)} [V]	Frequency[Hz]	Peak current [A]
Hi7-N30	Max. 4.4 KVA	220/380/400/415/440V	50/60	15 A
Hi7-N00	Max. 7.8 KVA	220/380/400/415/440V	50/60	30 A
Hi7-N80	Max. 10.5 KVA	220/380/400/415/440V	50/60	50 A
Hi7-N30U	Max. 4.4 KVA	460/480V	50/60	15 A
Hi7-N00U	Max. 7.8 KVA	460/480V	50/60	30 A
Hi7-N80U	Max. 10.5 KVA	460/480V	50/60	50 A

Note 1) Power capacity: Refers to the power supply capacity of the controller. For the power capacity of each robot, refer to the manipulator maintenance manual.

Note 2) Voltage range: $\pm 10\%$ (at the power terminal of the controller)

2.2. Exterior of the Controller

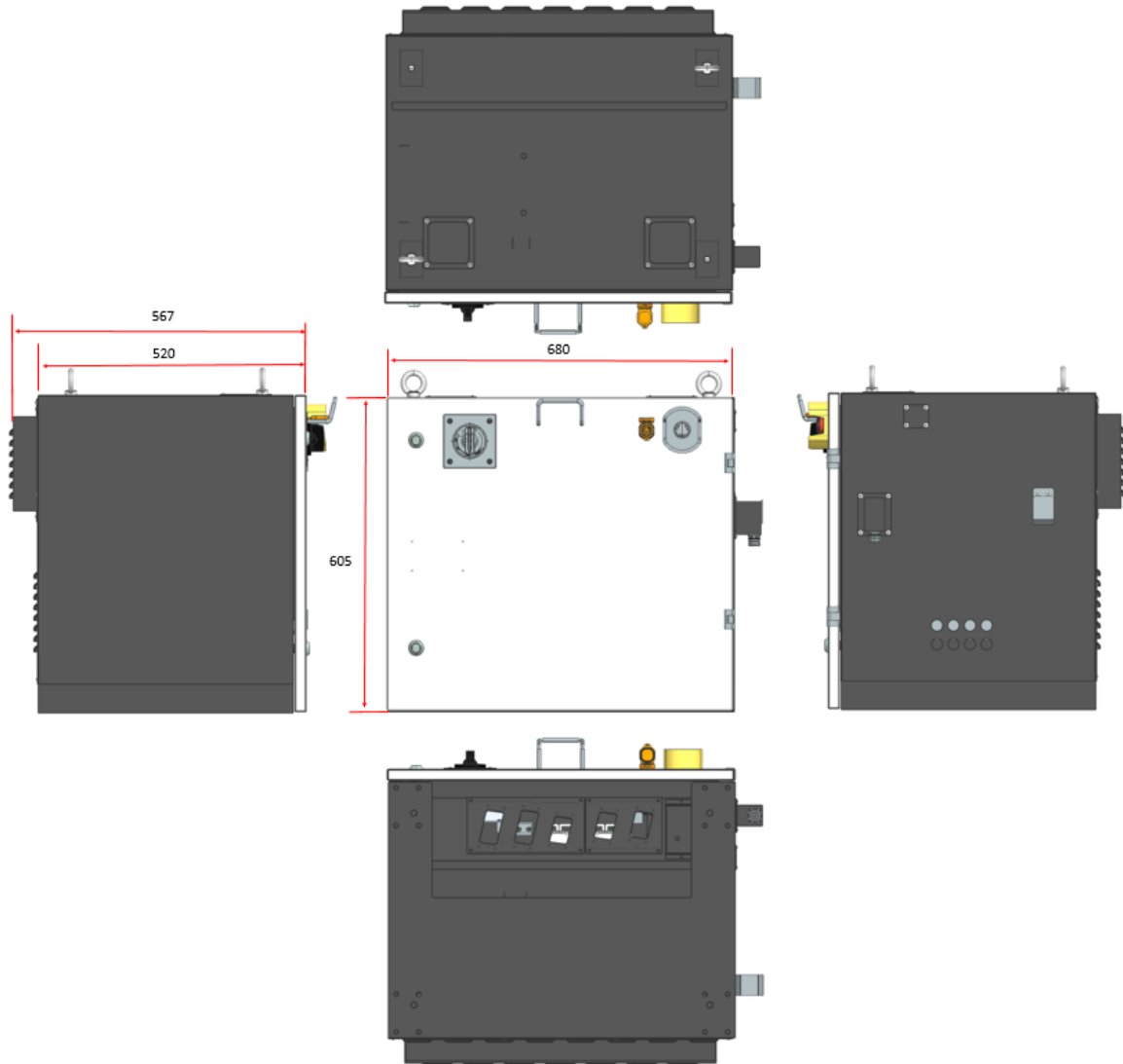


Figure 2.1 Front Exterior of the Controller



3

Installation of the
Controller



3. Installation of the Controller

If you properly install, transport, and store the product in consideration of the position and direction of the installation and the size of the surrounding space, you can secure the service life of the product and prevent the degradation of its performance.

- Checking of the installation and use environments
- Transport of the controller
- Storage of the controller
- Discarding of the controller



Caution

Before installing the product, you must perform a risk assessment sufficiently and then set the safety function based on the assessment results. For details on safety functions, refer to “1. Safety”.

3.1. Configuration

3.1.1. Basic Configuration

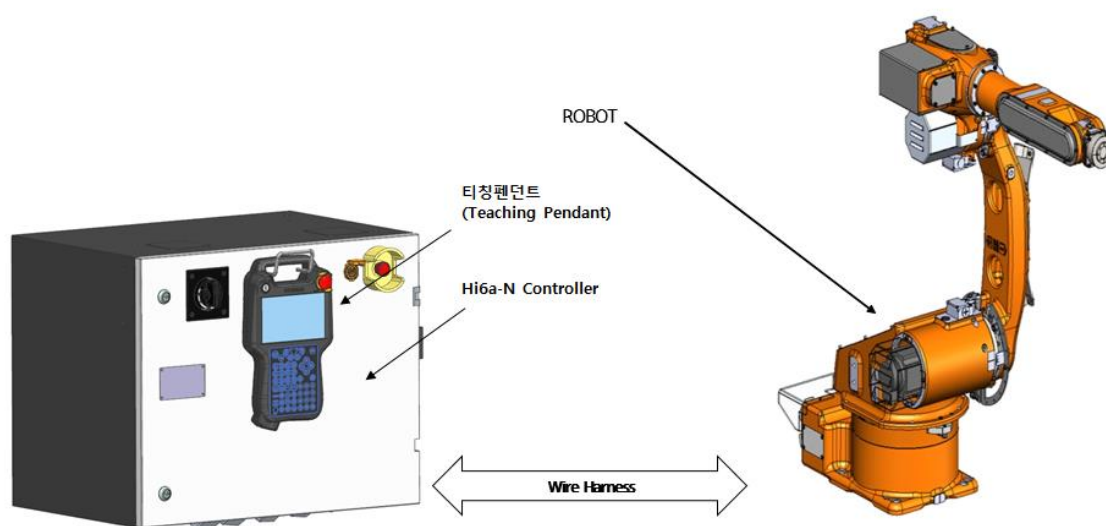
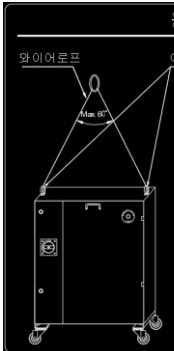
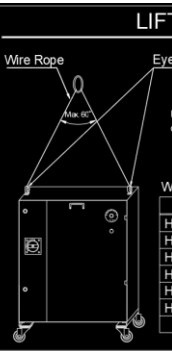


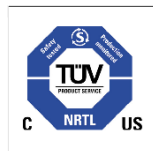
Figure 3.1 Basic Installation Configuration of the Industrial Robot

- Hi7 controller
- Teach Pendant
- Wire harness (Hi7 controller ↔ robot)

No.	Korean	English																												
4	경 고 고전압  - 고전압에 의해 중대한 인명사고가 일어날 수 있으므로 다음을 준수해 주십시오. - 제어기 문을 열 때는 반드시 전원을 OFF해 주십시오. - 수리시에는 전원을 OFF한 후 자물쇠를 이용하여 전원스위치를 OFF로 잠가주십시오.	WARNING High Voltage  - High voltage can cause injury or death. - Control cabinet must be turned to "OFF" before opening cabinet door. - The Robot System must be switched off before any maintenance, exchange, repair. - Padlock must be used to lock the power switch to "OFF" .																												
5	주 의  - 설치작업 전에 조작설명서 및 안전지침서를 주의깊게 읽어주십시오. - 조작 중에는 로봇 작업영역 내로 들어가지 마십시오. - 케이블을 연결하기 전에 로봇 본체와 제어기의 일련번호가 동일한지 확인하여 주십시오. 일련번호가 다른 경우, 비정상적인 동작을 일으킬 수 있습니다.	CAUTION  - Carefully read the operation manual and the safety manual before installation and using application. - Do not enter the working range of the Robot system under operation. - Before cables connecting, check that the S/N is identical on the controller and on the manipulator. - If the S/N is different, robot may be operated abnormally.																												
6	운 반  와이어로프 아이볼트(M16) 제어기를 들어올릴 때에는 외함에 2개의 아이볼트를 사용하십시오. 중량 (근사치) <table><thead><tr><th>모델</th><th>근사 중량(kg)</th></tr></thead><tbody><tr><td>Hi6-N00-00</td><td>185</td></tr><tr><td>Hi6-N30-00</td><td>165</td></tr><tr><td>Hi6-N80-00</td><td>195</td></tr><tr><td>Hi6-N00U-00</td><td>185</td></tr><tr><td>Hi6-N30U-00</td><td>165</td></tr><tr><td>Hi6-N80U-00</td><td>195</td></tr></tbody></table>	모델	근사 중량(kg)	Hi6-N00-00	185	Hi6-N30-00	165	Hi6-N80-00	195	Hi6-N00U-00	185	Hi6-N30U-00	165	Hi6-N80U-00	195	LIFTING  Wire Rope Eyebolts(M16) Use the two Eyebolts on the cabinet when lifting the controller. Weight (approx.) <table><thead><tr><th>Model</th><th>Approx. weight(kg)</th></tr></thead><tbody><tr><td>Hi6-N00-00</td><td>185</td></tr><tr><td>Hi6-N30-00</td><td>165</td></tr><tr><td>Hi6-N80-00</td><td>195</td></tr><tr><td>Hi6-N00U-00</td><td>185</td></tr><tr><td>Hi6-N30U-00</td><td>165</td></tr><tr><td>Hi6-N80U-00</td><td>195</td></tr></tbody></table>	Model	Approx. weight(kg)	Hi6-N00-00	185	Hi6-N30-00	165	Hi6-N80-00	195	Hi6-N00U-00	185	Hi6-N30U-00	165	Hi6-N80U-00	195
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Hi6-N30U-00	165																													
Hi6-N80U-00	195																													
7	Hi7-N00-00 Hi7-N00U-00	Hi7-N00-00 Hi7-N00U-00																												
8	알 림 제어기를 이동하지 않을 때에는 반드시 바퀴의 고정장치를 '고정' 위치로 해 주십시오.	NOTICE Lock the caster of control cabinet unless it needs to be moved.																												
9	 AC220 VOLTS	 AC220 VOLTS																												

3. Installation of the Controller

No.	Korean	English				
10	경고  고전압 ! 충전 에너지! DC400V의 충전된 에너지가 존재합니다. 완전히 방전 시키기 위해 전원 OFF후 5분 이상 기다리십시오.	WARNING  High Voltage ! Stored Energy! Be careful of stored energy of DC 400V. Wait more than 5 minutes for deenergizing after power off.				
11	Controller Ass'y No. DATE : Ass'y No. :  PJ-RY00123-1211-001  HD HYUNDAI ROBOTICS Tel : 82-53-670-7114 Fax : 82-53-615-6517	Controller Ass'y No. DATE : Ass'y No. :  PJ-RY00123-1211-001  HD HYUNDAI ROBOTICS Tel : 82-53-670-7114 Fax : 82-53-615-6517				
12	주의 공기 순환용 흡/배기구를 막지 마십시오. 제어기에 심각한 손상을 입힐 수 있습니다.	CAUTION Ensure no interference for air circulation of ventiduct. Interference may cause controller damage.				
13	Part : H6PSM30 Type : V10A S/N : BK-2008-0001	Part : H6PSM30 Type : V10A S/N : BK-2008-0001				
14	CNRTP	CNRTP				
15	H6D6XV20-3X3Y-111111	H6D6XV20-3X3Y-111111				
16	NFB1	NFB1				
17	PE	PE				
18	-	PASSED <table><tr><td>LINE QC</td><td>QM</td></tr><tr><td> </td><td> </td></tr></table> Worker : (sign)	LINE QC	QM		
LINE QC	QM					

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Warning

Any act of damaging the safety labels, such as relocating the name plates, warning markings, safety symbols, name markings and wire markings or painting over them or blocking them with a cover is prohibited.



Caution

Mark the installation and dangerous areas of the robot in a way that they can be differentiated from other facilities and devices in terms of type, color and style.

3.2. Installation and Use Environments

Install the product in an appropriate place in consideration of the installation and use environments and conditions.

- The proper use temperature of the product is 0°C - 45°C, and the proper storage humidity is 20 to 85%RH.
- Do not drop the product or apply a strong impact to it while moving or using it.
- Transport and install the product in a correct way based on the weight of the product while paying attention to the safety.
- Install and use the product in a solid, flat and vibration-free area where the product will not turn over easily.
- Do not install and use the product in an area with a lot of foreign substances such as water, moisture, gas, dust, or in a dirty place.
- Do not install or use the product in an area with flammable and corrosive substances or gases, or in an area where heat is generated, or near heat of fire.
- Do not install or use the product in an area that has a source of strong electrical noise or is affected by it.
- Install the controller in a safe area by referring to “1.10 Safety Measures When Installing”.
- Carry out the controller maintenance work by referring to “1.13. Safety Measures When Maintaining and Inspecting”.
- When installing the product in an area where welding work is performed, install the product in a location where there will be no effect from welding spatter and cooling water.
- When installing the controller, keep a distance of at least 500mm if there is a wall or obstacle nearby.
- For the matters related to the installation of the robot, refer to each robot maintenance manual.



If the product is not installed in the recommended locations, the performance and service life of the product may be reduced. Install and use the product according to the recommendations.



The robot should be installed and operated according to the guidelines of ISO 10218-2. In addition, it is required to comply with the relevant requirements of international standards and national laws.

Our company (or the manufacturer) will not be responsible for any accidents that occur due to not complying with the relevant requirements of international standards and national laws or due to not reviewing the “**risk assessment**”.

3.3. Transport of the Controller

The following items describe the precautions to take in packing, transporting, and unpacking the Hi7 controller. Refer to the robot maintenance manual for the matters related to the packing and transport of the robot.

3.3.1. Packing

- Attach the model name plate to the box.
- Protect all exposed connectors with a dust cap or polyvinyl.
- When the teach pendant is packaged in a box, use an air-filled cushioning to prevent the LCD from getting damaged due to external impact.
- Attach the waterproofed packing list to the outside of the box.

3.3.2. Transport (Modifying the Weight)

- Check whether the front door of the controller is completely locked.
- Remove anything that is not fixed onto the controller.
- Check if the eye bolts on the controller are surely fastened.
- As the controller is a precision device, pay attention to the transport of it to prevent any strong impact from being applied to it.
- The weight of the controller is maximum 170kg. When using a crane, take precautions to prevent an object on the controller from being damaged by the wire.
 - ※ For the weight of the controller, refer to “2. Details of Specifications’.
- When using a forklift, fix the controller in a way to prevent the controller from shaking.
- When moving the product by vehicle, fix the manipulator and controller by using squids.
- When transporting the product, fully understand the contents related to the packing and transport, and follow the instructions. Our company will not be responsible for any damage to or breaking of the product due to customer’s carelessness, inexperience in operation, or negligence.
- Check the following items when transporting the controller by using a crane.
 - In general, the controller should be transported by using crane wires for which eye bolts are used.
 - Check whether the wires have sufficient strength to withstand the weight of the controller.
 - Check whether the eye bolts are fastened tightly.

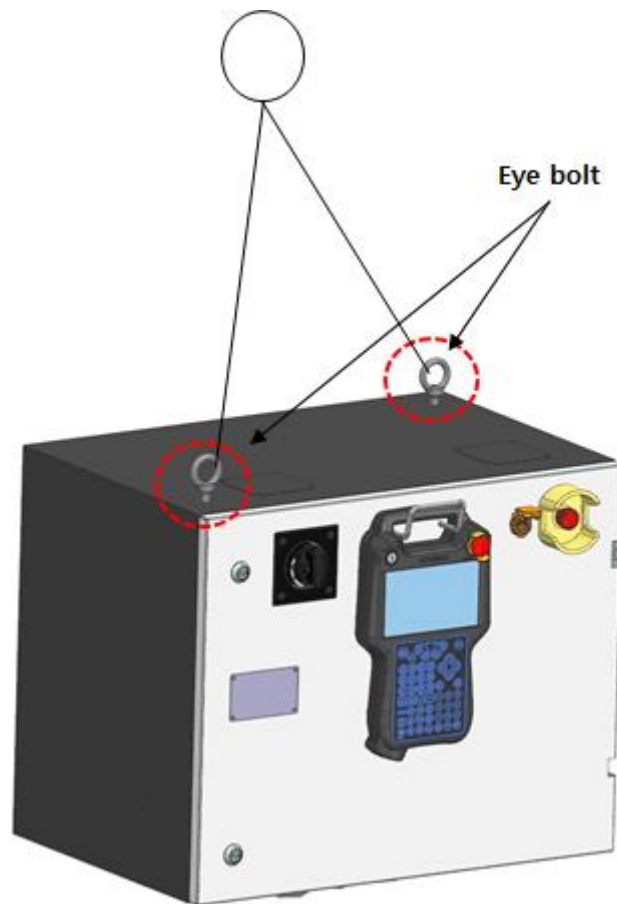


Figure 3.3 Controller Wire Connection Position

- Check the following items when transporting the controller by using a forklift,
 - When transporting the product by using wire ropes, use a wire that can withstand the weight of the controller.
 - Check whether the eyebolts are firmly fixed.
 - Transport the controller while keeping it as low as possible.



Figure 3.4 Transport of the Controller by Using a Forklift



If you transport the product by using lifting equipment, you should comply with the relevant national and local safety regulations and equipment usage guidelines. When moving the product by using a crane, you must make sure that no workers are under the product. Also, never work or walk under the crane or the product.

3.3.3. Unpacking

- Fully understand the safety regulations and other guidelines carefully before unpacking and installing the robot.
- Unpack the product according to the unpacking instructions.
- Check whether the location is an area where the robot and controller can be safely installed.
- Check if a path that allows the robot and controller to move safely is secured.
- Transport of the robot should be performed by a qualified person.
- When unpacking the product, check that whether there is any damage that might have occurred during transport or unpacking.

3.4. Storage of the Controller

When storing the controller, instead of installing it, refer to the following items.

- Store the controller while keeping it in the packaged state and seal the power and communication connection parts tightly.
- When storing the controller for a long time, you must take safety measures against the risk of it falling over.
- When storing the controller wrapped in packing material, pack it with a desiccant or store it in a dry place. If it is stored in a highly humid place, moisture may form inside the packing material, damaging the product.
- Avoid places where temperature and humidity may change easily (where condensation occurs) and store the controller in a cool, dry place where the ambient temperature ranges from -15 °C to 40 °C.
- Do not store the controller in locations where there are chemical products, acid and alkali products, batteries, circuit breakers, etc.

3.5. Discarding of the Controller

In order to ensure user safety and protect the environment, certain parts should be managed and discarded according to specified methods, and if they contain industrial waste materials, they must never be discarded together with general industrial or household waste. When discarding all or part of the robot system, you must comply with the relevant national or local regulations and laws. For details on the discarding and disposal of the product, please contact our customer support team.

3.6. Connection



Caution

1. Before connecting the cables, turn “off” the controller's main power switch and lock it by using a padlock.
2. The controller has DC400V charged energy. Be careful.
Turn “off” the power switch and then wait 5 minutes at least to discharge the charged energy.
3. When handling the PCB, take precautions not to allow static electricity to damage it.
4. Wiring and connection of wires must be performed by qualified personnel.

3.6.1. Connection of the Teach Pendant

Connect the cable connector of the Teach Pendant to the CNRTP receptacle of the controller.

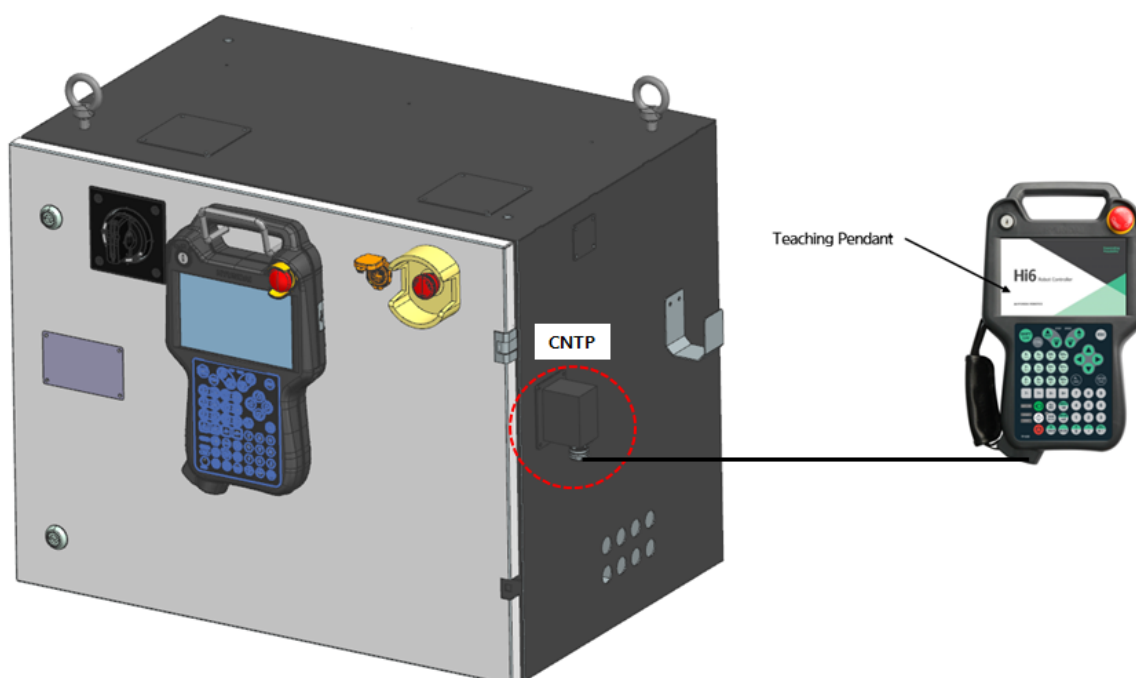


Figure 3.5 Connection of Hi7-N**(U) Teach Pendant

3.6.2. Connection of the Manipulator and Controller

Connect between the manipulator and controller by using a wire harness. Check the names of individual receptacles while connecting them.

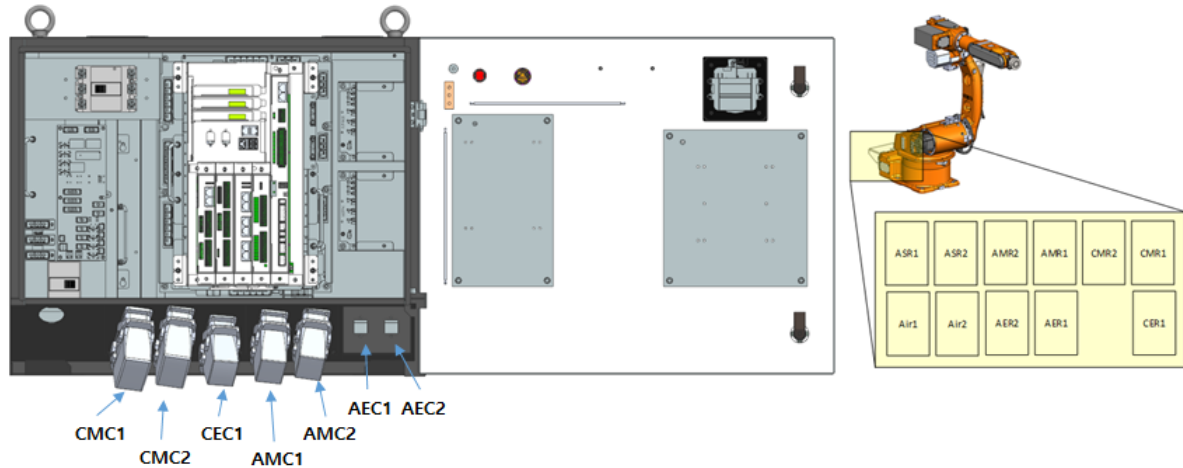


Figure 3.6 Connection of the Manipulator and Controller (Hi7-N**(U))

Wire Harness connection diagram

Hi7-N Controller	ROBOT
CMC1	CMR1
CMC2	CMR2
CEC1	CER1
AMC1	AMR1
AMC2	AMR2
AEC1	AER1
AEC2	AER2



Caution

Since the contact part of the connector for each robot may be different from the figure above, you must read the relevant robot maintenance manual carefully before connecting the wire harness.

3.6.3. Connectin of the Controller and Primary Power

Check whether the power has been removed from the primary power and breaker (NFB).

In the case of a Hi7-N** controller, insert the power cable through the power inlet and then connect it to the breaker (NFB).

At this time, use a terminal receptable of an appropriate size for the end side of the primary power supply cable.

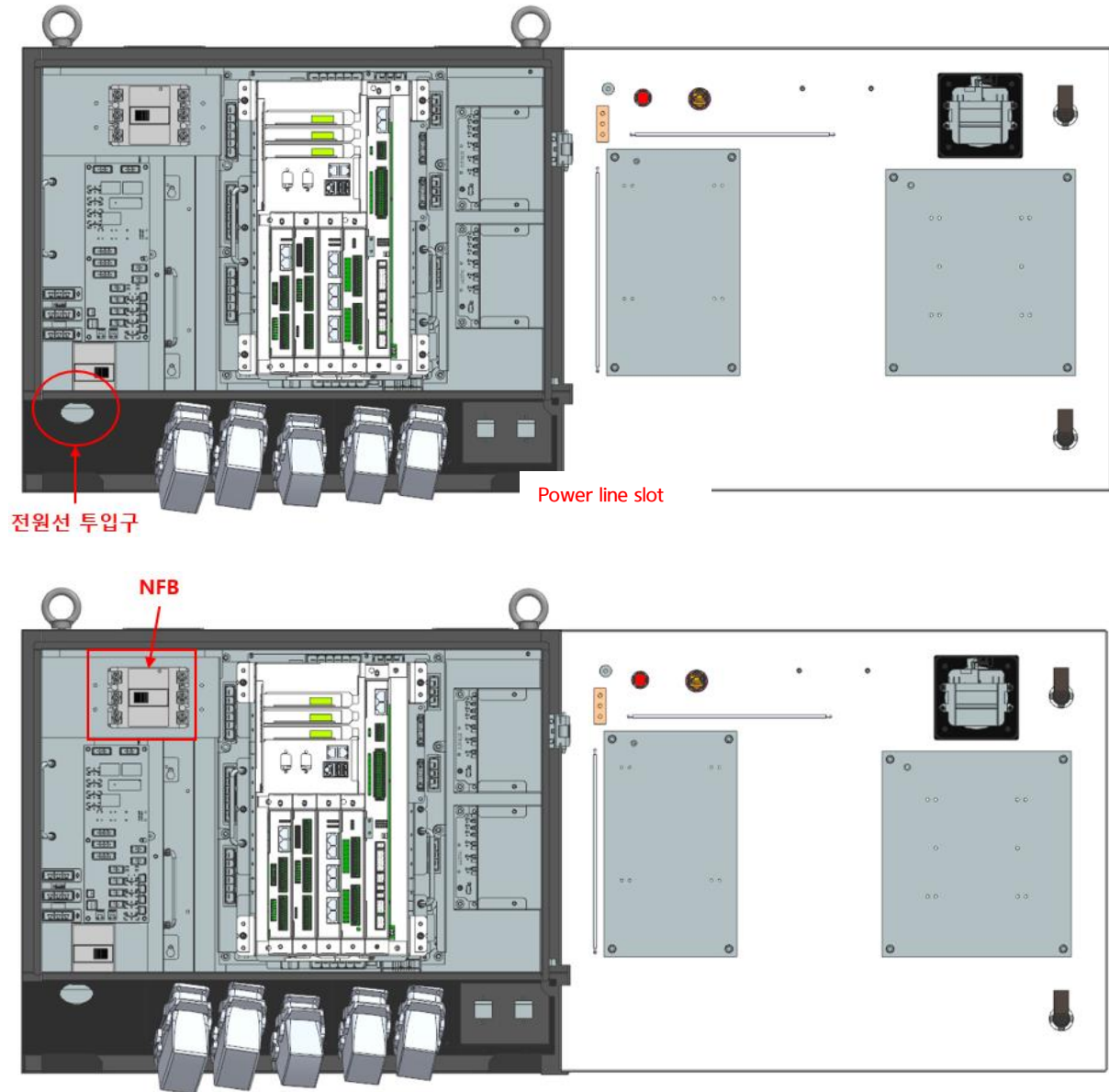


Figure 3.7 Primary Power Connection Part of Hi7-N**(U) Controller

3. Installation of the Controller

3.6.3.1. Power Requirements

Table 3-2 Power Requirements

No.	Controller type	Capacity* ¹⁾ [KVA]	Input voltage * ²⁾ [V]	Frequency [Hz]	Peak current [A]
1	Hi7-N00	Max. 7.8KVA	220V/380V/400V/440V	50/60	30A
2	Hi7-N30	Max. 4.4KVA	220V/380V/400V/440V	50/60	15A
3	Hi7-N80	Max. 10.5KVA	220V/380V/400V/440V	50/60	50A
4	Hi7-N00U	Max. 7.8KVA	460V/480V	50/60	30A
5	Hi7-N30U	Max. 4.4KVA	460V/480V	50/60	15A
6	Hi7-N80U	Max. 10.5KVA	460V/480V	50/60	50A

Note 1) Power capacity: Refers to the power supply capacity of the controller. For the power capacity of each robot, refer to the “Manipulator Maintenance Manual”.

Note 2) Voltage range: $\pm 10\%$ (at the power terminal of the controller)

3.6.3.2. Power Cable Thickness

Table 3-3 Recommended Minimum Cable Thickness

No.	Cable length m	Cable thickness (Hi7-N00(U), Hi7-N80(U))		Cable thickness (Hi7-N30(U), Hi7-N20(U))	
		mm ²	AWG	mm ²	AWG
1	0 - 50	5.5	10	3.5	12
2	50 - 100	5.5	10	3.5	12
3	100 - 180	8	8	5.5	10
4	180 - 300	8	8	5.5	10

3.6.4. The Controller and Grounding

For using the controller safely, connect the grounding wire to the controller. Use a grounding wire of 5.5 mm² or more. (Grounding of Category 3).

The ground terminal location of the controller is shown in Figure 3.9.

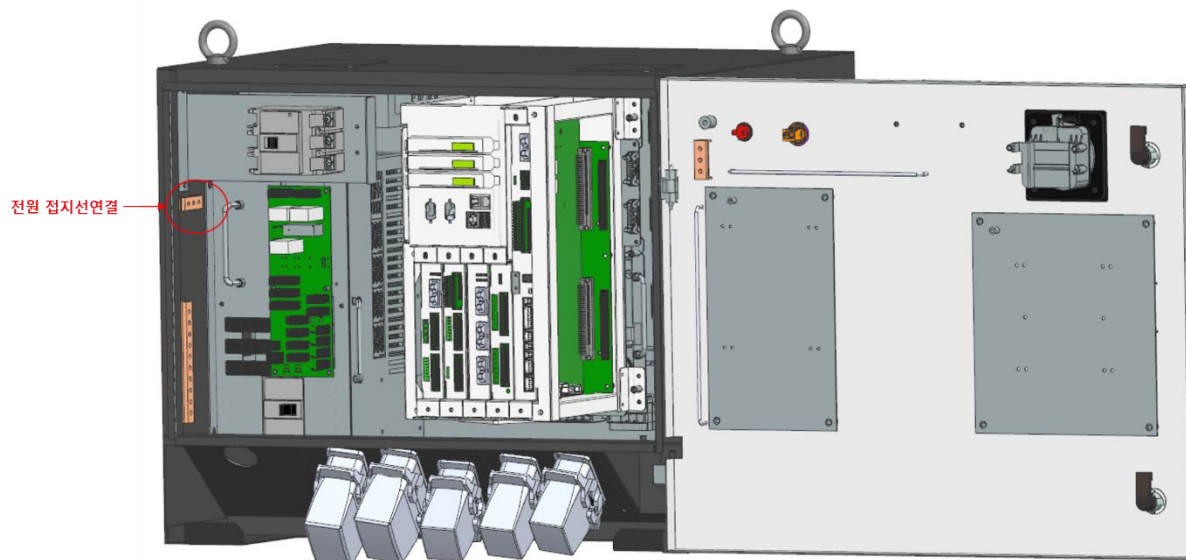


Figure 3.8 Power Ground Terminal of the Controller

3.6.5. Other Cautions



1. When wiring the controller and manipulator, separate the signal line and the power line.

In addition, use separate ducts respectively for high power lines and signal lines.

2. Use a protective cover for the wires, as a measure to prevent the wires from getting damaged when people are passing.

3. Before supplying the primary power, you must check again the relationship in terms of connection, the power specification and the power supply specification of the controller.

3.6.6. Connection of the Ethernet Port for the User

The Ethernet port for the user is located on the front door of the controller. The pin description and connection with a Pc are as follows.

Table 3-4 Pin Description (RJ45 Connector Specification: RJ 45P Shield)

RJ45 Pin No.	Name	Abbreviation	Direction
1	Transmit Data +	TX +	Out
2	Transmit Data -	TX -	Out
3	Receive Data +	RX +	In
6	Receive Data -	RX -	In



4

Basic Configuration of the Controller



4. Basic Configuration of the Controller



Caution

The person in charge of maintenance should work after understanding the placement of various devices and parts and their functions inside the controller.

4.1. Configuration

The controller consists of the main body and Teach Pendant.



Figure 4.1 Hi7-N**(U) Controller



Figure 4.2 Teach Pendant TP630

4.2. Placement of Parts

The main components of the Hi7-N00/N30/N80 controller and their individual names are shown below in Table 4-1 and are arranged as shown in Figure 4.3 to Figure 4.5.

Table 4-1 Names of Individual Parts of the Hi7-N00/N30/N80 Controller

No.	Type	Part name
1	H6COM-T	Main control module
2	BD642	Servo-safety board
3	BD604	Backplane board
4	H7PSM	Power supply module
5	H7D6X(Large/Medium-sized) /H7D6A(Small-sized)	Drive module for Large/Medium/Small-sized 6 axes
6	H7D1X (Optional)	Drive module for 1 axis of 100A
6-1	H7D1Z (Optional)	Drive module for 1 axis of 50A
7	OP EM. SW.	Emergency switch on the controller panel
8	NFB	No fuse breaker
9	FAN2	Drive module internal cooling fan
10- 12	FAN3~5	Drive module external cooling fan
13	NFT1	Line noise filter
14	RDR1	Small/Medium/Large-sized regenerative discharge resistor (CE/UL certified article)
15	TR	Transformer for the input power for options
16	CMC1	Motor drive power cable inlet connection
17	CMC2	Motor drive power cable inlet connection (Small controllers without a CMC2 mounted)
18	AMC1(Optional)	Inlet connector for the motor-driving power cable for options 1

No.	Type	Part name
19	AMC2(Optional)	Inlet connector for the motor-driving power cable for options 2
20	CEC1	Encoder communication cable inlet connection
21	AEC1	Inlet connector for the motor encoder cable for options 1
22	AEC2	Inlet connector for the motor encoder cable for options 2
23	CN RTP	Teach Pendant cable inlet connector



Figure 4.3 Placement of Parts on the Front Exterior of the Hi7-N00(U)/N30(U)/N80(U) Controller

4. Basic Configuration of the Controller

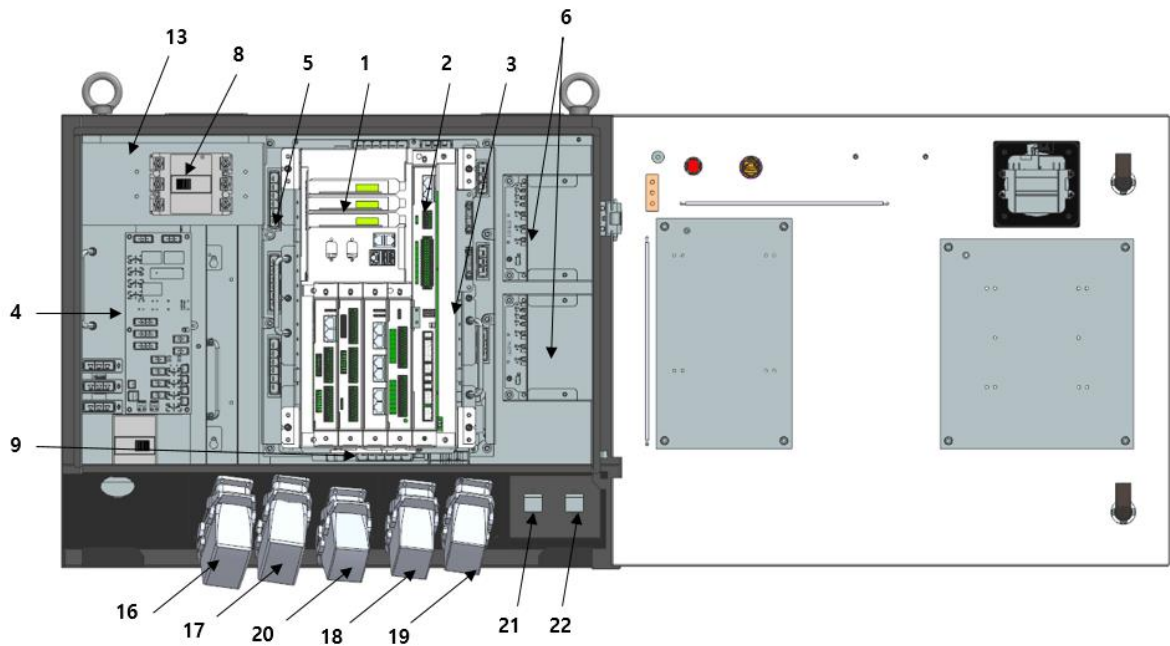


Figure 4.4 Placement of Parts on the Front Interior of the Hi7-N00(U)/N30(U)/N80(U) Controller

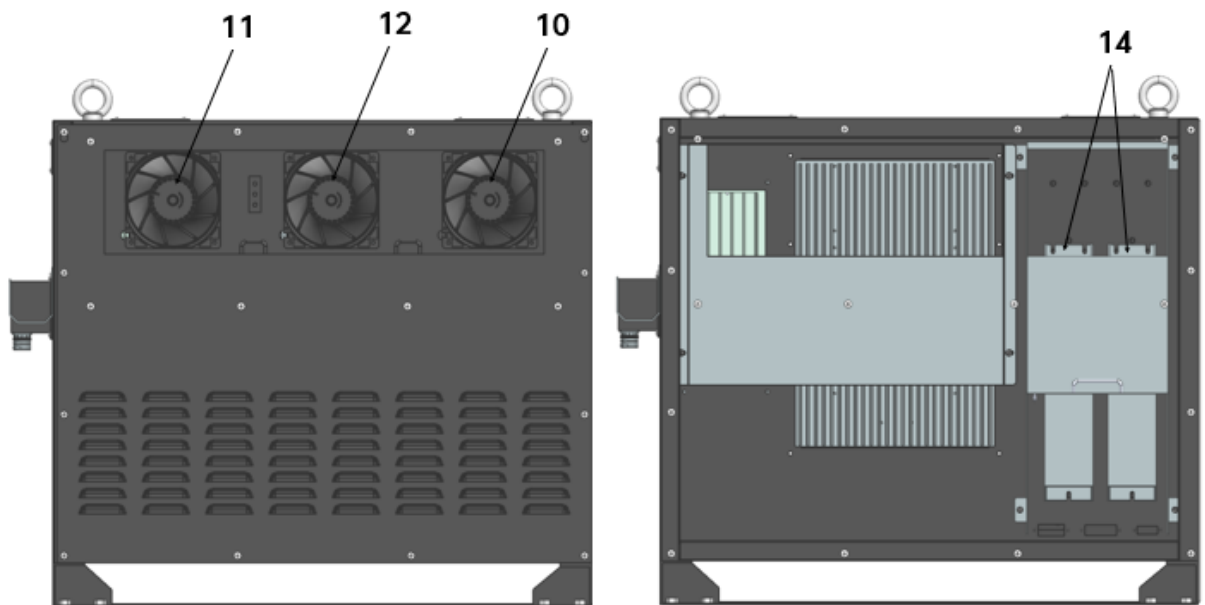


Figure 4.5 Placement of Parts on the Back of the Hi7-N00(U)/N30(U)/N80(U) Controller

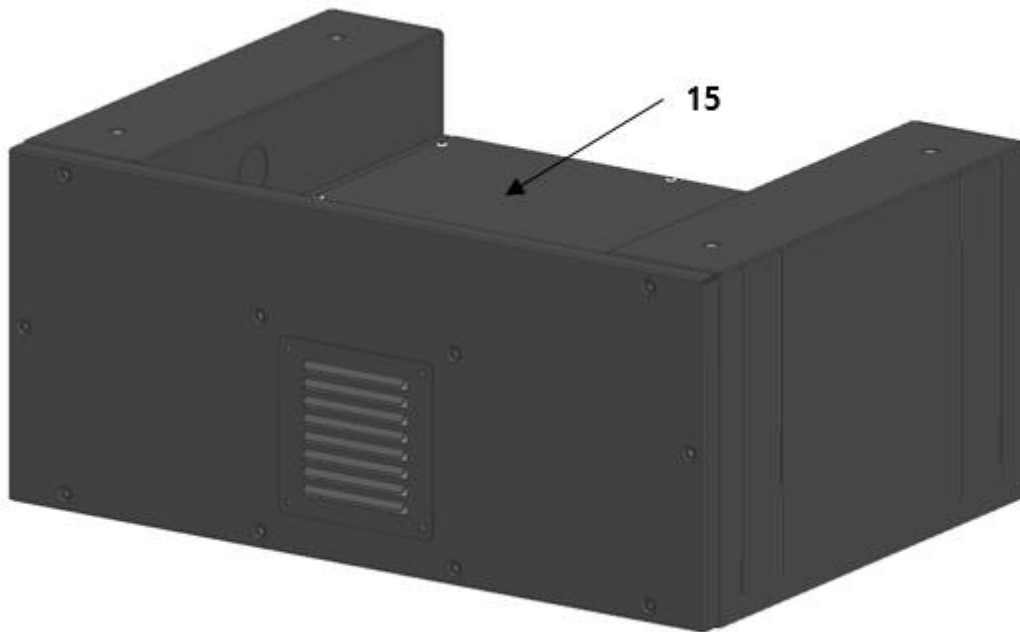


Figure 4.6 Hi7-N00(U),N30(u),N80(U) Controller Transformer Box

4.3. Functions of the Individual Components

Table 4-2 Summary of Functions of the Individual Components

Components		Functions
Control module	Main control module (H6COM-T)	● Recording the record points and calculating the operation paths ● Preservation of the programs and robot integers ● Teach pendant (T/P) communication ● Connection of the LAN, USB, and the serial (RS232) communication
	Servo board (BD642)	● DSP for servo control ● Encoder connection (Serial I/F) ● Open/close outputs for the servo motor ● Sequence control
	Safety module (BD604)	● I/O in the controller (I/O for the system) ● Processing of various input signals from the manipulator ● Safety chain circuit
Drive module	Large/Medium-sized 6 axes: H7D6X Small-sized 6 axes: H7D6A Additional axis: H7D1X, H6D1Z	● Generation of the motor drive power ● Regenerative discharge ● Servo motor power amplification circuit ● Various error outputs
T/P (Teach Pendant)	TP630	● Display of various information (LCD) ● Button inputs and switch inputs (function/jog, etc.) ● Emergency stop, enable, and T/P On/Off inputs
Cooling device	Fan	● Air circulation inside the panel ● Cooling of the drive module
Power supply module	H7PSM	● Opening/closing of the motor drive power ● Distribution of various power

※ For the types of components of each controller, refer to “2.1 Details of Specifications of Each Controller Model.”

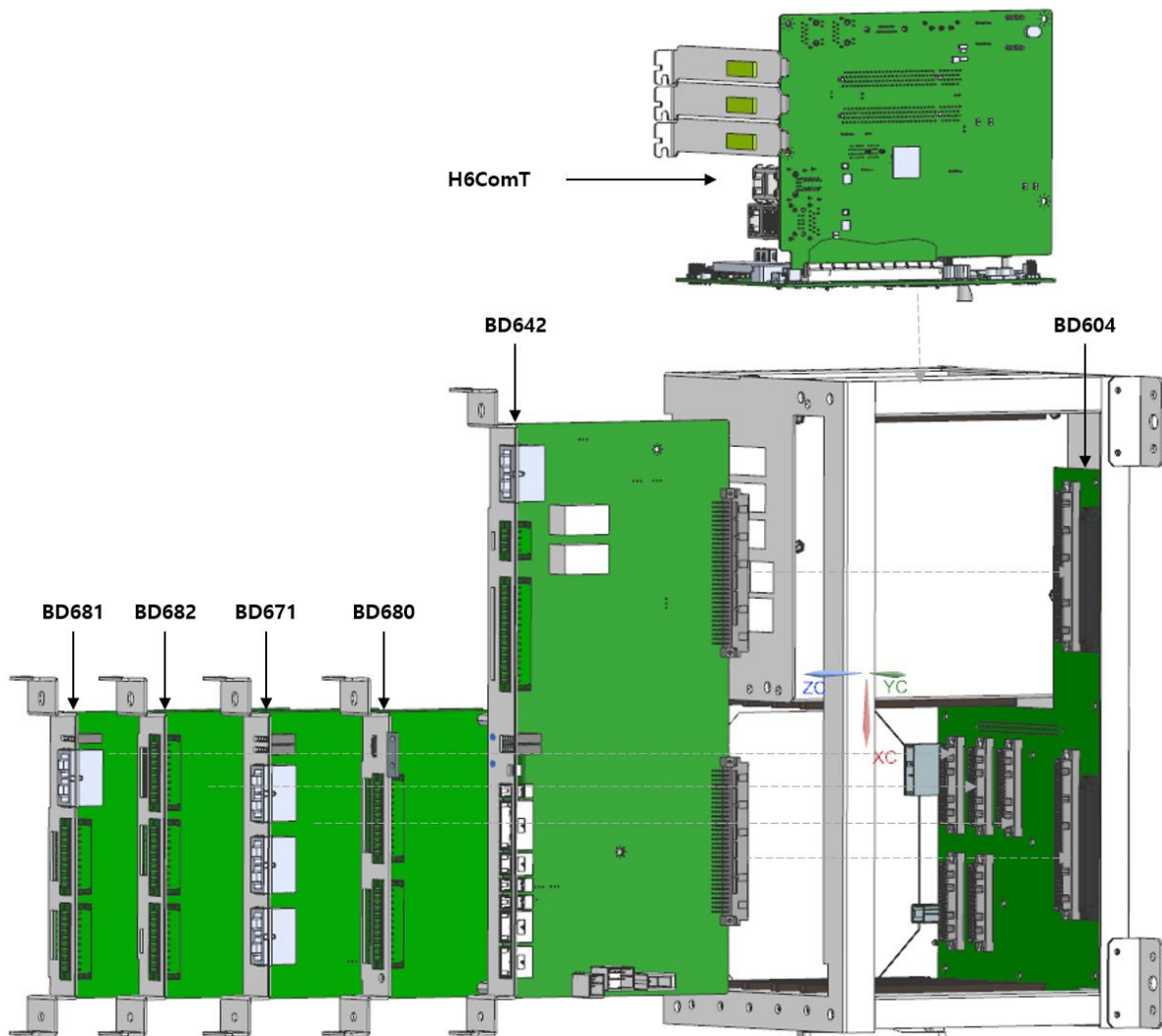


Figure 4.7 Components of the Control module

4.3.1. Main Module (H6COM-T)

4.3.1.1. Overview

H6COM-T is structure as shown in Figure 4.9, in which the main CPU board and the carrier board are combined. The main CPU board consists of an SSD slot, a CPU slot, a memory card slot, a USB port, a COM port, and a bus connector that is to be connected to the carrier board. The carrier board contains three LAN ports for external systems, two LAN ports for internal systems, two USB ports, one GPIO port, two PCI connectors, one PCI-e connector and one DC 24V power connector. The LAN ports for internal systems are used for EtherCAT communication, as well as for interface with the teach pendant, and the GPIO port is used to detect a power failure signal from the power system. The SB is used for debugging. One PCI expansion slot and three spare LAN ports for external systems are provided to support other universal bus interfaces. The connection to other communication interfaces than EtherCAT can be made via the relevant slots.

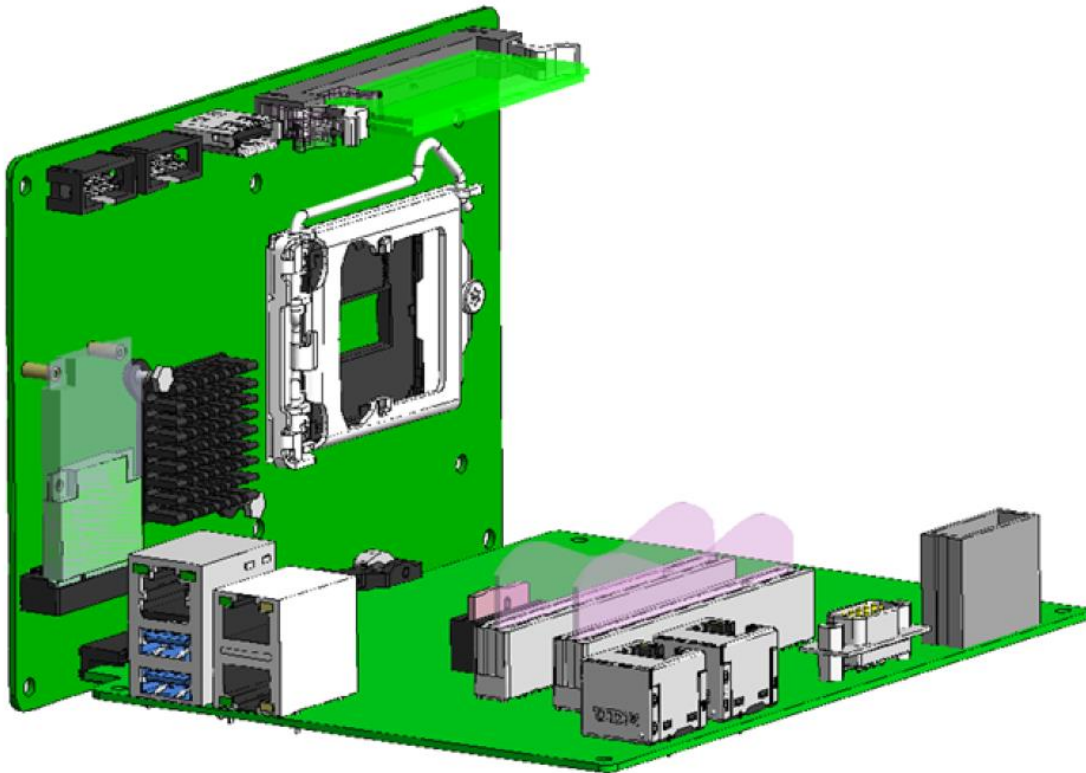


Figure 4.8 H6COM-T

4.3.1.2. Connectors

Table 4-3 describes the usage of the connector and the connection of external devices.

Table 4-3 Types and Usage of the Connectors of Hi7COM-T

Name	Usage	Connection of external devices
DC IN 12-24V	DC24V main power supply	-
COM 1,2	Serial port (RS232/RS422/RS485)	-
GIO	Application of the power failure of the power unit	-
LAN 4	EtherCAT master connector port	EtherCAT connector
LAN 5	Ethernet port: For communication between teach pendants	TP connector
LAN 1	Ethernet port: For the user (PC I/F)	Optional EtherCAT connector
LAN 2	Ethernet port: For the user (PC I/F)	
LAN 3	Ethernet port: For the user (PC I/F)	
PCI, PCIe	Optional expansion board slot	-
USB1,2	USB port: For the user (PC I/F)	

4.3.2. Safety Module (BD642)

4.3.2.1. Overview

For meeting the requirement of PLr=d cat3 (SIL2) in compliance with IOS 13849-1, the safety module (BD642) is designed in a dualized safety electric circuit and continuously monitors the status of safety-related inputs. If a system error or safety-related input is detected, this safety module renders the robot into a safe state by cutting off the motor power and brake power according to the classification of the stop types determined through risk assessment.

4.3.2.2. Connectors

The following figure shows the positions and usage of various connectors installed on BD642 (Safety IO Module).

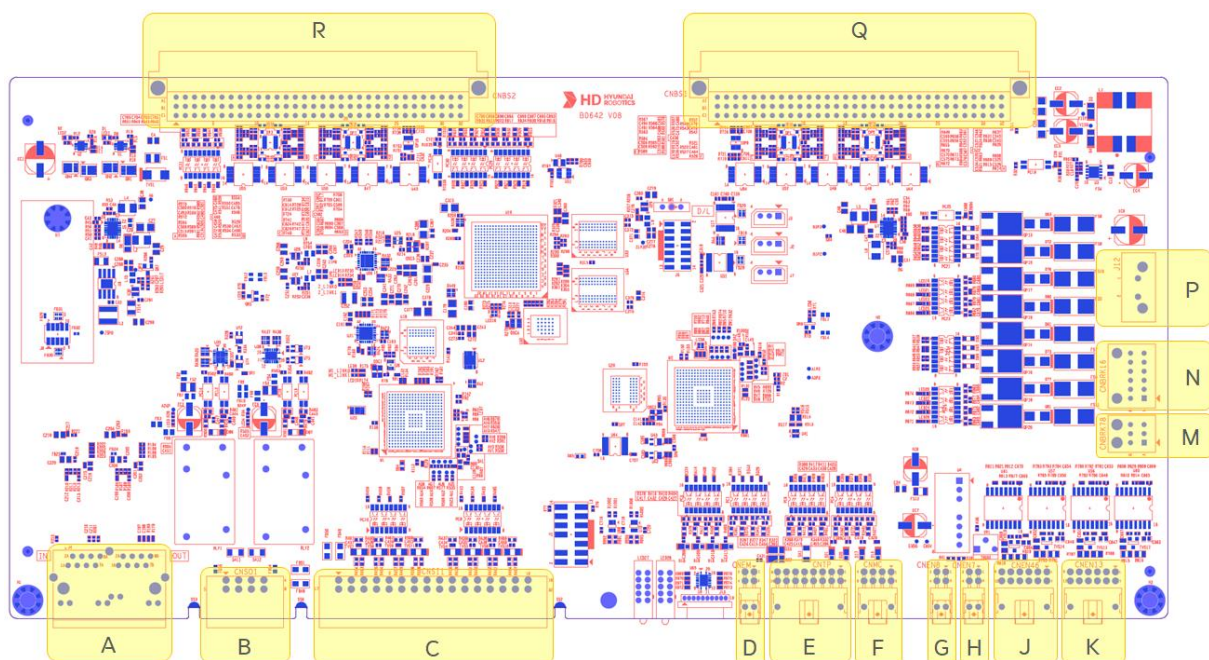


Figure 4.9 Placement of the Connectors and Switches of the BD642 (Safety IO Board)

Table 4-4 Types and Usage of the Connectors of the BD642 (Safety IO Board)

No.	Name	Usage	Connection of external devices
A	J4	EtherCAT communication	Hi6COM LAN4
B	CNSO1	Safety output terminal	External device
C	CNSI1	Safety input terminal	External device
D	CNEM	External emergency switch	Emergency switch
E	CNTP	Teaching pendant (power, emergency stop, mode switch, enable switch)	Connector CNRTP
F	CNMC	Magnet contact input/output signals	Power distribution board (BD6C2) CNMC
G	CNEN8	Encoder signal for additional axis 8	Connector AEC2
H	CNEN7	Encoder signal for additional axis 7	Connector AEC1
J	CNEN46	Encoder signal for axes 4–6	Connector CEC1
K	CNEN13	Encoder signal for axes 1–3	Connector CEC1
M	CNBRK78	Brake signal for additional axes 7 and 8	Connectors AMC1 and AMC2
N	CNBRK16	Brake signal for axes 1–6	Connectors CMC1 and CMC2
P	J12	Brake power	Power distribution board (BD6C2) CNOBK
Q	CNBS1	Drive device interface signals	Backplane board (BD604) CNBS1
R	CNBS2	Drive device interface signals	Backplane board (BD604) CNBS2


Caution

When a safety-related input is connected and activated, you must check whether the function is operating normally by referring to “1.11 Safety Measures When Operating the Robot.”

4. Basic Configuration of the Controller

4.3.2.3. Display Devices

(1) Top surface display of the board

The figure below shows the display devices (LED, 7-segment) on the servo/safety module (BD642). The table below describes the information of each display.

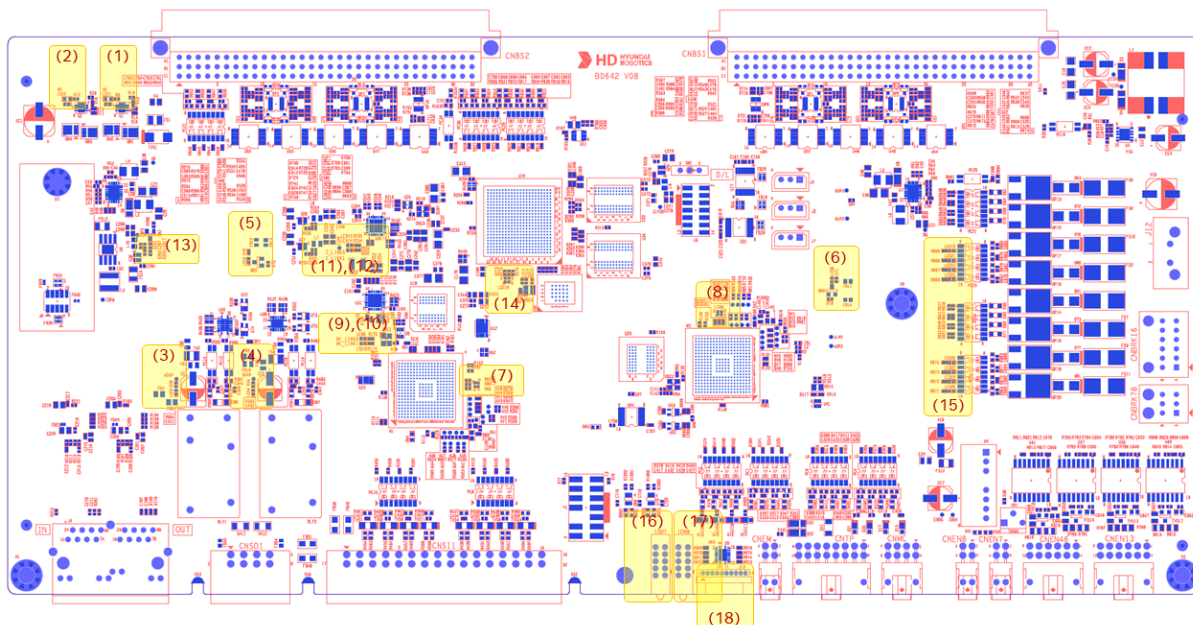


Figure 4.10 Top Board Display Layout of Servo/Safety Module (BD642)

Table 4-5 Top Board Display Description of Servo/Safety Module (BD642)

Number	Name	Description	Color	Normal Condition	Actions against Abnormal Condition
(1) (2)	LED1 LED2	Input power limiting function	Red	OFF	Symptom: Red LED ON Cause: Input voltage is under-voltage or over-voltage. Action: Check the input voltage (24 V).
(3)	LED3	External A channel power	Yellow	ON	Symptom: Yellow LED OFF Cause: External A channel power has overcurrent or external wiring fails, etc. Action: Check the fuse (FS2).
(4)	LED4	External B channel power	Yellow	ON	Symptom: Yellow LED OFF Cause: External B channel power has overcurrent and external wiring fails, etc.

					Action: Check the fuse (FS3).
(5)	LED5	A channel MCU power	Yellow	ON	Symptom: Yellow LED OFF Cause: A channel MCU power (3.3 V, 1.2 V) is abnormal. Action: Replace the board (BD642).
(6)	LED6	B channel MCU power	Yellow	ON	Symptom: Yellow LED OFF Cause: B channel MCU power (3.3 V, 1.2 V) is abnormal. Action: Replace the board (BD642).
(7)	LED7	A channel MCU state display	Red Green Blue	RGB blink	Symptom: All LED OFF and no blinking Cause 1: A channel MCU power (3.3 V, 1.2 V) is abnormal. Cause 2: A channel MCU program error, etc. occurs. Action: Replace the board (BD642).
(8)	LED8	B channel MCU state display	Red Green Blue	RGB blink	Symptom: All LED OFF and no blinking Cause 1: B channel MCU power (3.3 V, 1.2 V) is abnormal. Cause 2: B channel MCU program error, etc. occurs. Action: Replace the board (BD642).
(9) (10)	LED9 LED10	A channel MCU EtherCAT LINK0 state A channel MCU EtherCAT LINK1 state	Green Green	Green blink Green blink	Symptom: No blinking Cause: A channel MCU EtherCAT is abnormal. Action: Replace the board (BD642).
(11) (12)	LED13 LED14	FPGA EtherCAT LINK0 state FPGA EtherCAT LINK1 state	Green Green	Green blink Green blink	Symptom: No blinking Cause: FPGA EtherCAT is abnormal. Action: Replace the board (BD642).
(13)	LED17	FPGA power state	Yellow	ON	Symptom: Yellow LED OFF Cause: FPGA power (5 V, 3.3 V, 1.8 V, 1.35 V, and 1 V) is abnormal.

4. Basic Configuration of the Controller

					Action: Replace the board (BD642).
(14)	LED18	FPGA state display	Red Green Blue	RGB blink	Symptom: All LED OFF and no blinking Cause 1: FPGA power (5 V, 3.3 V, 1.8 V, 1.35 V, and 1 V) is abnormal. Cause 2: FPGA program error, etc. occurs. Action: Replace the board (BD642).
(15)	LED19 LED21 LED23 LED25 LED20 LED22 LED24 LED26	Axis 1 brake state Axis 2 brake state Axis 3 brake state Axis 4 brake state Axis 5 brake state Axis 6 brake state Axis 7 brake state Axis 8 brake state	Orange	Brake release (ON) Brake hold (OFF)	Symptom: Brake state mismatch Cause 1: Brake power is abnormal. Cause 2: Harness failure, etc. occurs. Action: Replace the board (BD642).
(16) (17) (18)	LED27 LED28 SEG1	Axis 1 brake state			Refer to the next front display device item

(2) Front display device of the board

The figure below shows the front display device of the servo/safety module (BD642). The table below describes the information of each display.

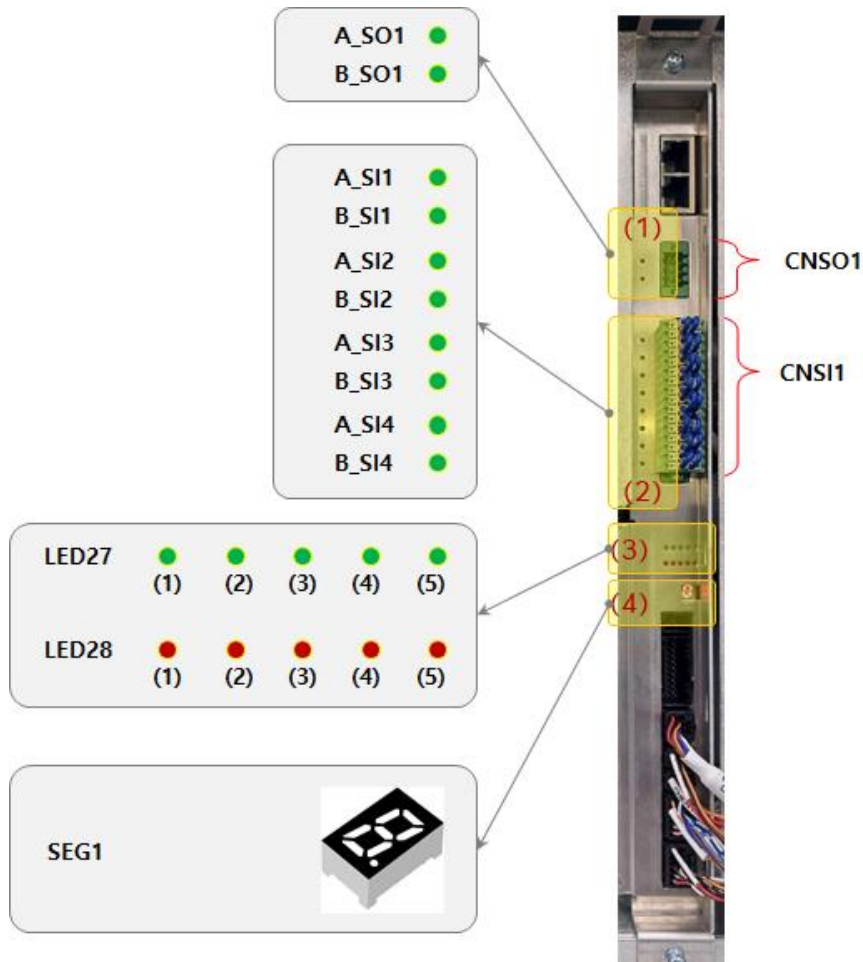


Figure 4.11 Front Display Layout of Servo/Safety Module (BD642)

Table 4-6 Front Display Description of Servo/Safety Module (BD642)

No.	Name	Description	Color	Display State	Display Description
(1)	A_SO1	A channel safety output 1 state	Green	ON	Each channel safety output 1 ON
	B_SO1	B channel safety output 1 state		OFF	Each channel safety output 1 OFF
(2)	A_SIx (x=1~4)	A channel safety input x state	Green	ON	Each channel safety input x ON

4. Basic Configuration of the Controller

	B_SIn (n=1~4)	B channel safety input x state		OFF	Each channel safety input x OFF
(3)	LED27 (1)	LED27 (1) display	Green		LED27 (1) MCU_A MOD
	LED27 (2)	LED27 (2) display			LED27 (2) MCU_B MOD
	LED27 (3)	LED27 (3) display			LED27 (3) ZYNQ MOD
	LED27 (4)	LED27 (4) display			LED27 (4) DSP_RUN
	LED27 (5)	LED27 (5) display			LED27 (5) ZYNQ_RUN
	LED28 (1)	LED27 (1) display	Red		LED28 (1) MCU_A STA
	LED28 (2)	LED27 (2) display			LED28 (1) MCU_B STA
	LED28 (3)	LED27 (3) display			LED28 (3) ZYNQ STA
	LED28 (4)	LED27 (4) display			LED28 (4) DSP ERR
	LED28 (5)	LED27 (5) display			LED28 (5) ZYNQ ERR
(4)	SEG1	BD642 board state	Red		Boot state

Table 4-7 Front LED Description of Servo/Safety Module (BD642)

LED	Color	Status	Description
MOD (Safety)	Green	OFF	Power_OFF
			INIT STATE
		GREEN Blinks(2s)	WAIT_ROBOT_INFO
		GREEN Blinks(1s)	WAIT_SAFE_PAFAMETER
		GREEN Blinks(0.5s)	INIT_PARAMETER
		On	OPERATING
STA (Safety)	Red	OFF	No error
		RED Blinks(2s)	MAINTENANCE
		RED Blinks(1s)	VIOLATION
		ON	FAULT
MOD (Servo)	Green	OFF	Powe_OFF
		GREEN Blinks(2s)	WAIT_DSP_BOOTING
		GREEN Blinks(0.5s)	WAIT_ETHERCAT_COMMUNICATION
		ON	OPERATING
STA (Servo)	Red	RED Blinnks(1s)	F/W Update
		On	Error / Fault
		OFF	Normal Operating

4. Basic Configuration of the Controller

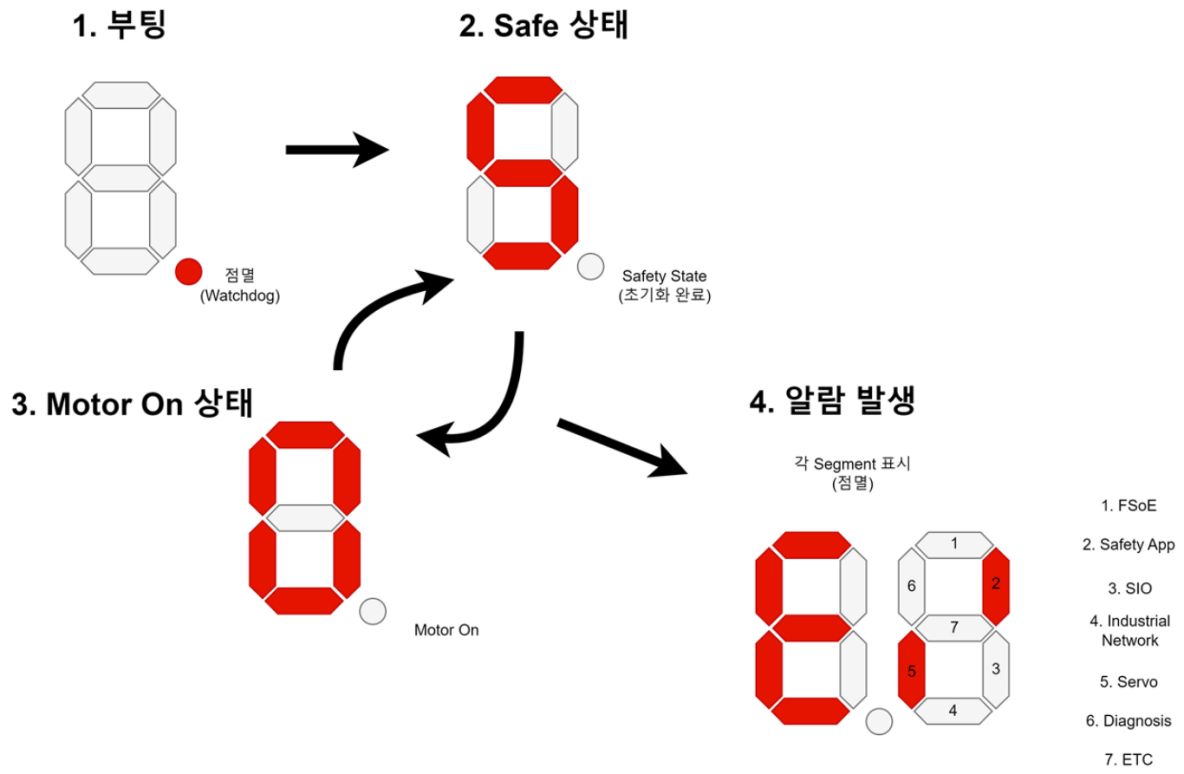


Figure 4.12 Segment State Display Description

4.3.2.4. Setting Devices

The figure below shows the positions of the setting (switch) devices on the servo/safety module (BD642). The table below describes the purpose of each setting.

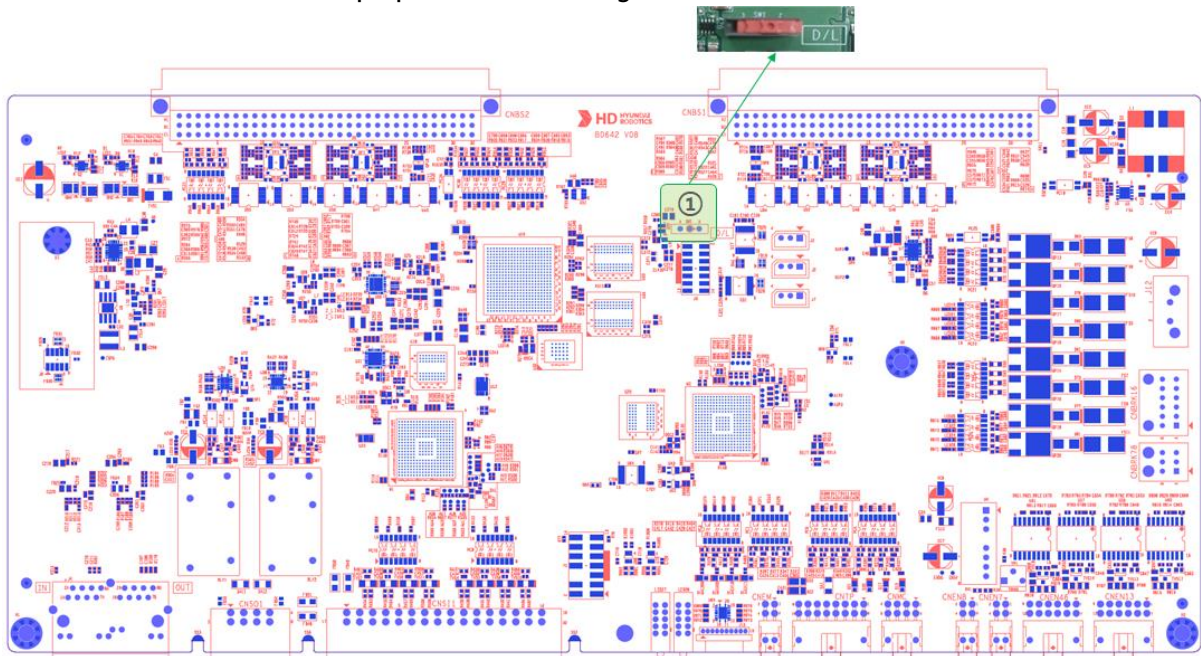


Figure 4.13 Description of the Setting Devices of the BD642 (Safety IO Board)



Warning

The following cannot be changed arbitrarily by the user and should only be referred to when FPGA JTAG reprogramming is required.

Table 4-8 Description of the SW1 setting devices of BD642 (Safety IO Module)

No.	Name	Setting State	Description	Remarks
①	SW1		Flash memory boot mode	Factory default settings
			JTAG program download mode	-

4. Basic Configuration of the Controller

4.3.2.5. Safety Output Wiring

(1) External Emergency Stop of Contact Input Type



Before installing the external wiring, you must turn off the main power.



Figure 4.14 Actual Photo of Servo/Safety Module (BD642), and Position of Safety Output Connector (CNSO1)

The wiring of safety outputs differs depending on whether to use an internal power supply or an external power supply, and also according to the NPN/PNP type. The following shows the wiring for each case.

(1) When using the internal power supply

● NPN-TYPE(: Active Low)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the internal power of A channel, connect pins 1–2 of CNSO1 as shown in the figure below.

When using the internal power of B channel, connect pins 3–4 of CNSO1 as shown in the figure below.

For connection with external devices, refer to the wiring examples below:

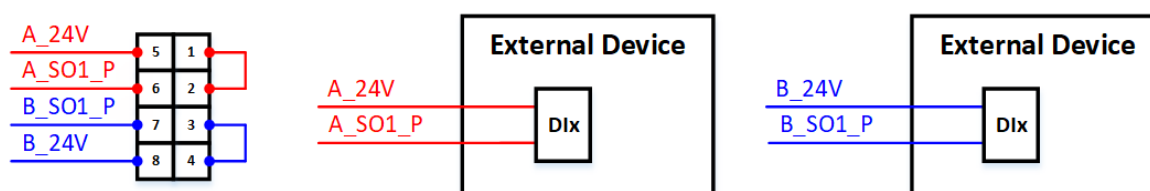


Figure 4.15 Safety Output Wiring with Internal Power Supply (NPN Type) for Servo/Safety Module (BD642)

● PNP-TYPE(: Active High)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the internal power of A channel, connect pins 5–6 of CNSO1 as shown in the figure below.

When using the internal power of B channel, connect pins 7–8 of CNSO2 as shown in the figure below.

For connection with external devices, refer to the wiring examples below:

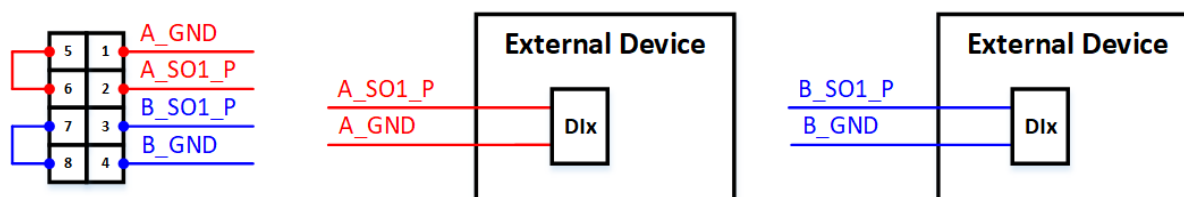


Figure 4.16 Safety Output Wiring with Internal Power Supply (PNP Type) for Servo/Safety Module (BD642)

4. Basic Configuration of the Controller

(2) When using the external power supply

- NPN-TYPE(: Active Low)

In the figure below, red lines represent A channel and blue lines represent B channel.

Do not connect pins 1, 4, 5, and 8 of CNSO1.

When using the external power of A channel, connect EX_AG (GND) to pin 2 of CNSO1 as shown in the figure below.

When using the external power of B channel, connect EX_BG (GND) to pin 3 of CNSO1 as shown in the figure below.

For connection with external devices, refer to the wiring examples below

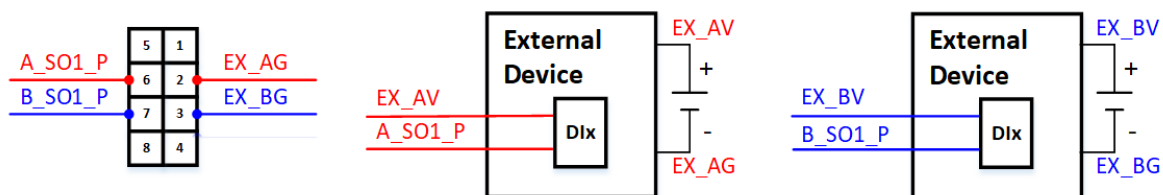


Figure 4.17 Safety Output Wiring with External Power Supply (NPN Type) for Servo/Safety Module (BD642)

- PNP-TYPE(: Active High)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the external power of A channel, connect EX_AV (24 V) to pin 2 of CNSO1 as shown in the figure below.

When using the external power of B channel, connect EX_BV (24 V) to pin 3 of CNSO1 as shown in the figure below.

For connection with external devices, refer to the wiring examples below:

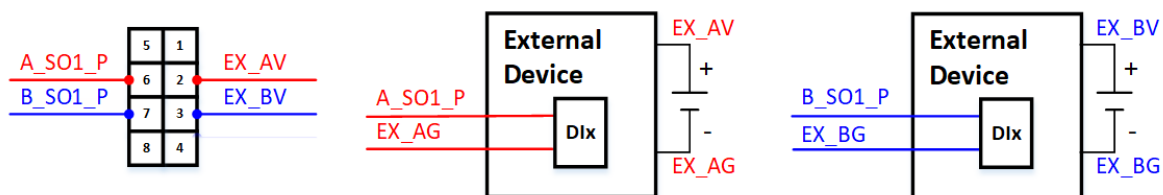


Figure 4.18 Safety Output Wiring with External Power Supply (PNP Type) for Servo/Safety Module (BD642)

4.3.2.6. Safety Input Wiring



Before installing the external wiring, you must turn off the main power.

The figure below shows the actual photo of the servo/safety module (BD642) and the position of the safety input connector (CNSI1) when viewed from the front during installation.

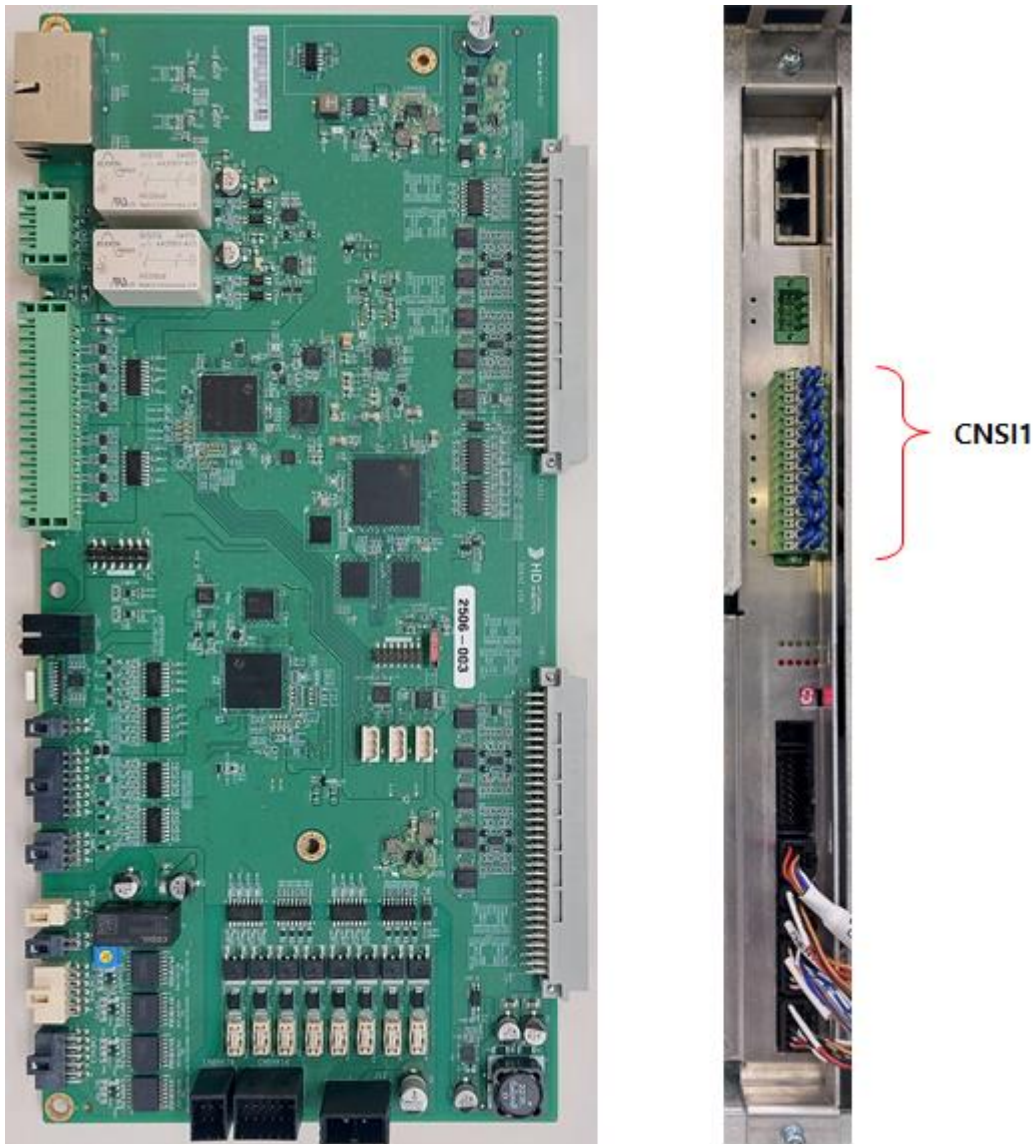


Figure 4.19 Actual Photo of Servo/Safety Module (BD642), and Position of Safety Input Connector (CNSI1)

4. Basic Configuration of the Controller

(1) Factory setting of safety input (When not in use)

When safety input signals are not used, they should be connected in the default NC (Normal Close, B contact) configuration. The following shows the wiring when not in use (factory setting connection).

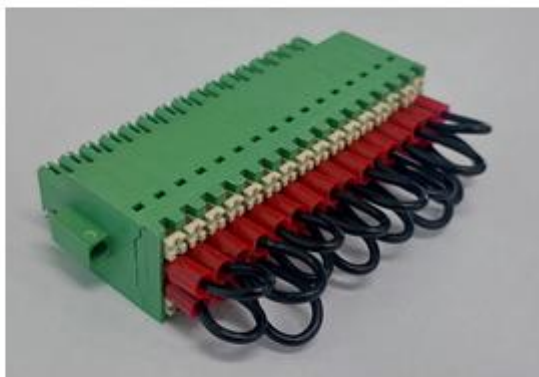
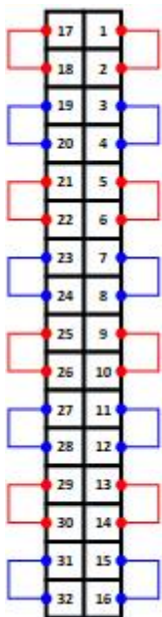


Figure 4.20 Safety Input Factory Setting of Servo/Safety Module (BD642)

The wiring of safety inputs differs depending on whether to use an internal power supply or an external power supply, and also according to the NPN/PNP type. The following shows the wiring for each case.

(2) When using the internal power supply

- NPN-TYPE(: Active Low)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the internal power of A channel, connect pins 17–18, 21–22, 25–26, and 29–30 of the channel's CNSI1 as shown in the figure below.

When using the internal power of B channel, connect pins 19–20, 23–24, 27–28, and 31–32 of the channel's CNSI1 as shown in the figure below.

For connection with external devices, refer to the wiring examples below:

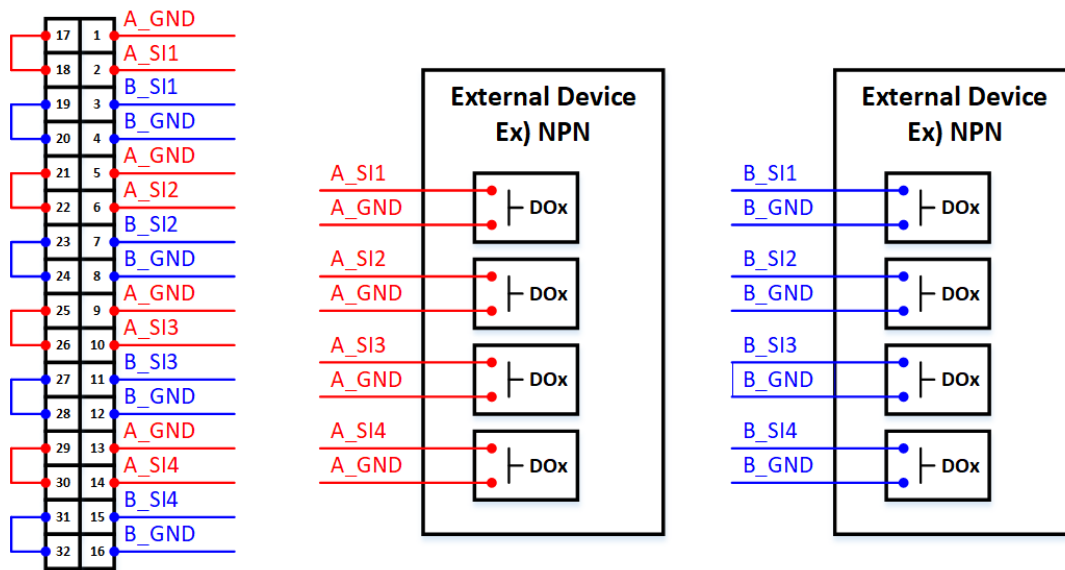


Figure 4.21 Safety Input Wiring with Internal Power Supply (NPN Type) for Servo/Safety Module (BD642)

● PNP-TYPE(: Active High)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the internal power of A channel, connect pins 1–2, 5–6, 9–10, and 13–14 of the channel's CNSI1 as shown in the figure below.

When using the internal power of B channel, connect pins 3–4, 7–8, 11–12, and 15–16 of the channel's CNSI1 as shown in the figure below.

For connection with external devices, refer to the wiring examples below

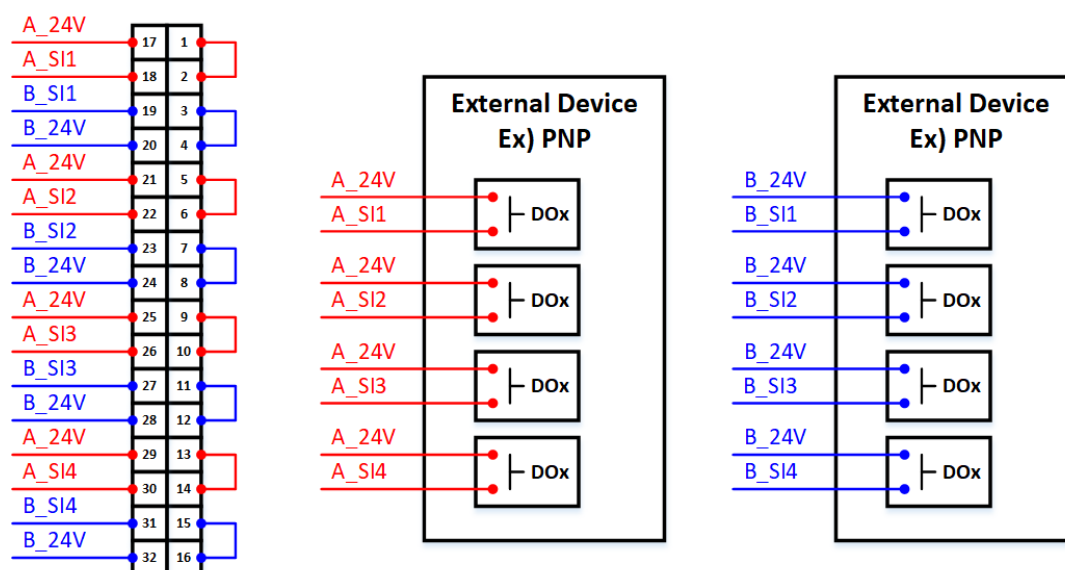


Figure 4.22 Safety Input Wiring with Internal Power Supply (PNP Type) for Servo/Safety Module (BD642)

4. Basic Configuration of the Controller



Warning When connecting the internal power supply to external devices, do not use it as the power source for them.

(3) When using the external power supply

● NPN-TYPE(: Active Low)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the external power of A channel, do not connect pins 1, 17, 5, 21, 9, 25, 13, and 29 of CNSI1.

When using the external power of B channel, do not connect pins 4, 20, 8, 24, 12, 28, 16, and 32 of CNSI1.

For connection with external devices, refer to the wiring examples below:

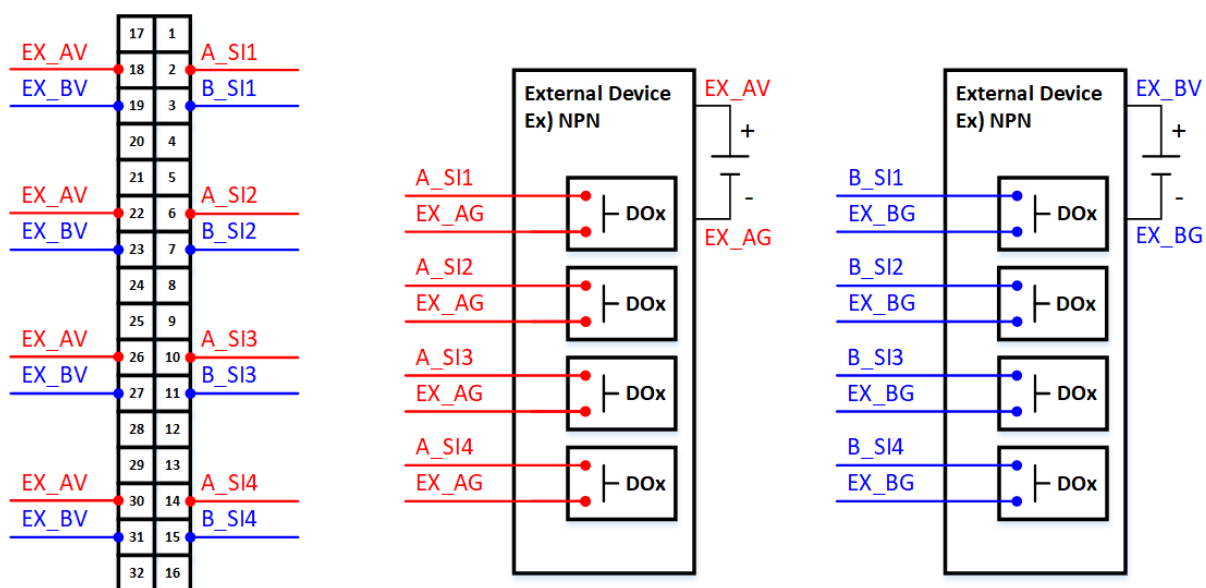


Figure 4.23 Safety Input Wiring with External Power Supply (NPN Type) for Servo/Safety Module (BD642)

● PNP-TYPE(: Active High)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the external power of A channel, do not connect pins 1, 17, 5, 21, 9, 25, 13, and 29 of CNSI1.

When using the external power of B channel, do not connect pins 4, 20, 8, 24, 12, 28, 16, and 32 of CNSI1.

For connection with external devices, refer to the wiring examples below:

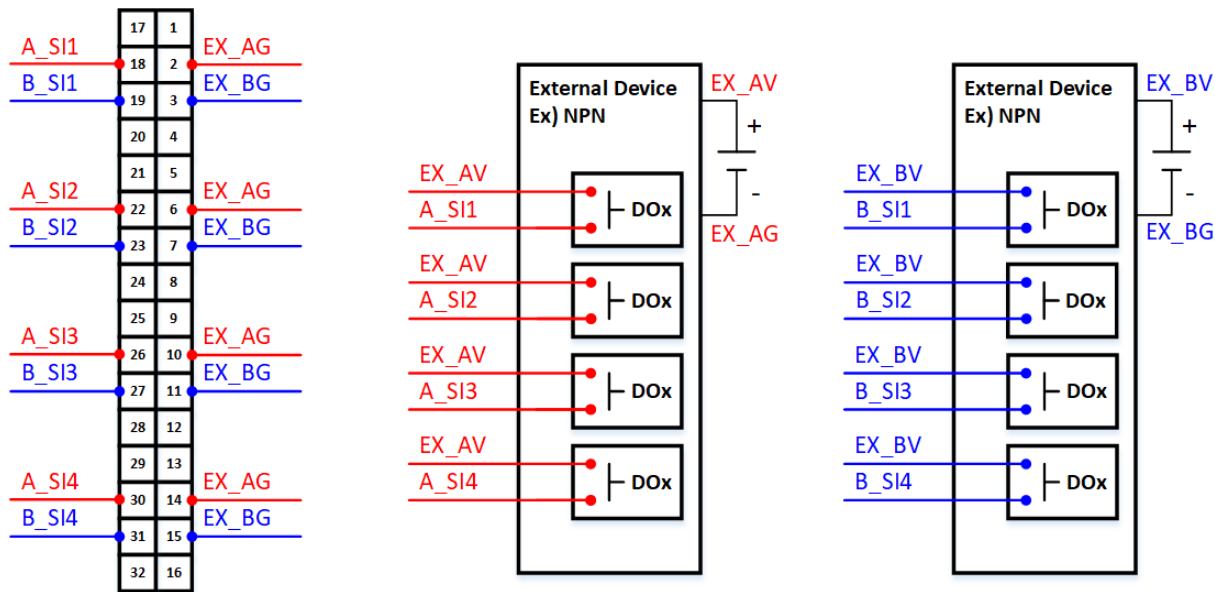


Figure 4.24 Safety Input Wiring with External Power Supply (PNP Type) for Servo/Safety Module (BD642)

4.3.3. Backplane Board (BD604)

4.3.3.1. Overview

The backplane board (BD604), as shown in Figure 4.50, supplies control power to Hi6a boards and transmits AMP-related signals generated from BD642 through the AMP interface board (BD652/BD654). It also serves to mount major optional boards and transmit signals between them.

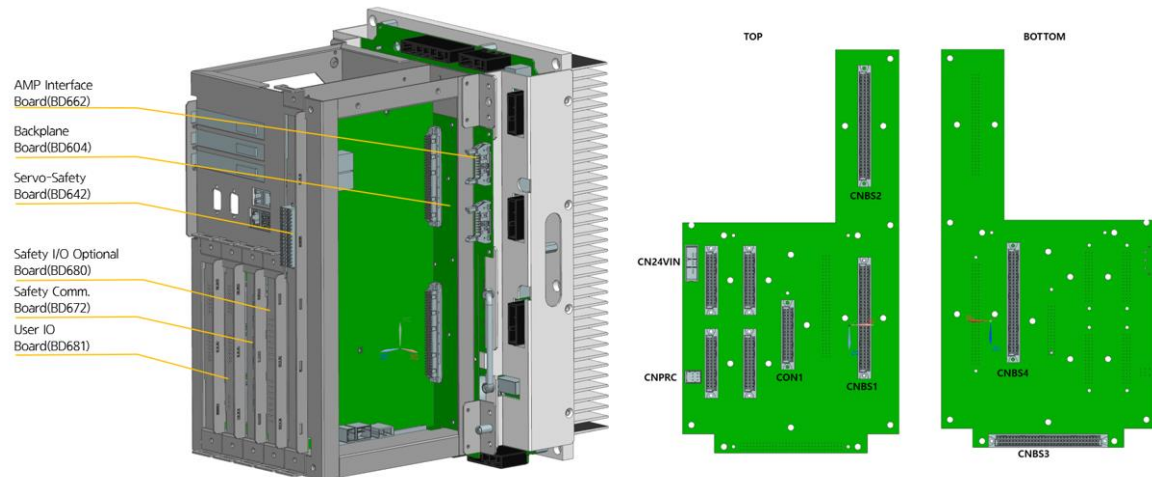


Figure 4.25 Backplane Board Structure

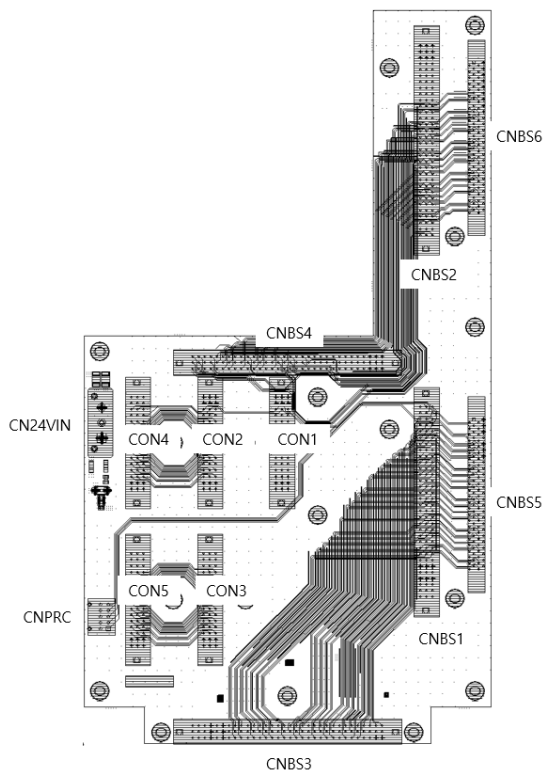


Figure 4.26 Backplane Board Connectors

4.3.3.2. Connectors

Table 4-9 below describes the function of each connector and its connection to external devices.

Table 4-9 Types and Functions of Backplane Board Connectors

Name	Function	External Device Connection
CN24VIN	DC 24 V main power supply	CNOCM connector on the power distribution board (BD6C3)
CNPRC	Precharge relay and fan relay control and monitoring	CNPRC connector on the power distribution board (BD6C3)
CNBS1, CNBS2	Power and signal connector for servo safety board (BD642)	CNBS1 and CNBS2 connectors on servo safety board (BD632)
CNBS3, CNBS4	Power and signal connector for AMP interface board (BD652/BD654)	CNBS1 and CNBS2 connectors on AMP interface board (BD652/BD654)
CNBS5, CNBS6	Power and signal cable connector for AMP interface board (BD652/BD654)	-
CON1	Power and signal connector for safety communication board (BD671)	CN1 connector on safety communication board (BD671)
CON2, CON3	Power and signal connector for user IO board (BD681)	CN3 and CN4 connectors on user IO board (BD681)
CON4, CON5	Power and signal connector for user IO expansion board (BD682)	CN4 and CN5 connectors on user IO expansion board (BD682)

The above connector configuration is shown in Figure 4.27.

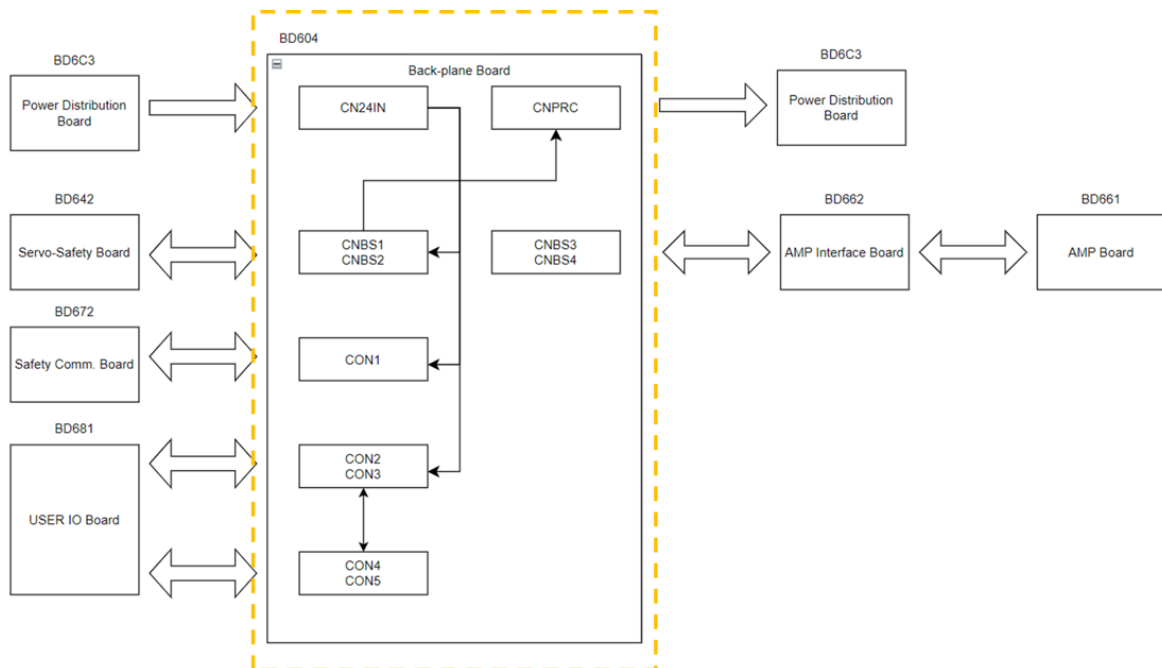


Figure 4.27 Connector Configuration of Backplane Board

4.3.4. Drive Module

4.3.4.1. H7D6X (Medium-Sized 6 Axes Integrated Drive Module)

The drive module performs a power amplification function that allows the current to flow to the individual phases of the motor according to the current command from the servo board. The six axes integrated drive module can drive six motors at the same time and is configured as follows.

The three-phase current entered from the power supply module is rectified through a diode module and then converted into direct current and stored in a smoothing capacitor. When the motor speed of the robot is decelerated, the electric power generated from the motor will be consumed through IGBT and resistors. The relevant configuration is as follows.

Table 4-10 Configuration of H6D6X (Medium-Sized 6 Axes Integrated Drive Module)

Components		Functions
BD651 (Power board)	Gate drive circuit	Transmit the IPM gate signal
	Gate power module	Generates the IPM gate power
	Current detection part	Detects the current that flows through the motor
	Regenerative control	Drives the IGBT when the PN voltage rises
	Error detection part	Detects the PN overvoltage, regenerative discharge resistor overheating, and PN undervoltage errors
	High voltage capacitor	Smooths the direct current
BD652 (Interface board)	Sequence interlocking part	Interlocks between the sequence status and the servo on signal
	Dedicated IO Terminal Blocks	Reserved IO port unit inside the controller
Other parts	Heat sink	Releases the heat generated from power elements to the outside
	Rectification part	Rectify the AC input power to generate DC power for driving the motor
	Regenerative IGBT	Performs regenerative discharge
	IPM	Convert the power for driving a three-phase motor

■ Configuration of the Type Number of the Medium-Sized 6 Axes Integrated Drive Module

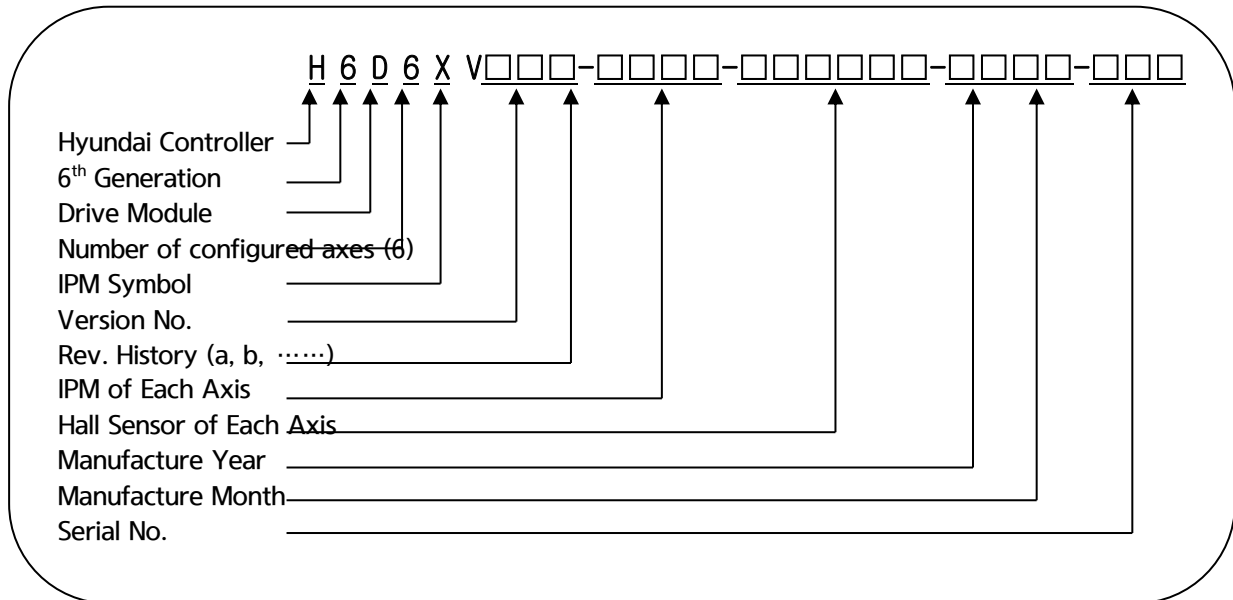


Table 4-11 Type Symbol of the Medium-Sized 6 Axes Integrated Drive Module

Category	Type symbol
Hi7 Medium-Sized 6 Axes drive module	H7D6X

Table 4-12 Specification of the Medium-Sized 6 Axes Integrated Drive Module

Configurati on	Classification		Application	
IPM capacity	3X	3Y	HS180, HS220, HH300, HH050	6 axes integrated
	4X	2Y	HC2502B2D, HC2503B2D	
Year	00~99		Production year: 2000~2099	
Month	01~12		Production month: January~December	
Serial No.	001~999		Number of units produced monthly: 1~999	

4. Basic Configuration of the Controller

Table 4-13 Symbols of the IPM of the medium-sized 6 axes drive module

Drive Model	IPM symbol	IPM Specification
Drive module of the medium-sized 6 axes	X	(IPM current rating) 100A
	Y	(IPM current rating) 75A
	Z	(IPM current rating) 50A

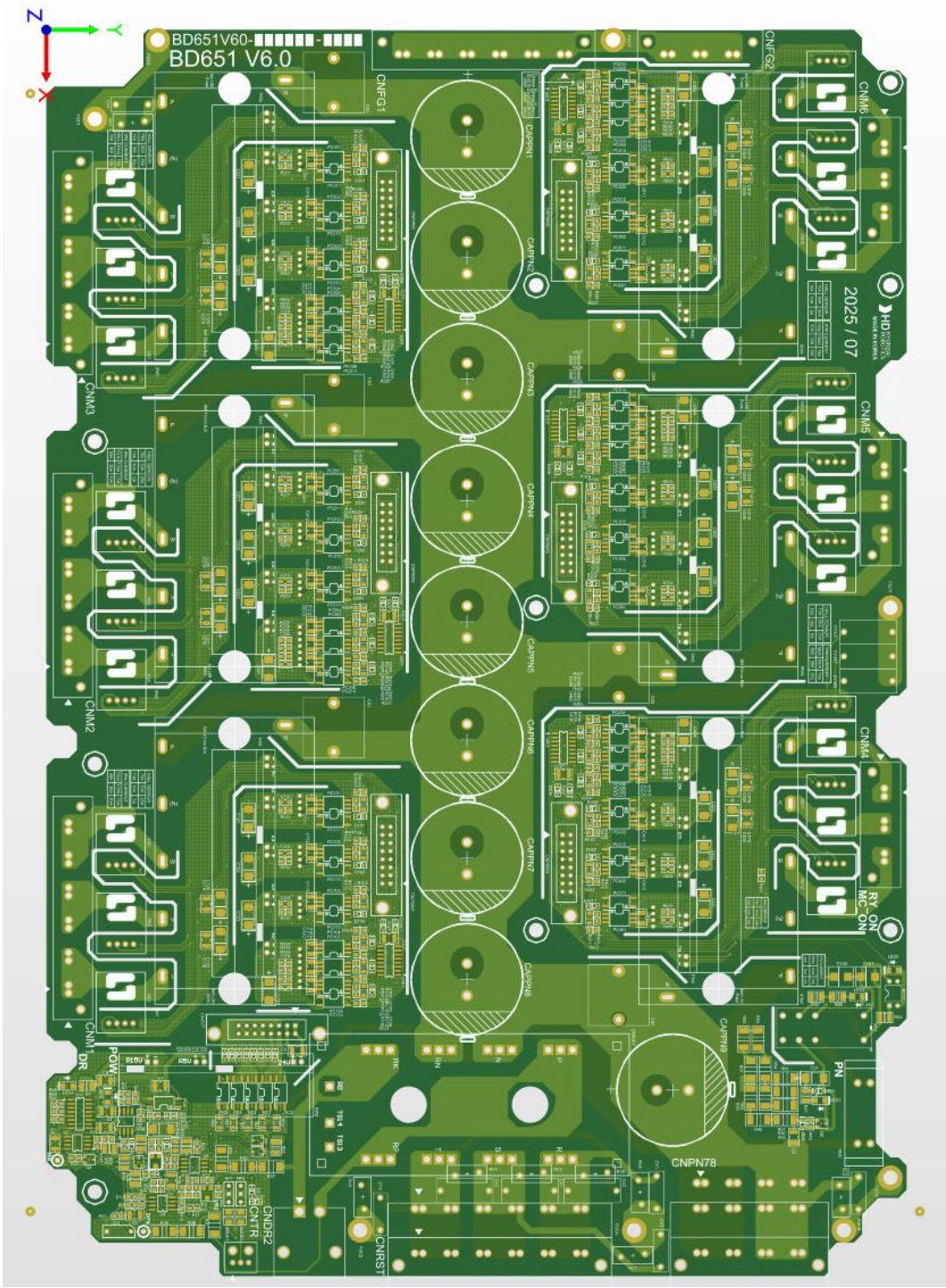
Table 4-14 Symbols of the Hall Sensors of the medium-sized 6 axes drive module

Drive model	Hall sensor symbol (Specification)	Full-scale current (Im)	IPM specification (Rated current)
Drive module of the medium-sized 6 axes	1 (4V/50A)	93.75 Apeak	PM100CG1APL065 202G (100A) PM75CG1APL065 202G (75A) PM50CG1APL065 202G (50A)
	2 (4V/25A)	46.87 Apeak	
	3 (4V/15A)	28.12 Apeak	
	4 (4V/10A)	18.75 Apeak	
	5 (4V/ 5A)	9.37 Apeak	



Caution

The drive module differs depending on the type of the robot, so you must check the type when replacing it.



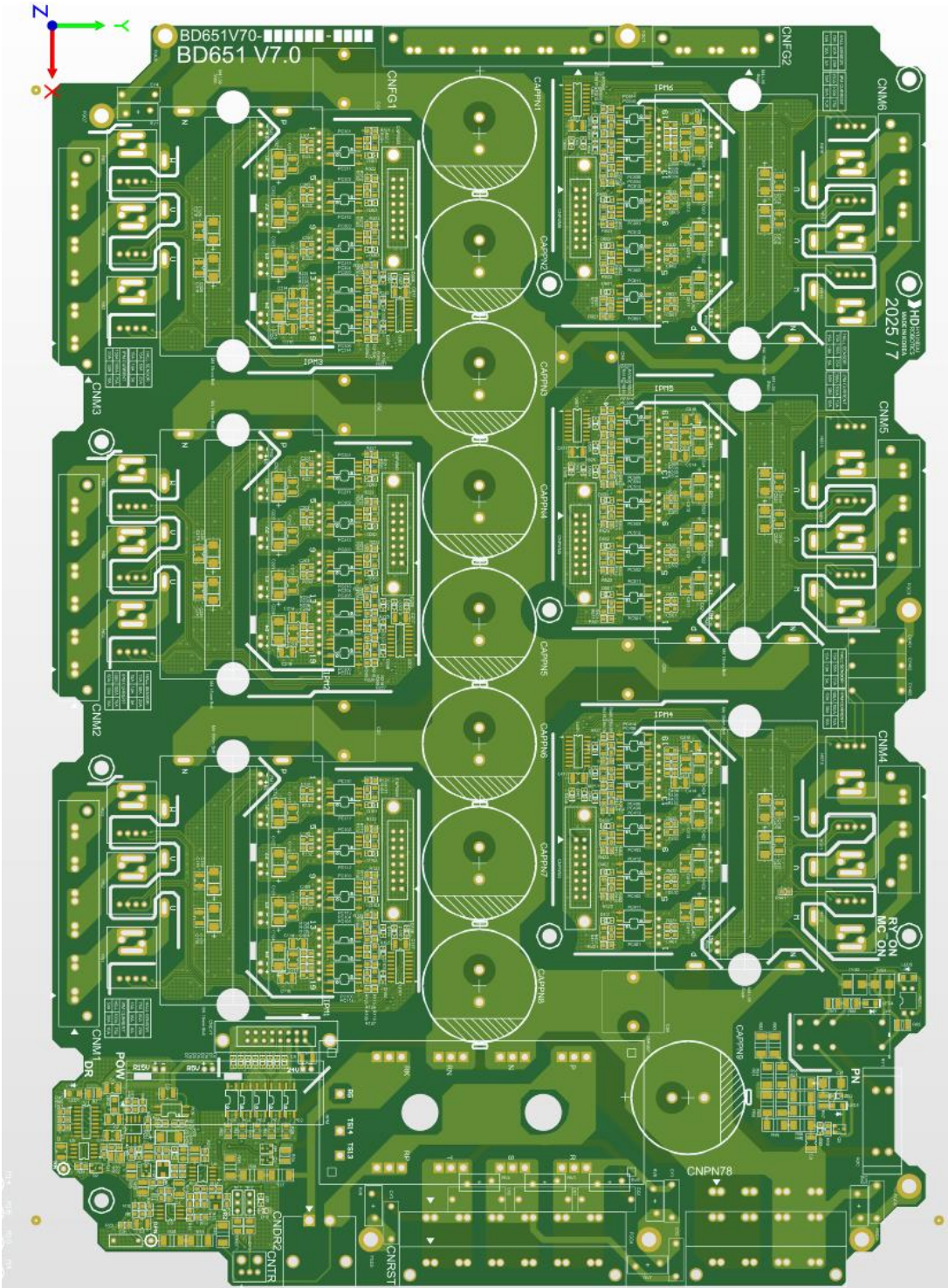


Figure 4.28 Component Layout of BD651V60 and BD651V70

Table 4-15 Description of the Connectors of BD651

Name	Usage	Connection of external devices
------	-------	--------------------------------

CNPWM1-6	PWM signals and IPM error signals	Board-to-board connectors of the BD652
CNRST	3-phase power input	CNRST for the electronic module
CNCVT	Converter part error signal	Board-to-board connectors of the BD652
CNDR	Regenerative discharge power output	Regenerative discharge resistor
CNTR	Regenerative discharge resistor overheating detection	Regenerative discharge resistor temperature sensor
CNM1~3	Motor drive output for axis 1 to axis 3	CMC1
CNM4~6	Motor drive output for axis 4 to axis 6	CMC2
CNPN7~8	Direct current power for the drive module of the additional axis	CNPN for the drive module for an optional additional axis.
CNFG1	Frame ground for axis 1 to axis 3	CMC1
CNFG2	Frame ground for axis 4 to axis 6	CMC2

Table 4-16 Description of LEDs of BD651

Name	Color	Status Display
MC ON	Yellow	Will be turned on when the magnet contact is driving
POW	Green	Will be turned on when the control voltage of the converter part is normal
DR	Red	Will be turned on the regenerative discharge is operating
PN	Red	Will be turned on when the PN voltage is higher than 42V
RYON	Red	Will be turned off when the PN discharge is operating

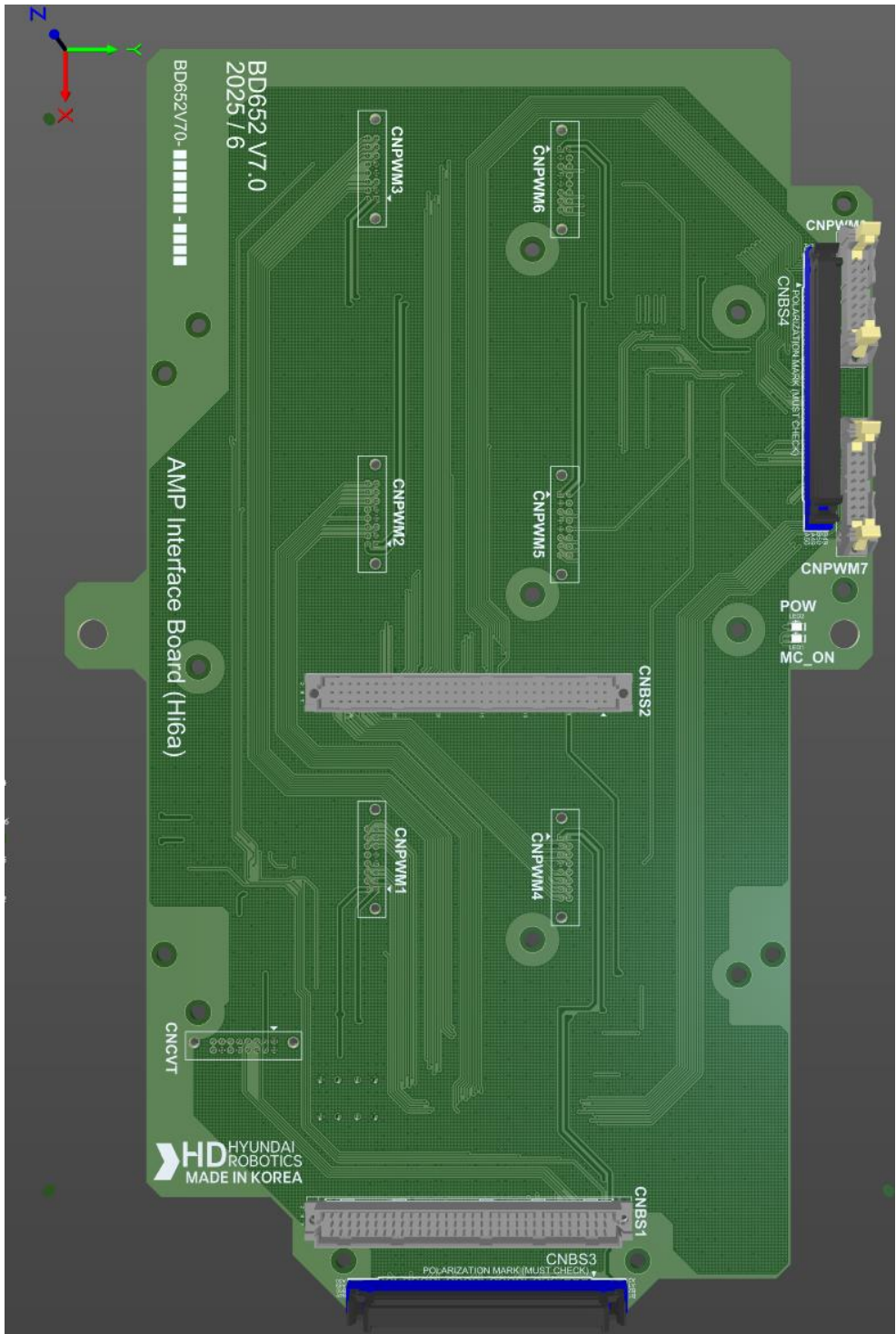


Figure 4.29 Component Layout of BD652V60 and BD652V70

4. Basic Configuration of the Controller

Table 4-17 Description of the Connectors of BD652

Name	Usage	Connection of external devices
CNBS1~3	PWM signals and IPM error signals for 8 axes Converter part error signal	Board-to-board connectors of BD640
CNPWM1~6	PWM signals and IPM error signals for individual axes	Board-to-board connectors of BD651
CNPWM7~8	PWM signal and IPM error signal for the additional axis	CNPWM of the drive module (BD658 or BD659) of the additional axis
CNCVT	Converter part error signal	Board-to-board connectors of BD651

Table 4-18 Description of the LEDs of BD652

Name	Color	Status Display
MC	Yellow	Will be turned on when the magnet contact is driving
POW	Green	Will be turned on when the control power is normal

4.3.4.2. H7D6A (Small-Sized 6 Axes Integrated Drive Module)

The drive module performs a power amplification function that allows the current to flow to individual phases of the motor according to the current command from the servo board. The six axes integrated drive module can drive six motors at the same time and is configured as follows.

The three-phase current entered from the power supply module is rectified through a diode module and then converted into direct current and stored in a smoothing capacitor. When the motor speed of the robot is decelerated, the electric power generated from the motor will be consumed through IGBT and resistors. The relevant configuration is as follows.

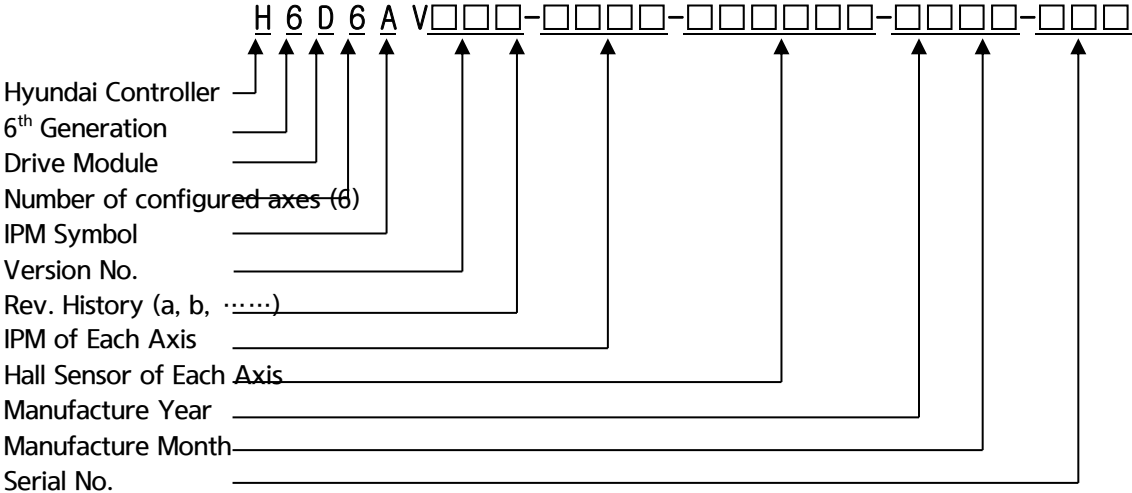
Table 4-19 Configuration of H6D6A (Small-Sized 6 Axes Integrated Drive Module)

Components		Components
BD653 (Power Board)	Gate drive circuit	Generates the IPM gate signal
	Gate power module	Generates the gate power
	Current detection part	Detects the current that flows through the motor
	Regenerative control	Turn on the IGBT when the PN voltage rises
	Error detection part	Detects the overvoltage, regenerative resistor overheating, and undervoltage errors
	High voltage capacitor	Smooths the direct current
BD654 (Interface Board)	Sequence interlocking part	Interlocks between the sequence status and the servo on signal
	Dedicated IO Terminal Blocks	Reserved IO port unit inside the controller
Other Parts	Heat sink	Releases the heat generated in power elements to the outside
	Rectification part	Rectify the AC input power to generate DC power for driving the motor
	Regenerative IGBT	Performs regenerative discharge
	IPM	Convert the power for driving a three-phase motor



Caution

The drive module differs depending on the type of the robot, so you must check the type when replacing it.



Category	Type symbol
Hi7 Small-Sized 6 Axes drive module	H7D6A

Configuration	Classification		Application	
IPM capacity	3A	3D	HA006B, HH020	6 axes integrated
Year	00~99		Production year: 2000~2099	
Month	01~12		Production month: January~December	
Serial No.	001~999		Number of units produced monthly: 1~999	

Table 4-22 Symbols of the IPM of the small-sized 6 axes drive module

Drive Model	IPM symbol	IPM Specification
Drive module of the small-sized 6 axes	A	(IPM allowable current rating) 30A
	D	(IPM allowable current rating) 10A

Table 4-23 Symbols of the Hall Sensors of the small-sized 6 axes drive module

Drive model	Hall sensor symbol (Specification)	Full-scale current (Im)	IPM specification (Allowable current rating)
Drive module of the small-sized 6 axes	3 (4V/15A)	27.27 Apeak	6MBP50VAA060 (30A)
	4 (4V/10A)	18.18 Apeak	
	5 (4V/5A)	9.19 Apeak	6MBP20VAA060 (10A)
	6 (4V/3A)	5.45 Apeak	
	7 (4V/6A)	10.91 Apeak	6MBP50VAA060 (30A)
	8 (4V/2A)	3.64 Apeak	6MBP20VAA060 (10A)
	9 (4V/1A)	1.82 Apeak	



Caution

The drive module differs depending on the type of the robot, so you must check the type when replacing it.

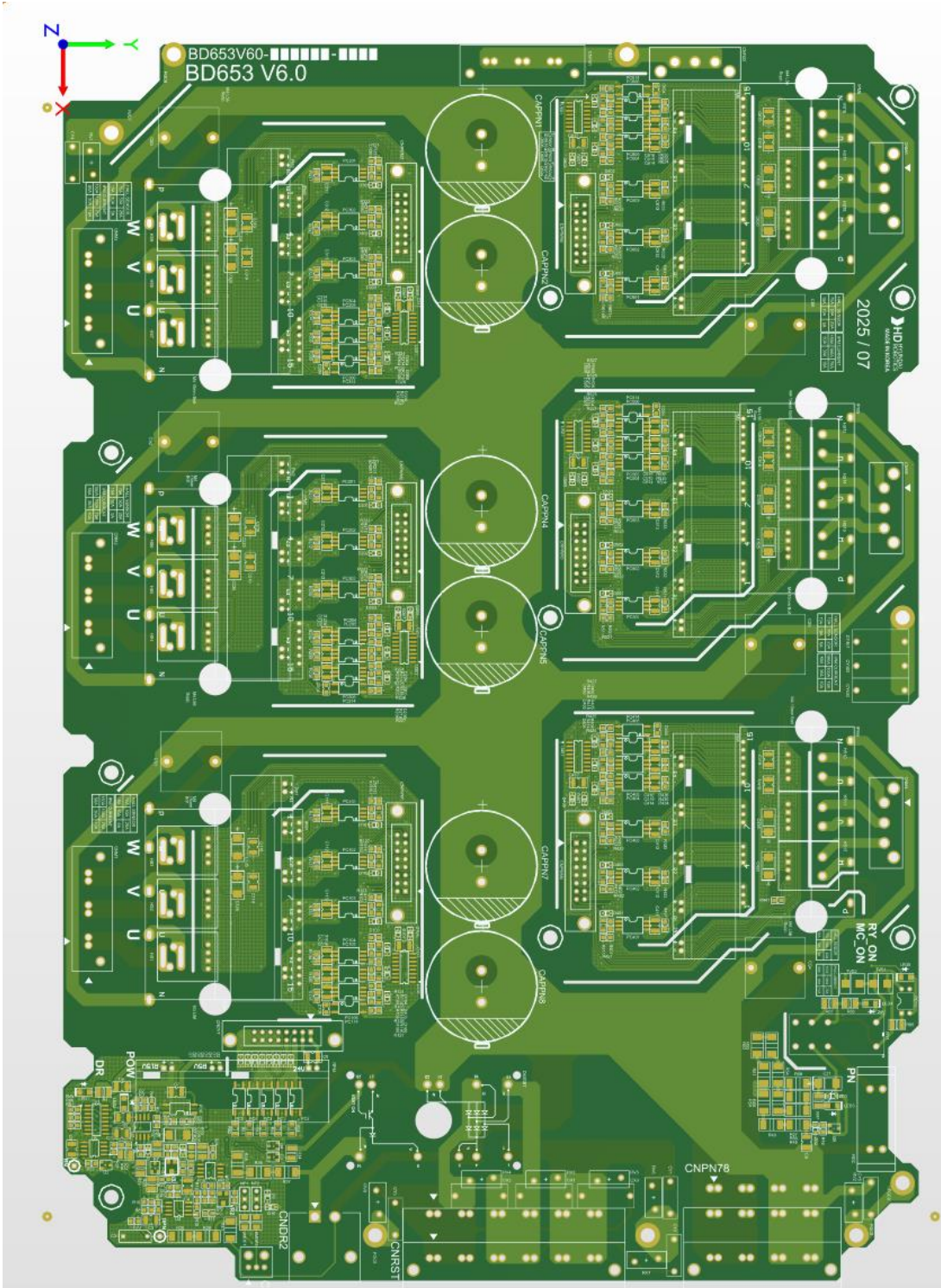


Figure 4.30 Component Layout of BD653V60

Table 4-24 Description of the Connectors of BD653

Name	Usage	Connection of external devices
CNPWM1~6	PWM signals and IPM error signals	Board-to-board connectors of BD654
CNRST	3-phase power input	CNRST for the electronic module
CNCVT	Converter part error signal	Board-to-board connectors of BD654
CNDR	Regenerative discharge power output	Regenerative discharge resistor
CNTR	Regenerative discharge resistor overheating detection	Regenerative discharge resistor temperature sensor
CNM16	Motor drive output	CMC1
CNPN7~8	Direct current for the drive module of the additional axis	CNPN for the drive module for an optional additional axis
CNFG1, CNFG4	Frame ground for motors	CMC1

Table 4-25 Description of the LEDs of BD653

Name	Color	Status display
MC ON	Yellow	Will be turned on when the magnet contact is driving
POW	Green	Will be turned on when the control voltage of the converter part is normal
DR	Red	Will be turned on the regenerative discharge is operating
PN	Red	Will be turned on when the PN voltage is higher than 42V
RYON	Red	Will be turned off when the PN discharge operation starts

4. Basic Configuration of the Controller

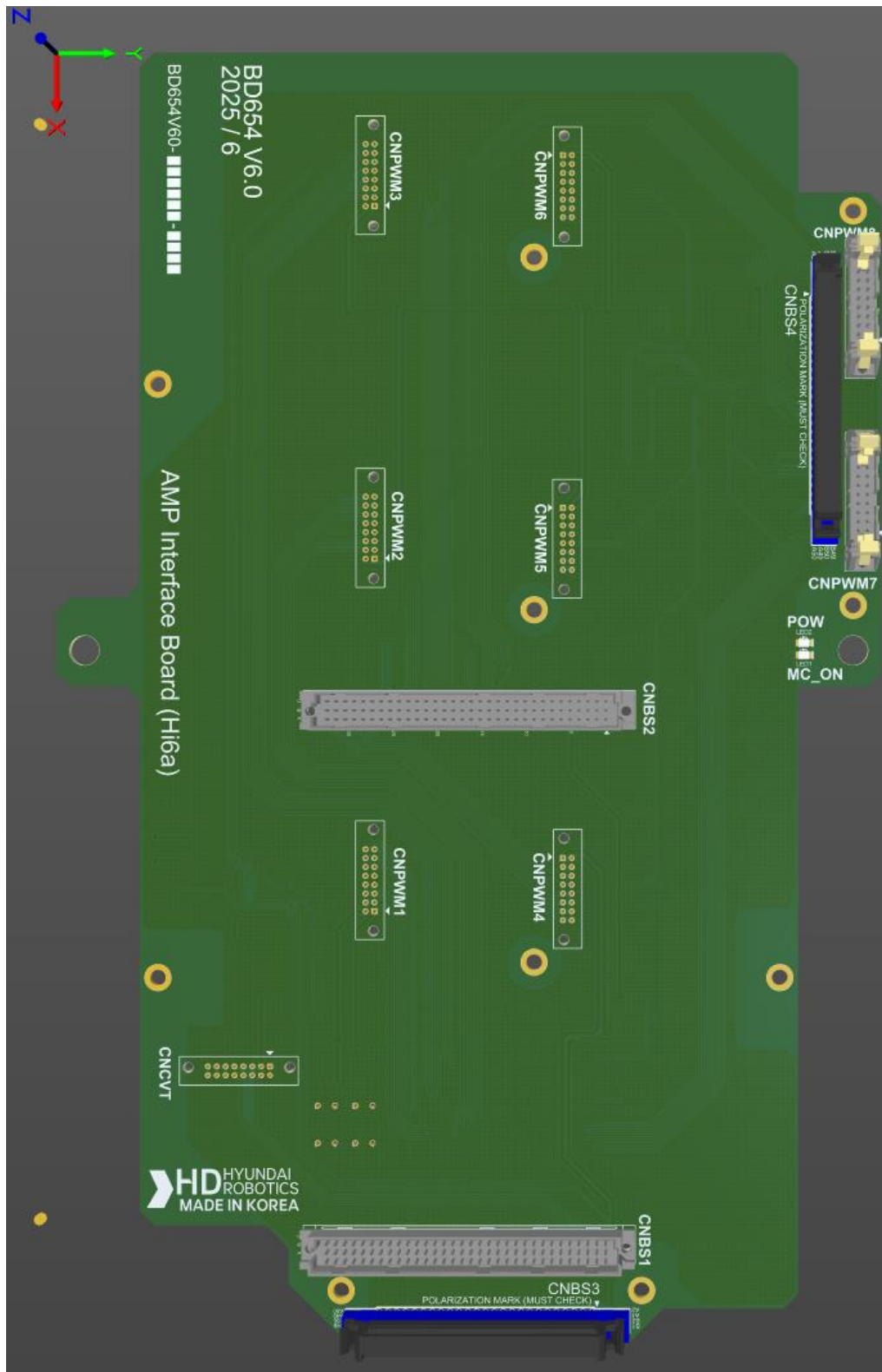


Figure 4.31 Component Layout of BD654V60

Table 4-26 Description of the Connectors of BD654

Name	Usage	Connection of external devices
CNBS1-3	PWM signals and IPM error signals for 8 axes Converter part error signal	Board-to-board connectors of BD640
CNPWM1-6	PWM signals and IPM error signals for individual axes	Board-to-board connectors of BD653
CNPWM7-8	PWM signal and IPM error signal for the additional axis	CNPWM of the drive module (BD658 or BD659) of the additional axis
CNCVT	Converter part error signal	Board-to-board connectors of BD653

Table 4-27 Description of the LEDs of BD654

Name	Color	Status Display
MC	Yellow	Will be turned on when the magnet contact is driving
POW	Green	Will be turned on when the control power is normal

4. Basic Configuration of the Controller

4.3.4.3. Specification of the Optional Drive Module

■ Configuration of the Type Number of the Optional Drive Module

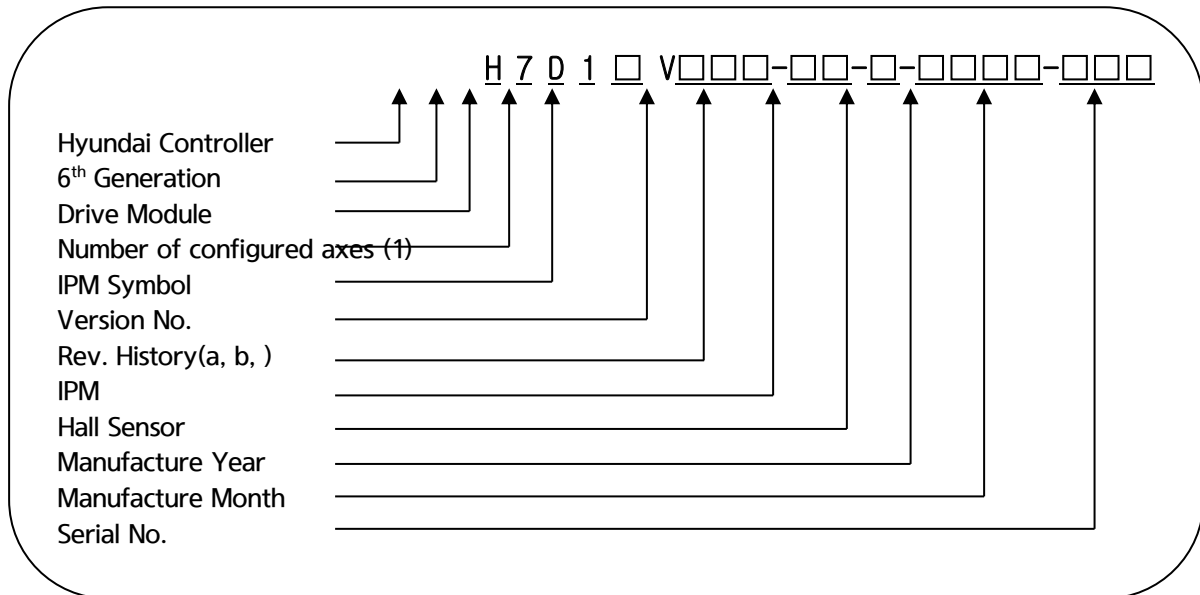


Table 4-28 Type Symbol of the Optional Drive Module

Category	Type symbol
Hi7 1 Axes drive module	H7D1

Table 4-29 Symbols of the IPM of the Optional Drive Module

Drive Model	IPM symbol	IPM Specification
Drive module of the additional axis	X	(IPM current rating) 100A
	Y	(IPM current rating) 75A
	Z	(IPM current rating) 50A

Table 4-30 Symbols of the Hall Sensors of the Optional Drive Module

Drive model	Hall sensor symbol (Specification)	Full-scale current (Im)	IPM specification (Rated current)
Drive module of the additional axis	1 (4V/50A)	93.75 Apeak	PM100CG1APL065 202G (100A) PM75CG1APL065 202G (75A) PM50CG1APL065 202G (50A)
	2 (4V/25A)	46.87 Apeak	
	3 (4V/15A)	28.12 Apeak	
	4 (4V/10A)	18.75 Apeak	
	5 (4V/5A)	9.37 Apeak	

4. Basic Configuration of the Controller

4.3.4.4. H7D1X (Carriage Drive Module: Optional)

The drive module performs a power amplification function that allows the current to flow to the individual phases of the motor according to the current command from the servo board. The carriage drive module can drive one motor of 100A or below and is configured as follows.

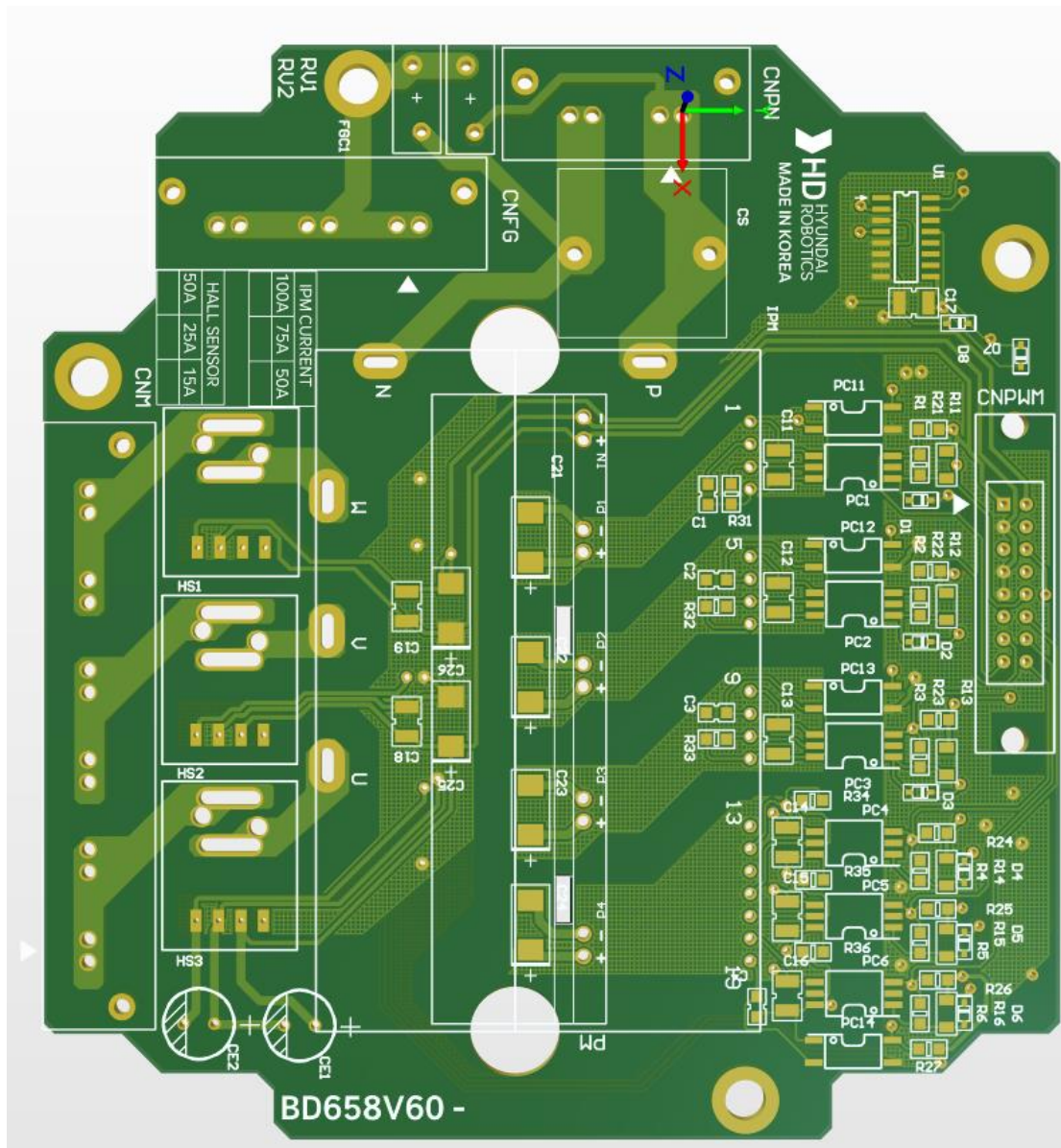


Figure 4.32 Component Layout of BD658V60 for H6AD1X

Table 4-31 Configuration of H6D1X

Components		Functions
BD658 (IPM board)	Logic part	Converts the PWM signals received from the drive module for the 6 axes into the upper and lower sides drive signals of the IPM, and processes errors
	Gate power module	Generates the IPM gate power
	Current detection part	Detects the current that flows through the motor
Other parts	Heat sink	Releases the heat generated in IPM to the outside
	IPM	Convert the power for driving a three-phase motor

Table 4-32 Description of the Connectors of H6D1X

Name	Usage	Connection of external devices
CNPWM	PWM signal and error signal	CNPWM7 or CNPWM8 of the drive module (BD652 or BD654) for the 6 axes
CNM	Motor drive output	AMC1 or AMC2
CNFG	Frame ground for the motor	AMC1 or AMC2
CNPN	Drive direct current power input	CNPN7 or CNPN8 of the drive module (BD651 or BD653) for the 6 axes

4. Basic Configuration of the Controller

4.3.4.5. H7D1Z (Servo Gun Drive Module: Optional)

The drive module performs a power amplification function that allows the current to flow to the individual phases of the motor according to the current command from the servo board. The servo gun drive module can drive one motor of 50A or below and is configured as follows.

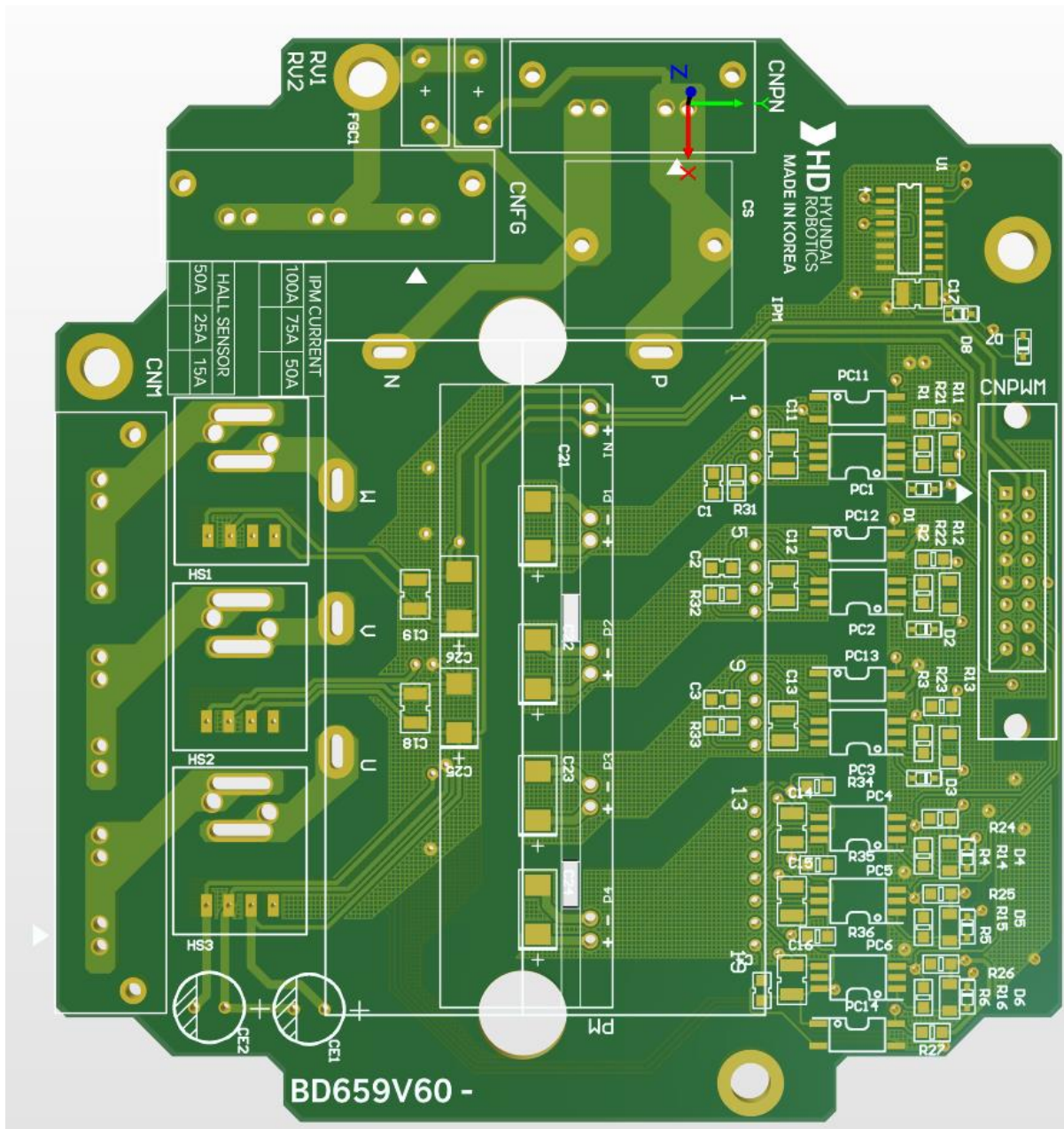


Figure 4.33 Component Layout of BD659V60 for H6AD1Z

Table 4-33 Configuration of H6D1Z

Components		Functions
BD659 (IPM Board)	Logic part	Converts the PWM signals received from the drive module for the 6 axes into the upper and lower sides drive signals of the IPM, and processes errors
	Gate power module	Generates the IPM gate power
	Current detection part	Detects the current that flows through the motor
Other parts	Heat sink	Releases the heat generated in IPM to the outside
	IPM	Convert the power for driving a three-phase motor

Table 4-34 Description of the Connectors of H6D1Z

Name	Usage	Connection of external devices
CNPWM	PWM signal and error signal	CNPWM7 or CNPWM8 of the drive module (BD652 or BD654) for the 6 axes
CNM	Motor drive output	AMC1 or AMC2
CNFG	Frame ground for the motor	AMC1 or AMC2
CNPN	Drive direct current power input	CNPN7 or CNPN8 of the drive module (BD651 or BD653) for the 6 axes

4.3.5. Power Supply Module (H7PSM)

4.3.5.1. H7PSM and Power Distribution Board (BD6C3)

The H7PSM (Hi7-N controller power supply module) module is responsible for the opening and closing and distribution of various power supplied to the controller. The following figures show the interior and exterior of the electrical module with diverse connectors and fuses.

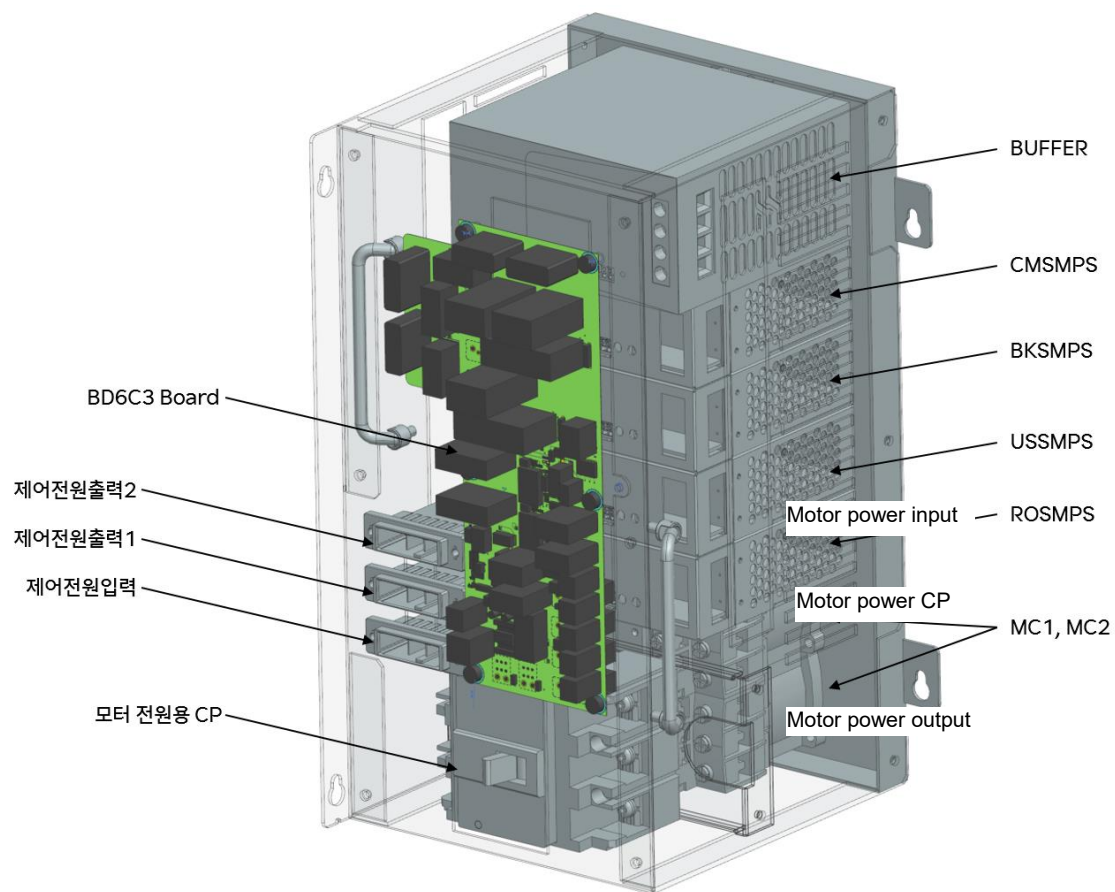


Figure 4.34 Exterior of H7PSM (Hi7-N Controller Power Supply Module)

The following figure shows the power system diagram for the AC control power related to the opening and closing of the 3-phase AC power for the motor power, the generation of the brake power, and the driving of the fan. The diagram in the figure also shows the power distribution, such as the SMPS power for the DC power supply to the control module. A circuit breaker (CP) or fuse is connected to each power to protect individual circuits against overcurrent.

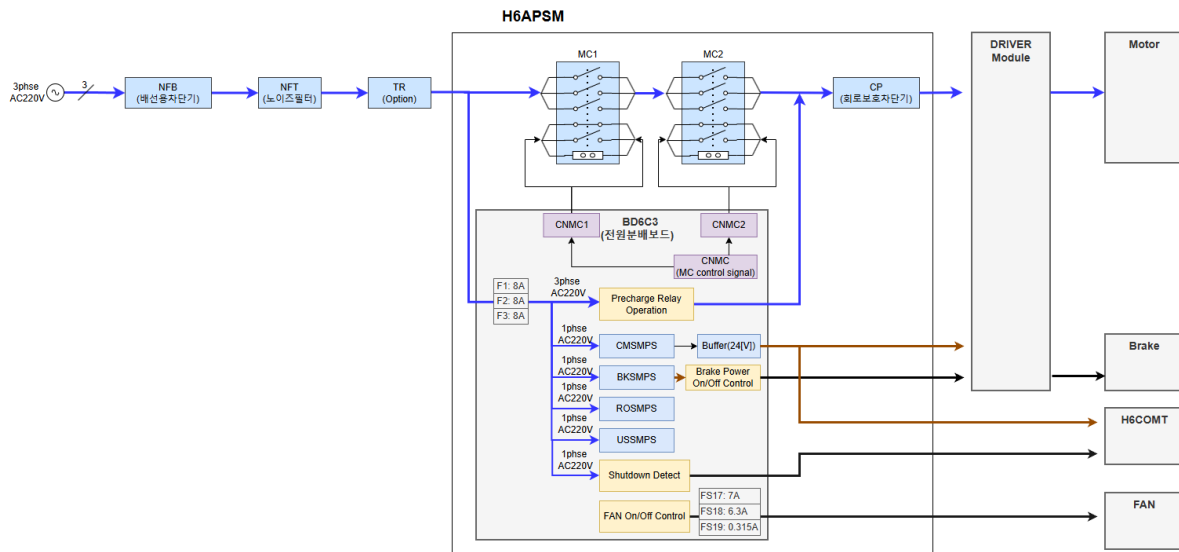


Figure 4.35 Power System of the Hi7-N Controller

Table 4-35 Types and Usage of the Fuses of the Electronic Module

Name	Function	Specification
F1, F2, F3	Fuse for overcurrent protection of control power (AC220 V)	AC220V 8A
FS17	Fuse for overcurrent protection of CMDCFAN and DCFAN2-5 GND	7VAC/60VDC 7A
FS18	Fuse for overcurrent protection of DCFAN2-5	125VAC/125VDC 6.3A
FS19	Fuse for overcurrent protection of the DC fan for control module cooling	125VAC/125VDC 0.315A

4. Basic Configuration of the Controller

4.3.5.2. Connectors of BD6C3

The connector layout of the electrical board (BD6C3) is shown in the following figure. The function and connection device for each are listed in Table 4-40.

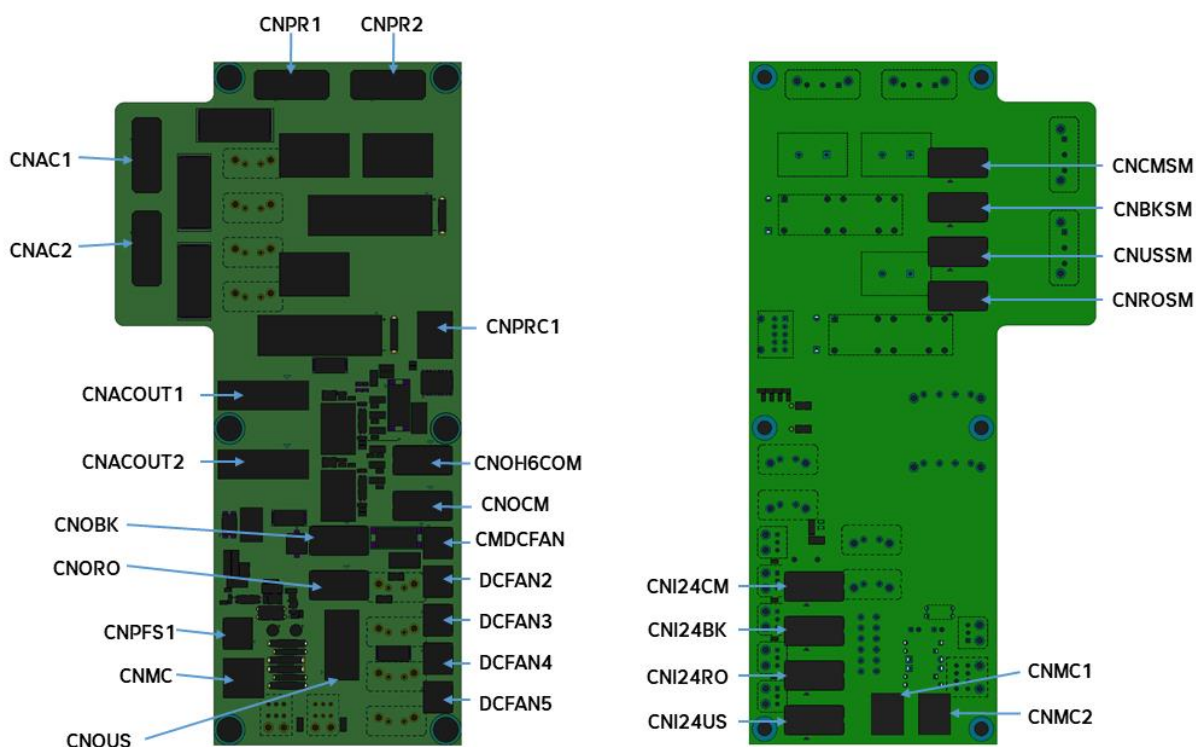


Figure 4.36 Connectors of the Electronic Board (BD6C3)

Table 4-36 Type and Function of BD6C3 Connector

Name	Fuction	Specification
CNAC1	3-phase power input for control	3-phase 220 V
CNAC2	3-phase power output for control of 16 axes or more	3-phase 220 V
CNPR1	Inrush current limiting circuit input	3-phase 220 V, MC1 input terminal
CNPR2	Inrush current limiting circuit output	3-phase 220 V, MC2 output terminal
CNACOUT1	220 VAC power output 1 for users	Single-phase 220 V
CNACOUT2	220 VAC power output 2 for users	Single-phase 220 V
CNPFS1	CMSMPS power failure detection signal output 1	H6COM DIO

CNMC	Contactor control and monitoring	BD642 CNMC
CNPRC	Control and monitoring of inrush current limiting circuit, fan fault and fan power	BD642 CNPRC
CNFN1	DC fan power output for control module	24VDC
CNFN2~5	DC fan power output	24VDC
CNOCM	SMPS 24VDC output for control modules	24VDC
CNOH6COM	SMPS 24VDC output for H6COMs	24VDC
CNOBK	SMPS 24VDC output for motor brakes	24VDC
CNORO	SMPS 24VDC output for robots	24VDC
CNOUS	SMPS 24VDC output for users	24VDC
CNCMSM	SMPS 220VAC input for control modules	Single-phase 220 V
CNBKSM	SMPS 220VAC input for motor brakes	Single-phase 220 V
CNUSSM	SMPS 220VAC input for users	Single-phase 220 V
CNROSM	SMPS 220VAC input for robots	Single-phase 220 V
CNI24CM	Input for SMPS 24VDC distribution for common modules	24VDC
CNI24BK	Input for SMPS 24VDC distribution for brakes	24VDC
CNI24RO	Input for SMPS 24VDC distribution for robots	24VDC
CNI24US	Input for SMPS 24VDC distribution for users	24VDC
CNMC1	Magnetic contactor 1 ON/OFF power input and feedback, and brake control signal control	MC1
CNMC2	Magnetic contactor 2 ON/OFF power input and feedback, and brake control signal control	MC2

4.3.5.3. BD6C3 LED

The LED layout of the electrical board (BD6C3) is shown in the following figure. The function, connected power, and LED color for each are listed in Table 4-41.

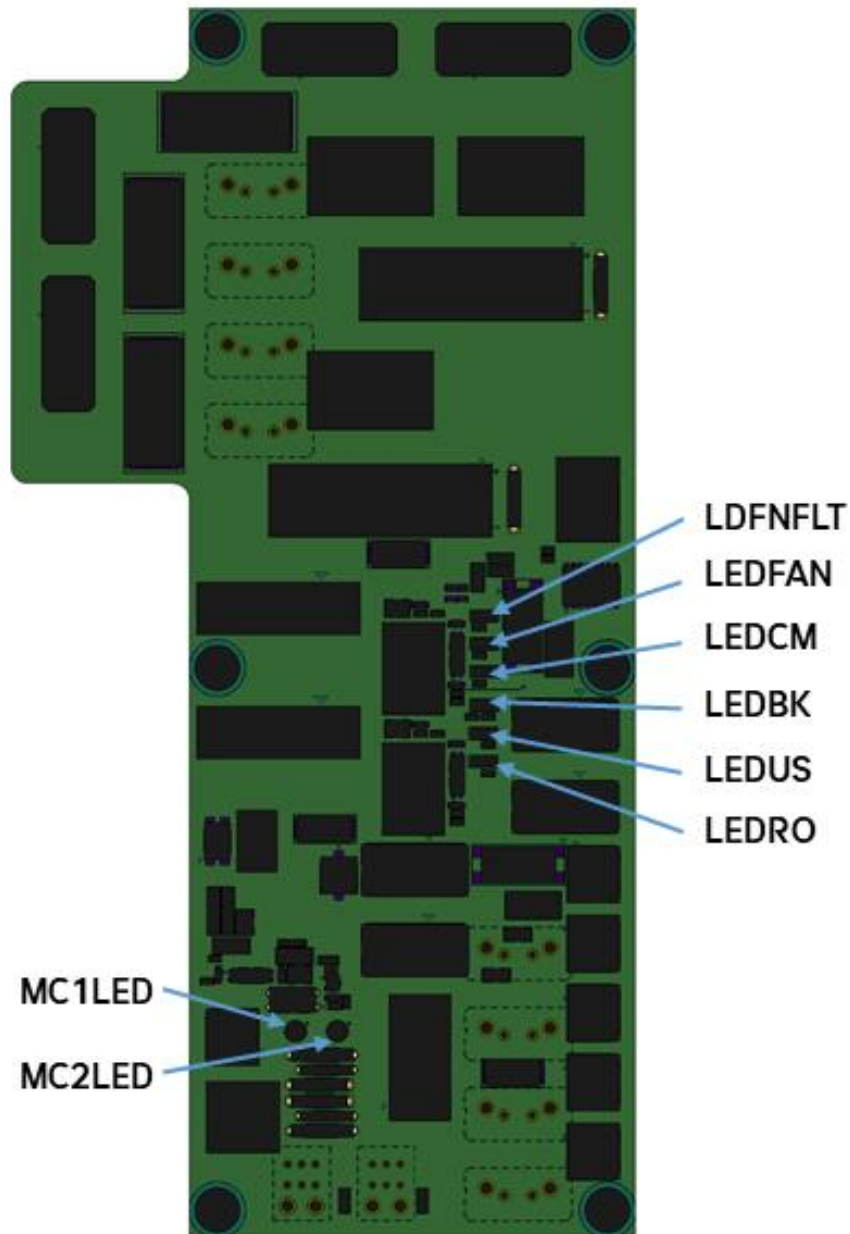


Figure 4.37 LEDs of BD6C3

Table 4-37 Types and Functions of BD6C3 LEDs

명칭	용도	사양
LDFNFLT	ON when a fault occurs in one or more of FAN 1-5	FAN Fault, Red
LEDFAN	ON when the relay for supplying 24VDC to the fan is ON	FAN Power Relay, Green
LEDCM	ON when SMPS 24VDC for control modules is normally supplied	CMSMPS, Green
LEDBK	ON when SMPS 24VDC for brakes is normally supplied	BKSMPS, Green
LEDUS	ON when SMPS 24VDC for users is normally supplied	USSMPS, Green
LEDRO	ON when SMPS 24VDC for robots is normally supplied	ROSMPS, Green
MC1LED	ON when the ON power for magnetic contactor 1 is normally supplied	BD642, Green
MC2LED	ON when the ON power for magnetic contactor 2 is normally supplied	BD642, Green
RYPRC1	ON inside relay when aprecharging relay ON signal is supplied	BD604, Green
RYPRC2	ON inside relay when a fan power relay ON signal is supplied	BD604, Green

4.3.6. Teach Pendant (TP630)

4.3.6.1. Overview

The teach pendant (TP630) communicates with the main module (H6COM-T) of the controller through Ethernet and allows the user to directly operate the following functions.

- Monitoring : Job program / Data of each axis / Input and output signals / Status of the robot, etc.
- Log management : System version, operation time, error log, stop log, etc.
- File management : Version and teach program up/down
- Setting of various variables : User environment / control / robot / application / automatic integer, etc.
- Robot teaching : Jog and teach program registration
- Robot operation : Motor on / start / stop / mode settings

The teach pendant is also equipped with a three-stage enable switch and emergency stop switch to ensure user safety.

In addition, a USB A type connector is mounted under the rubber cover at the bottom of the teach pendant, allowing the user to upload/download necessary files, such as data and teaching programs, as well as versions of diverse types of boards using a USB memory stick.



Figure 4.38 Exterior of the Teach Pendant TP630

4.3.6.2. USB Cover

There is a USB A type connector mounted under the rubber cover at the bottom of the teach pendant, allowing the user to upload/download necessary files, such as versions of diverse types of boards, as well as data and teaching programs, using a USB memory stick.

The background features a light gray grid of dots. A dark gray circle is positioned on the right side of the page, containing the number 5 and the title text.

5

Optional
Components of the
Controller



5. Optional Components of the Controller

Hi7-N Controller Maintenance Manual

Table 5-1 Summary of Optional Configurations

Component		Function
Control options	Extended safety signal board (BD680)	- 8-channel safety digital input - 8-channel safety digital output
	User DIO board (BD681)	- Up to 48-digital input channel
	Extended DIO board (BD682)	- Up to 48-digital output channel - NPN/PNP switchable - 8 relay output channels selectable among digital outputs - PROFINET communication: 240 bytes input, 240 bytes output
Communication options	Ethernet/IP Master/Slave	- Supports H6COM-T LAN port - Configurable in TP630
	2'nd EtherCAT Master	
	CIP Safety	
PCI communication card	15 types of PC cards CFX 50 series	- Ethernet Master/Slave - CC-Link Slave - DeviceNET Master/Slave - PROFIBUS Master/Slave - CC-Link IE Field
Brake release unit	-	- Used when motor brake release of each robot axis is required
Remote IO module	Communication module IO module	- Required for additional use beyond user DIO signals

5.1. PCI Communication Card

5.1.1. Overview

To use industrial communication in the Hi7 controller, you can use a PCI communication card. Descriptions below are provided based on a PCI communication card for Ethernet, which is a general model. For details, please refer to the contents for the PC Cards CIFS 50 model in the document “PC Cards CIFS 50 50E 70E 100EH UM 51 EN.”

Table 5-2 Part Names of PCI Communication Cards

No.	Model name	Communication type	Interface connector
1	CIFS 50-RE/ML-HRC	HRC Real-Time Ethernet Master PCI	RJ45
2	CIFS 50-RE-HRC	HRC Real-Time Ethernet Slave PCI	RJ45
3	CIFS 50E-RE/ML-HRC	HRC Real-Time Ethernet Master PCIe	RJ45
4	CIFS 50E-RE-HRC	HRC Real-Time Ethernet Slave PCIe	RJ45
5	CIFS 50-CC-HRC	CC-Link Slave PCI	CombiCon Male, 5 pins
6	CIFS 50E-CC-HRC	CC-Link Slave PCIe	CombiCon Male, 5 pins
7	CIFS 50-DN/ML-HRC	DeviceNet Master PCI	CombiCon Male, 5 pins
8	CIFS 50-DN-HRC	DeviceNet Slave PCI	CombiCon Male, 5 pins
9	CIFS 50E-DN/ML-HRC	DeviceNet Master PCIe	CombiCon Male, 5 pins
10	CIFS 50E-DN-HRC	DeviceNet Slave PCIe	CombiCon Male, 5 pins
11	CIFS 50-DP/ML-HRC	PROFIBUS Master PCI	Dsub Female, 9 pins
12	CIFS 50-DP-HRC	PROFIBUS Slave PCI	Dsub Female, 9 pins
13	CIFS 50E-DP/ML-HRC	PROFIBUS Master PCIe	Dsub Female, 9 pins
14	CIFS 50E-DP-HRC	PROFIBUS Slave PCIe	Dsub Female, 9 pins
15	CIFS 50E-CCIES-HRC	CC-Link IE Field PCIe	RJ45

5.1.2. Configuration of the PCI Communication Card

A PCI communication card is basically configured, as shown below (when Ethernet-based communication is used), and varies in the number of connectors and LEDs depending on the type of industrial communication.

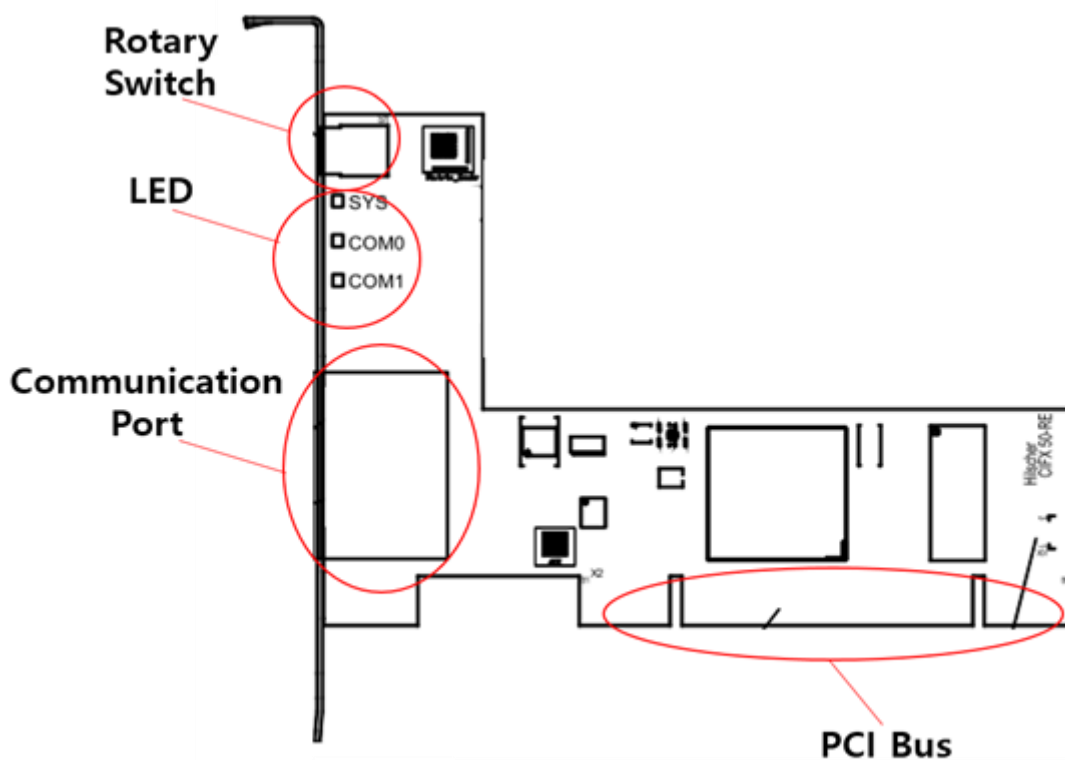


Figure 5.1 Exterior of the PCI Communication Card

Table 5-3 Description of the Exterior of the PCI Communication Card

Name	Usage
Rotary switch	Sets the communication according to the slot ID
LED	Displays the system and communication status
Communication port	Communication connection port
PCI bus	PC connection bus

5.1.3. Front Part of the PCI Communication Card

You can check the communication setting, communication cable connection, and communication status through the front part of the PCI communication card. Basically, you can use the card by setting the rotary switch to 1-4 in order according to the location of the H6COM-T PCI slot.

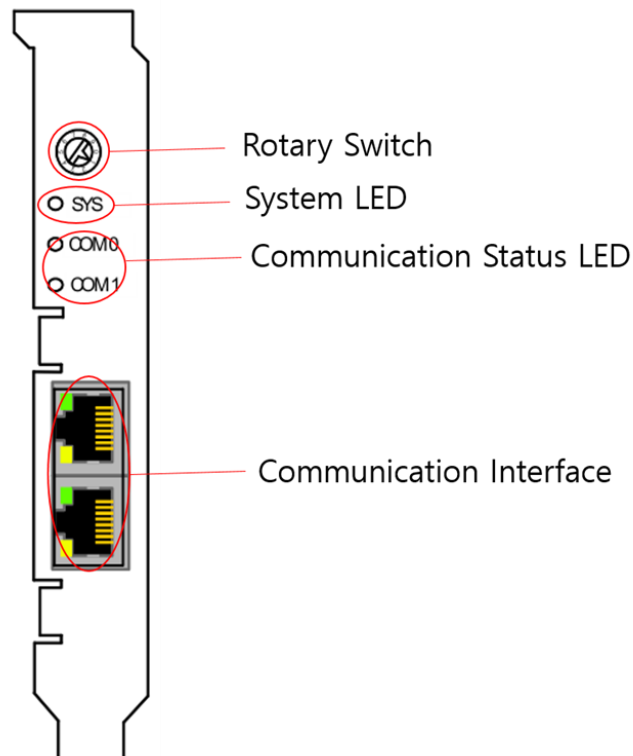


Figure 5.2 Front Part of the PCI Communication Card

Table 5-4 Configuration of the Front Part of the PCI Communication Card, and the Description of the Functions

Name	Usage	Description of Functions
Rotary switch	Sets the communication for each slot number	H6COM-T PCI slots are fixed as #1~#4 sequentially starting from the one at the top (Communication to be set from TP).
System LED	System status checking LED	Green: System in operation Yellow: Boot loader waiting
Communication status LED	Communication status checking LED	Green: Communication in operation Red: Communication error
Communication Interface	Communication cable connection port	Use of connectors suitable for communication

5.2. Brake Release Unit

5.2.1. Overview

The brake release unit can be used when it is necessary to release the motor brake for each axis of the robot. It can help to set the robot posture mainly during the initial installation of the robot. When releasing the brake, you must fully understand the safety information of “Release of the manual brake” in “1.8.2 Other Related Functions” before using the robot.



Caution

1. Do not release more than two axes at the same time.
2. Keep a safe distance from the robot first before using the brake release unit.
3. Use the brake release unit after preparing for the dropping of the robot's axis using equipment such as a crane.
4. Check the safety matters while working in a group of at least two people.



Warning

The robot should be installed and operated according to the guidelines of ISO 10218-2. In addition, it is required to comply with the relevant requirements of international standards and national laws.

Our company (or the manufacturer) will not be responsible for any accidents that occur because of not complying with the relevant requirements of international standards and national laws or not following the above “caution.”

5.2.2. Brake Release Switch

The placement of the switches of the brake release unit is as shown in Figure 5.3, and their individual usage and operation are described in Table 5-4. To release the brake of the targeted axis, first press the Enable button, and then, while pressing the Enable button, press one of the buttons B1– B8 at the same time. Then, the relevant axis will be released.

5. Optional Components of the Controller

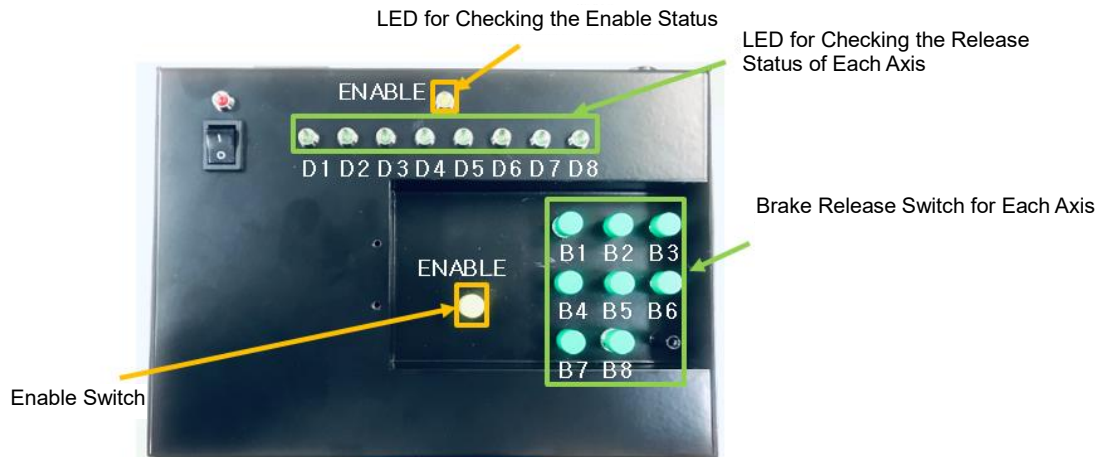


Figure 5.3 Switches and Status Checking LEDs of the Brake Release Unit

Table 5-5 Usage of the Switches of the Brake Release Unit

Name	Usage	During operation
E	Brake Release Enable	Yellow Enable LED ON
B1	Axis 1 Brake Release	Green D1 LED ON
B2	Axis 2 Brake Release	Green D2 LED ON
B3	Axis 3 Brake Release	Green D3 LED ON
B4	Axis 4 Brake Release	Green D4 LED ON
B5	Axis 5 Brake Release	Green D5 LED ON
B6	Axis 6 Brake Release	Green D6 LED ON
B7	Axis 7 Brake Release	Green D7 LED ON
B8	Axis 8 Brake Release	Green D8 LED ON

5.2.3. Power and Connectors

The placement of the power and connectors of the brake release unit is as shown in Figure 5.4 below, and their individual usage and connection devices are as shown in Table 5-5 below.

 Caution	- When using the brake release unit, follow the procedures below. 1. Turn off the AC220V power switch and check that the DC24V power switch is turned off. 2. Connect the AC power cable to the AC power connector. 3. Turn on the AC220V power switch. 4. Turn on the DC24V power switch.
	- When the use of the brake release unit is finished, follow the procedures below. 1. Turn off the DC24V power switch. 2. Turn off the AC220V power switch. 4. Disconnect the AC power cable.
	- Do not use AC220V power and DC24V battery power at the same time.

 Warning	Our company (or the manufacturer) will not be responsible for any accidents that occur because of not complying with the above “caution.”
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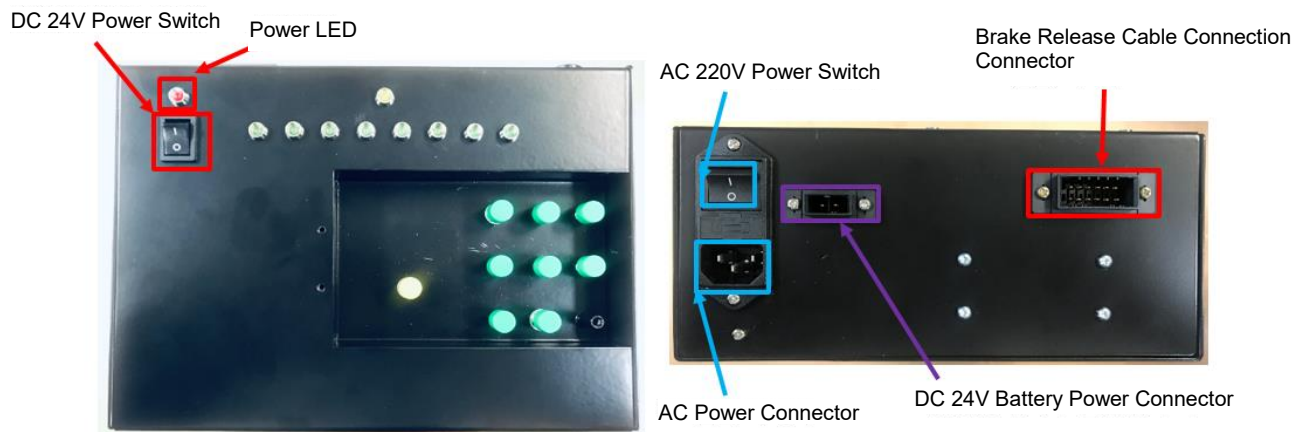


Figure 5.4 Switches and Connectors of the Brake Release Unit

5. Optional Components of the Controller

Table 5-6 Types and Usage of the Connectors of the Brake Release Unit

Name	Usage	Connection of external devices
AC 220V power connector and switch	Application of the AC power	100V AC~240V AC Single phase
Brake release cable connection connector	Connection of the brake release unit and controller	CNB1, CNB7, and CNB8 of BD640
DC24V battery power connector	Connection of the power of a portable 24V battery	Portable 24V battery
DC24V power switch	Brake release unit drive on/off	None

5.2.4. Brake Release Unit Status Display LEDs

The LEDs for displaying the status of the brake release unit are as shown in Figure 5.3, and their individual usage and operation status are as shown in Table 5-6 below.

Table 5-7 Usage and Operation of the LEDs for the Status of the Brake Release Unit

Name	Usage	LED On operation
Enable	Checking by pressing the Enable switch	When the Enable switch is pressed, the yellow Enable LED will be turned on.
D1	Checking by pressing the switch for axis 1	When the switch for axis 1 is pressed, the green D1 LED will be turned on.
D2	Checking by pressing the switch for axis 2	When the switch for axis 2 is pressed, the green D2 LED will be turned on.
D3	Checking by pressing the switch for axis 3	When the switch for axis 3 is pressed, the green D3 LED will be turned on.
D4	Checking by pressing the switch for axis 4	When the switch for axis 4 is pressed, the green D4 LED will be turned on.
D5	Checking by pressing the switch for axis 5	When the switch for axis 5 is pressed, the green D5 LED will be turned on.
D6	Checking by pressing the switch for axis 6	When the switch for axis 6 is pressed, the green D6 LED will be turned on.
D7	Checking by pressing the switch for axis 7	When the switch for axis 7 is pressed, the green D7 LED will be turned on.
D8	Checking by pressing the switch for axis 8	When the switch for axis 8 is pressed, the green D8 LED will be turned on.

5.3. Remote IO

5.3.1. Overview

To use general IO signals in the Hi7 controller, you need the commercial remote IO. Basically, the commercial remote IO can be used by connecting an “IO module” (selected by the user) to one “communication module.” The module introduced below is a commercial remote IO module of Crevis, and you may purchase and use a commercial remote of other companies. For details on how to use each module, you are required to ask the company about the IO module you purchased.



Caution

Fieldbus communication must be available to use the commercial remote IO. Therefore, you should configure the PCI communication card together by referring to “5.1 PCI Communication Card” mentioned above.

Communication Module

IO and Other Modules



Figure 5.5 Example of Configuration of Commercial Remote IO

5.3.2. Communication Module (of Crevis)

The types of communication modules are as shown below and can be used according to the desired communication.

Table 5-8 Communication Module (of Crevis)

No.	Model name	Specification
1	M9212	DeviceNet Network Adapter
2	M9287	ProfiNet Network Adapter
3	M9289	ModbusTCP/UDP, EthernetIP Network Adapter
4	M9386	EtherCAT ID Network Adapter, 1452 bytes

5.3.3. IO and Other Modules (of Crevis)

The types of IO and other modules are as shown below and can be configured and used according to the desired operation.

Table 5-9 IO Module (of Crevis)

No.	Model name	Specification
1	M12DF	Digital Input 16 Points, Universal (Sink or Source), 24VDC, 18RTB
2	M12FA	Digital Input 32 Points, Universal (Sink or Source), 24VDC, Hirose 40P
3	M225F	Digital Output 16 Points, Sink, 24VDC/0.3A, 18RTB
4	M226F	Digital Output 16 Points, Source, 24VDC/0.3A, 18RTB
5	M22BA	Digital Output 32 Points, Sink, 24VDC/0.3A, Hirose 40P
6	M2618	Digital Output 8 Points, Sink, 24VDC/1A, Max 8A, 18RTB
7	M2628	Digital Output 8 Points, Source, 24VDC/1A, Max 8A, 18RTB

5. Optional Components of the Controller

Table 5-10 Relay Module (of Crevis)

No.	Model Name	Specification
1	M2788	MOS Relay, 8 Points, 110VDC/AC, 1A, 18RTB

Table 5-11 Analogue IO Module (of Crevis)

No.	Model Name	Specification
1	M3534	Analog Input 4ch Voltage, -10 to 10VDC, 14 bits
2	M4534	Analog Output 4ch Voltage, -10 to 10VDC, 14 bits

Table 5-12 Pulse Measuring Module (of Crevis)

No.	Model Name	Specification
1	M5112	High-speed counter, 2 Channels, 24VDC, 18RTB (Open Collector)
2	M5102	High-speed counter, 2 Channels, 5VDC, 18RTB (RS422 Differential)

Table 5-13 Serial Communication Module (of Crevis)

No.	Model Name	Specification
1	M5212	RS232 Serial Interface, 2 Channels Full Duplex
2	M5232	RS485 Serial Interface, 2 Channels Full Duplex

5.4. Option Safety IO Module(BD680)

5.4.1. Overview

The optional safety IO module (BD680) is used when additional safety IO signals are required.

When applying the robot controller to the field, if additional safety IO signals are required, this module can be mounted on the servo/safety module (BD642). The BD680 cannot be used independently and can operate by connecting to the BD642 via a board-to-board connector through power supply, control signals, and interface connections.

Safety IO signals consist of 8-point digital input signals and 8-point digital output signals.

5.4.2. Connector

The figure below shows the position of each connector required for external connections to the optional safety IO module (BD680). Also, the table below describes the name and function of each connector.

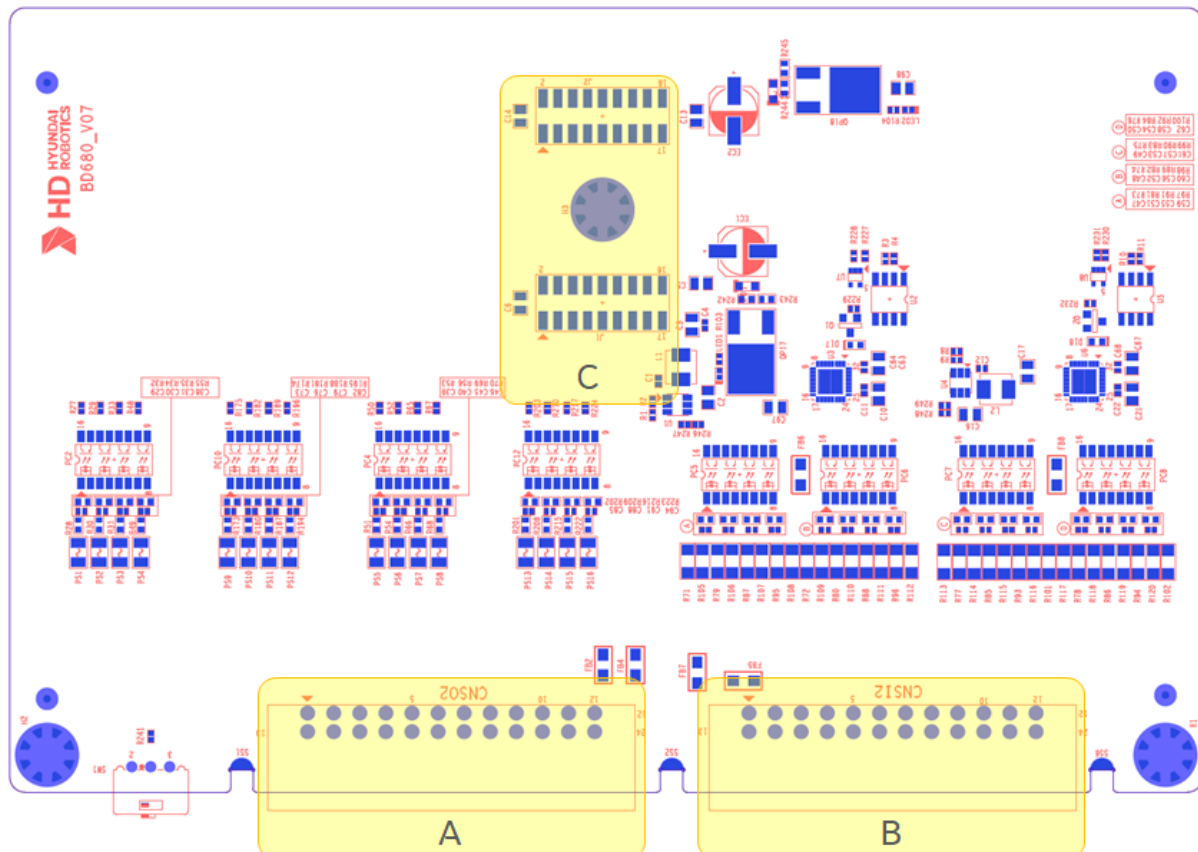


Figure 5.6 Connector Layout of Optional Safety IO Module (BD680)

5. Optional Components of the Controller

Table 5-14 Connector Name, Function, and External Connection Device of Optional Safety IO Module (BD680)

No.	Name	Usage	Connection
A	CNSO2	Safety output terminal	External device
B	CNSI2	Safety input terminal	External device
C	J1 J2	BD642 connection connector (board-to-board)	Servo/Safety module (BD642)



Caution

If activated by connecting safety-related inputs, be sure to check whether the functions operate normally by referring to "1.11. Safety Measures for Robot Operation."

5.4.3. Device for display

(1) Top surface display of the board

The figure below shows the position of each display device (LED) on the optional safety IO module (BD680). The table below describes the information of each display.

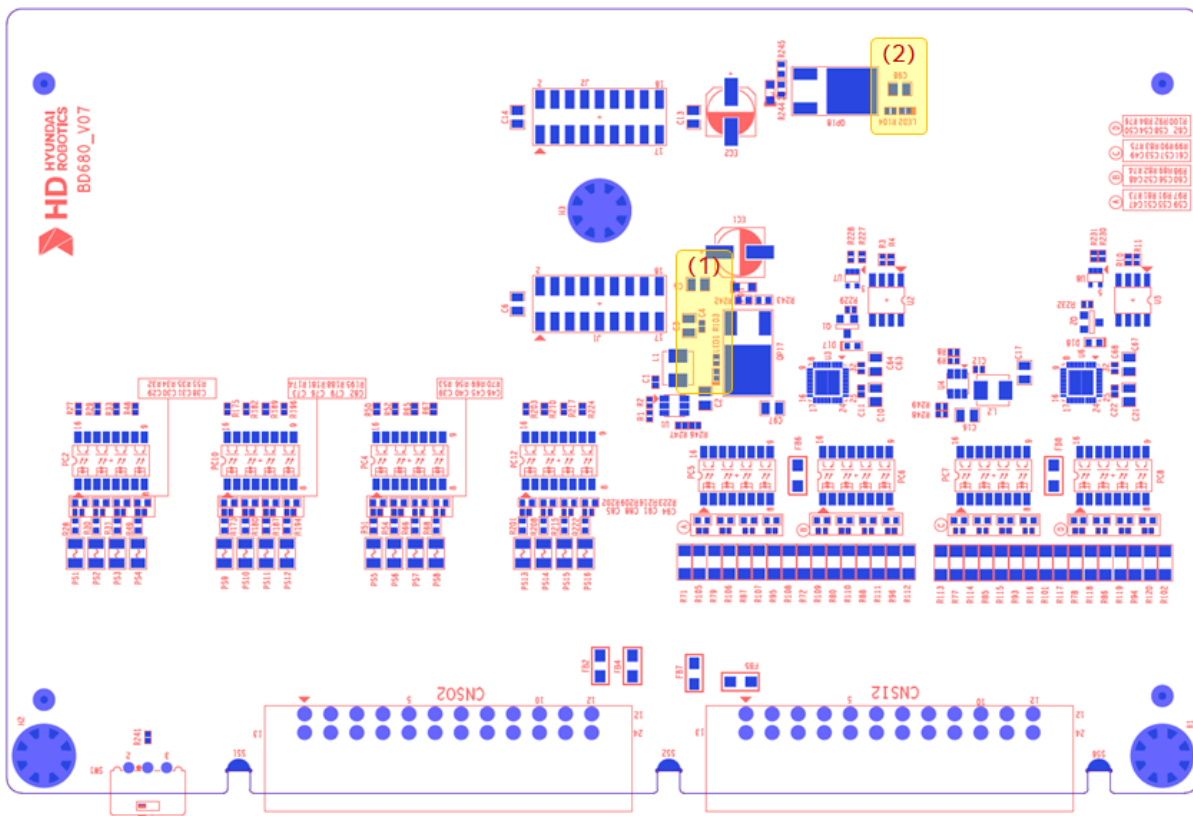


Figure 5.7 Display Device Layout of Optional Safety IO Module (BD680)

Table 5-15 Display Device Description of Optional Safety IO Module (BD680)

No.	Name	Description	Color	Normal Condition	Actions against Abnormal Condition
(1)	LED1	A channel power display	Yellow	ON	Symptom: Yellow LED OFF Cause: A channel input power is abnormal. Action 1: Check the A channel input power (24 V). Action 2: Check the fuse (FS1).
(2)	LED2	B channel power display	Yellow	ON	Symptom: Yellow LED OFF Cause: B channel input power is abnormal. Action 1: Check the B channel input power (24 V). Action2: Check the fuse (FS2).

(2) Front display device of the board

The figure below shows the front display device of the optional safety IO module (BD680). The table below describes the information of each display.

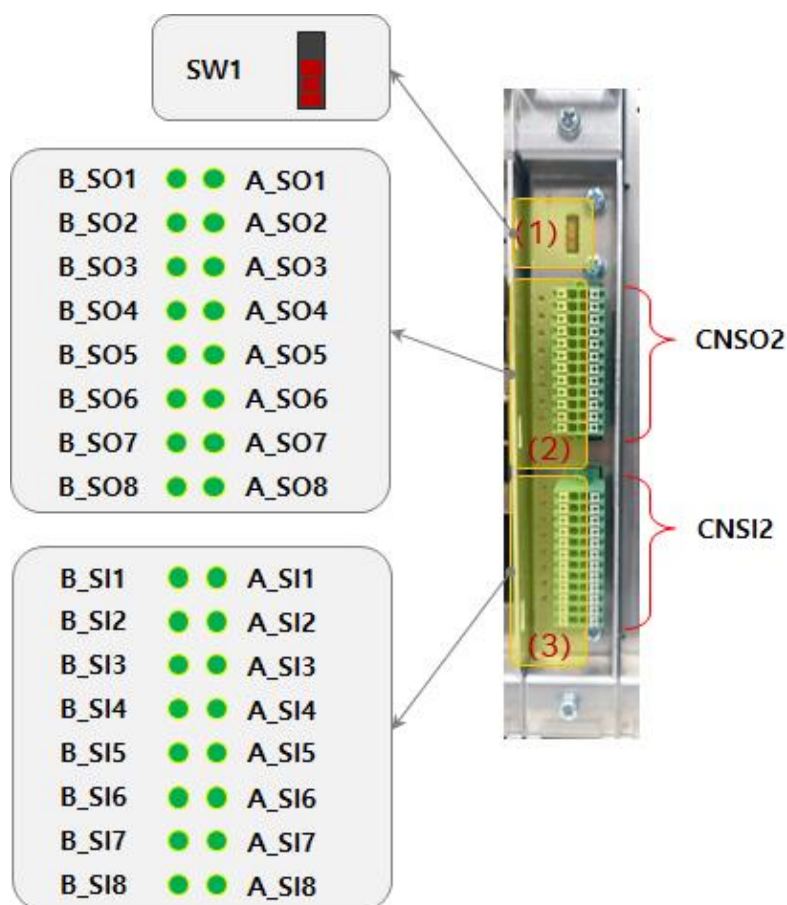


Figure 5.8 Front Display Layout of Optional Safety IO Module (BD680)

5. Optional Components of the Controller

Table 5-16 Front Display Description of Optional Safety IO Module (BD680)

No.	Name	Description	Color	Display State	Display Description
(1)	SW1				For future use
(2)	A_SOx (x=1~8)	A channel safety output x state display	Green	ON OFF	Each channel safety output x ON state
	B_SOx (x=1~8)	B channel safety output x state display			Each channel safety output x OFF state
(3)	A_SIx (x=1~8)	A channel safety input x state	Green	ON OFF	Each channel safety input x ON
	B_SIx (x=1~8)	B channel safety input x state			Each channel safety input x OFF

5.4.4. Device for setting

The figure below shows the position of each setting device (switch) on the optional safety IO module (BD680). The table below describes the function of each setting.

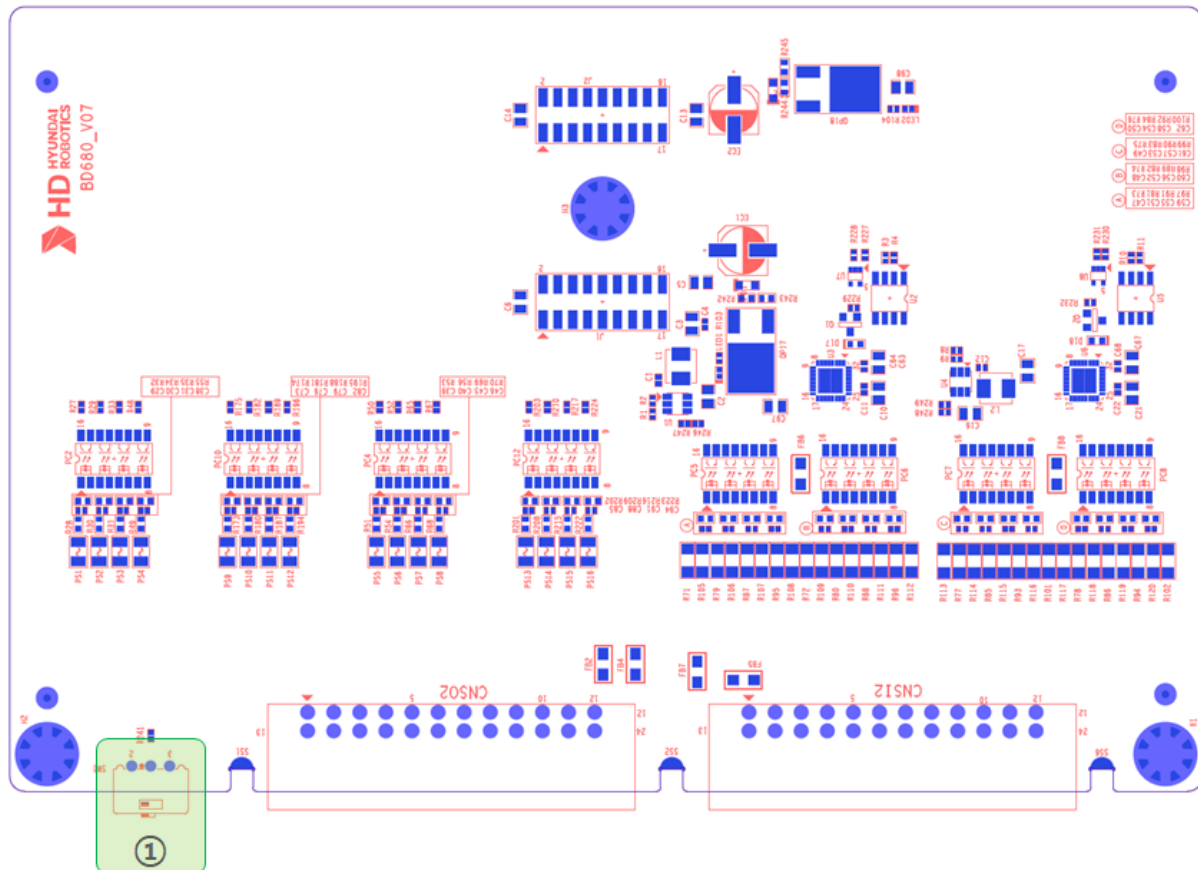


Figure 5.9 Setting Device Layout of Optional Safety IO Module (BD680)

Table 5-17 SW1 Setting Device Description of Optional Safety IO Module (BD680)

No.	Name	Setting State	Description	Remarks
①	SW1	-	-	For future use

5.4.5. Connection for safety output



Before performing safety output wiring work, turn the controller power OFF.

The figure below shows the actual photo of the optional safety IO module (BD680) and the position of each connector when viewed from the front during installation.

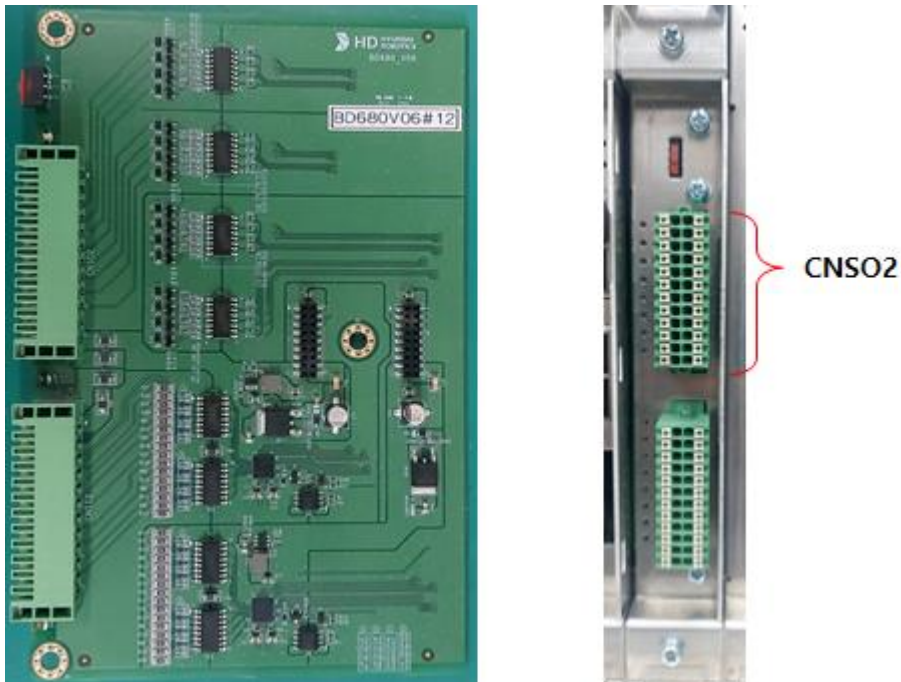


Figure 5.10 Actual Photo of Optional Safety IO Module (BD680), and Position of Safety Output Connector

The wiring of safety outputs differs depending on whether to use an internal power supply or an external power supply. The following shows the wiring for each case.

(1) When using the internal power supply

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the internal power of A channel, connect pins 1–2 and pins 11–12 of CNSO2 as shown in the figure below.

When using the internal power of B channel, connect pins 13–14 and pins 23–24 of CNSO2 as shown in the figure below.

For connection with external devices, refer to the wiring examples below:

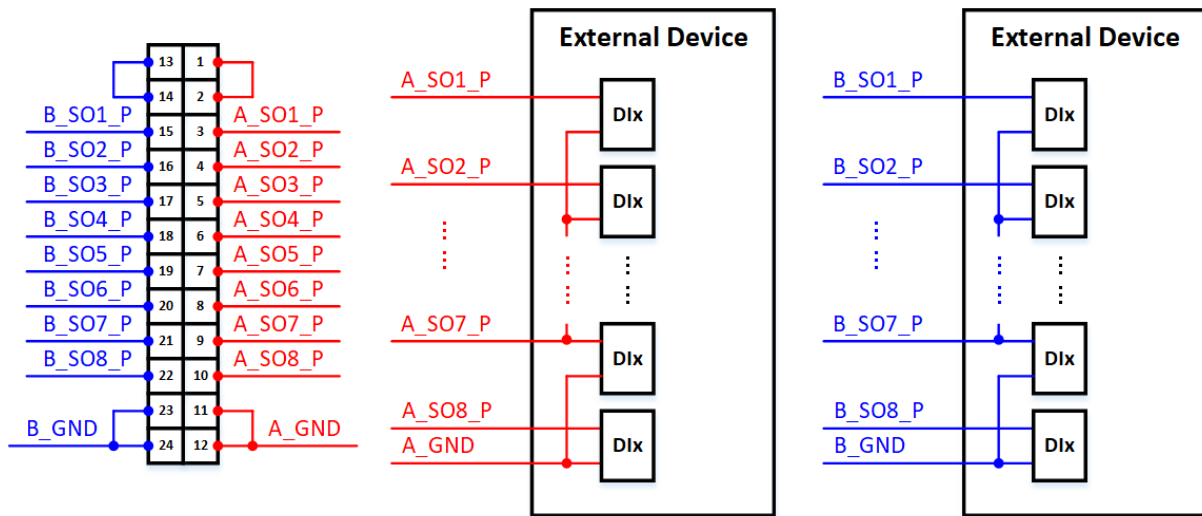


Figure 5.11 Safety Output Wiring with Internal Power Supply for Optional Safety IO Module (BD680)

(2) When using the external power supply

In the figure below, red lines represent A channel and blue lines represent B channel.

Do not connect pins 1, 12, 13, and 24 of CNS02.

Connect A channel external power EX_AV (24 V) to pin 2 and EX_AG (GND) to pin 11.

Connect B channel external power EX_BV (24 V) to pin 14 and EX_BG (GND) to pin 23.

For connection with external devices, refer to the wiring examples below:

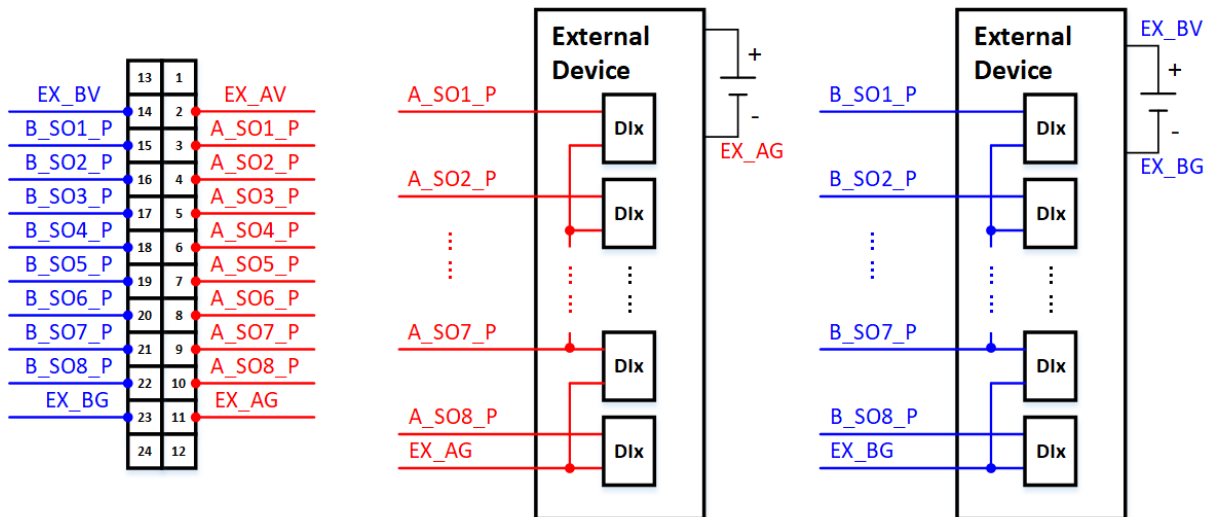


Figure 5.12 Safety Output Wiring with External Power Supply for Optional Safety IO Module (BD680)

5.4.6. Connection for safety input



Before performing safety input wiring work, turn the controller power OFF.

The figure below shows the actual photo of the optional safety IO module (BD680) and the position of each connector when viewed from the front during installation.



Figure 5.13 Actual Photo of Optional Safety IO Module (BD680), and Position of Safety Input Connector

The wiring of safety inputs differs depending on whether to use an internal power supply or an external power supply, and also according to the NPN/PNP type. The following shows the wiring for each case.

(1) When using the internal power supply

- NPN-TYPE(: Active Low)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the internal power of A channel, connect pins 1–3 of CNSI2 as shown in the figure below.

When using the internal power of B channel, connect pins 13–15 of CNSI2 as shown in the figure below.

For connection with external devices, refer to the wiring examples below:

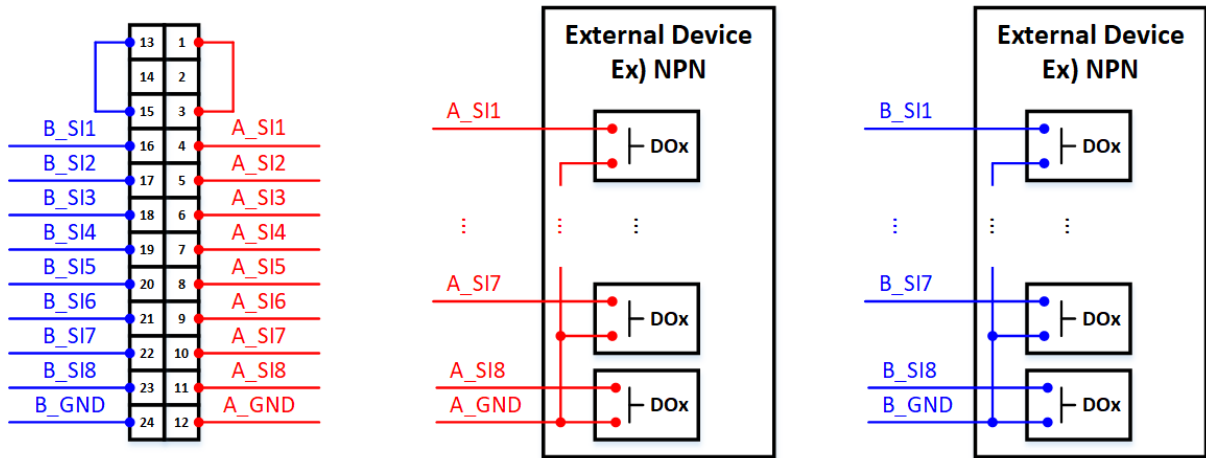


Figure 5.14 Safety Input Wiring with Internal Power Supply (NPN Type) for Optional Safety IO Module (BD680)

- PNP-TYPE(: Active High)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the internal power of A channel, connect pins 3–12 of CNSI2 as shown in the figure below.

When using the internal power of B channel, connect pins 15–24 of CNSI2 as shown in the figure below.

For connection with external devices, refer to the wiring examples below:

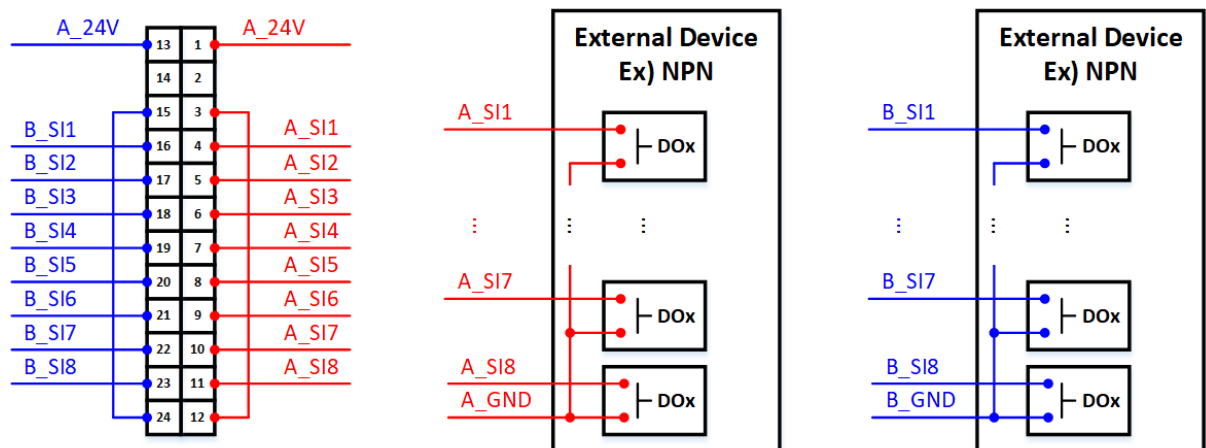


Figure 5.15 Safety Input Wiring with Internal Power Supply (PNP Type) for Optional Safety IO Module (BD680)



When wiring the internal power to external devices, do not use it as the power source for them.

5. Optional Components of the Controller

(2) When using the external power supply

- NPN-TYPE(: Active Low)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the external power of A channel, do not connect pins 1 and 12 of CNSI2, and connect the external power (EX_AV) to pin 3.

When using the external power of B channel, do not connect pins 13 and 24 of CNSI2, and connect the external power (EX_BV) to pin 15.

For connection with external devices, refer to the wiring examples below:

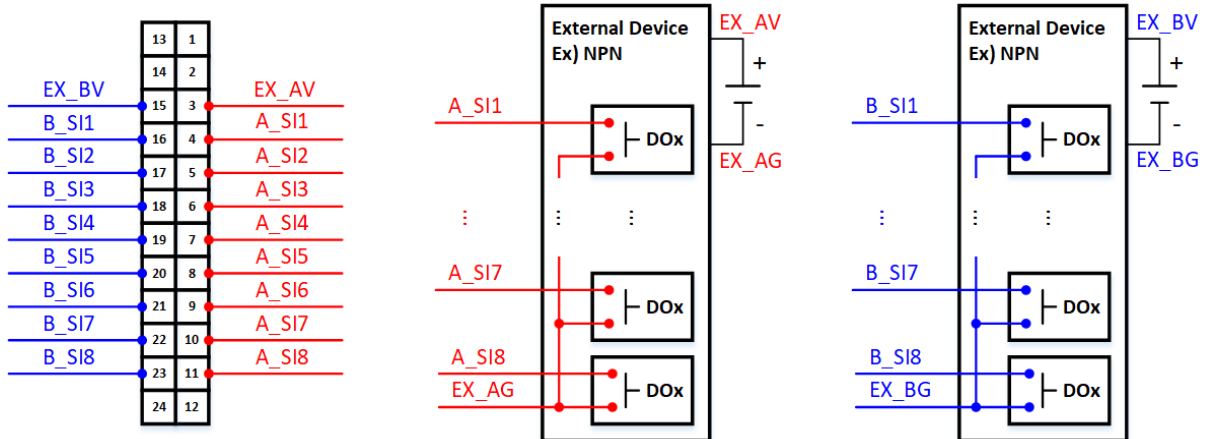


Figure 5.16 Safety Input Wiring with External Power Supply (NPN Type) for Optional Safety IO Module (BD680)

- PNP-TYPE(: Active High)

In the figure below, red lines represent A channel and blue lines represent B channel.

When using the external power of A channel, do not connect pins 1 and 12 of CNSI2, and connect the external power (EX_AG) to pin 3.

When using the external power of B channel, do not connect pins 13 and 24 of CNSI2, and connect the external power (EX_BG) to pin 15.

For connection with external devices, refer to the wiring examples below:

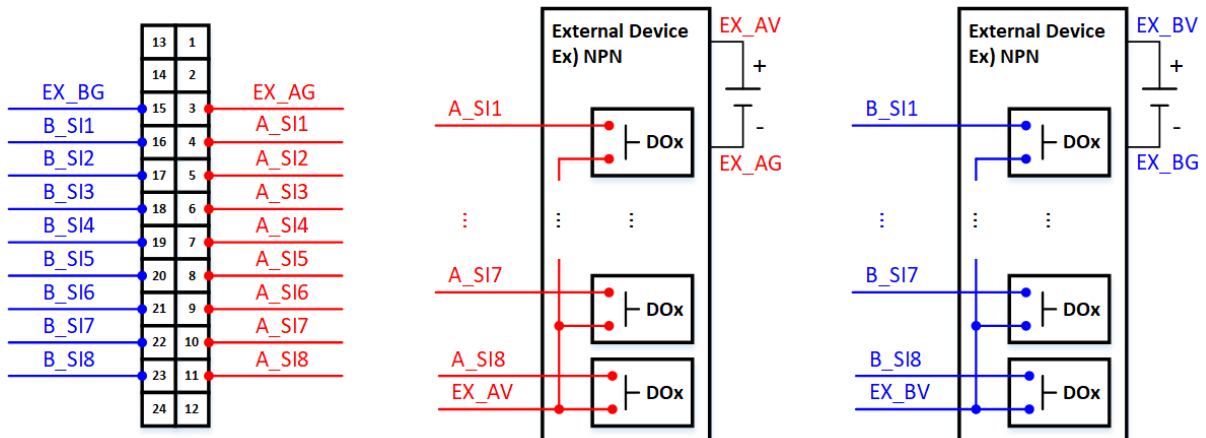


Figure 5.17 Safety Input Wiring with External Power Supply (PNP Type) for Optional Safety IO Module (BD680)

5.5. User DIO

5.5.1. Overview

In Hi7, the 'user DIO board (BD681)' and 'extended DIO board (BD682)' can be used to handle general-purpose I/O signals and synchronize conveyor encoders.



Caution

In this manual, DIO refers to Digital Input and Output.

The 'extended DIO board (BD682)' cannot be used independently and should be used together with the 'user DIO board (BD681)'.

Table 5-18 Board Specifications

No.	Board Name (Board Identifier)	Board Function
1	User DIO board (BD681)	- 16-channel general-purpose input - 16-channel general-purpose output - Can be used independently
2	User DIO board (BD682)	- 16-channel general-purpose input - 16-channel general-purpose output - 2-channel conveyor encoder - Cannot be used independently (should be used with BD681)

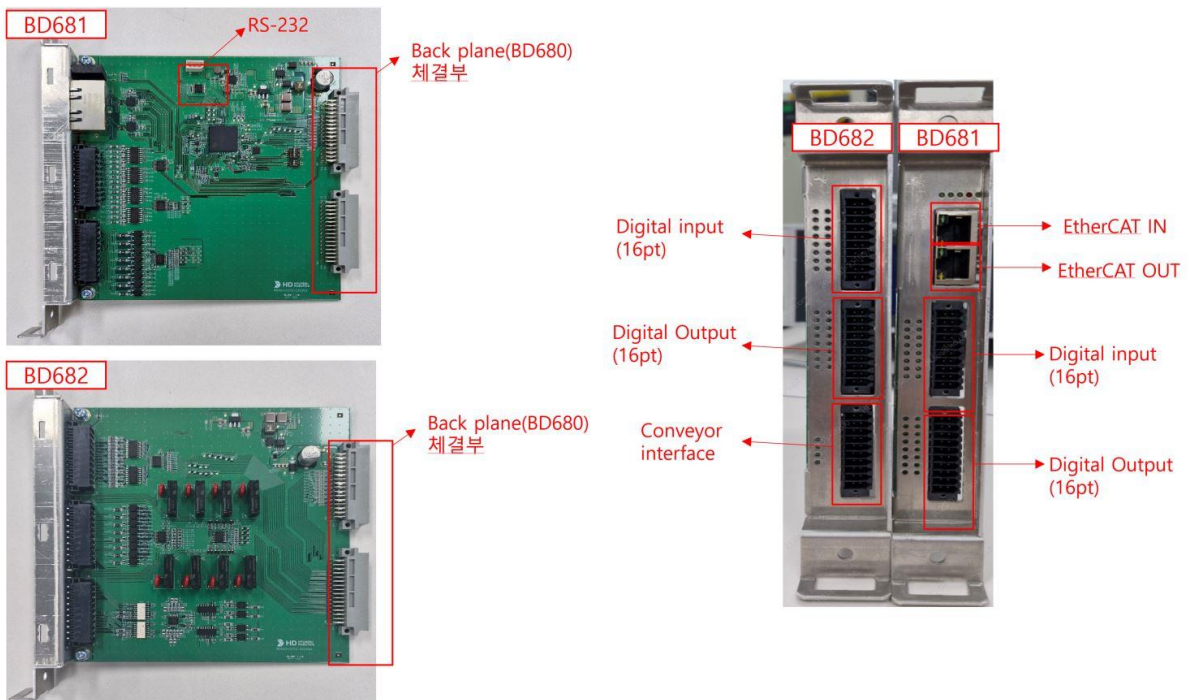
By using two BD681 and one BD682, up to 48 channels of input/output can be controlled.

5.5.2. Information of hardware

The user IO module (BD681) allows connection and configuration with various devices through digital input/output ports.

In addition, the extended IO module (BD682) allows the addition of digital I/O ports and synchronization with conveyor systems.

The basic hardware configuration of the boards is as follows:



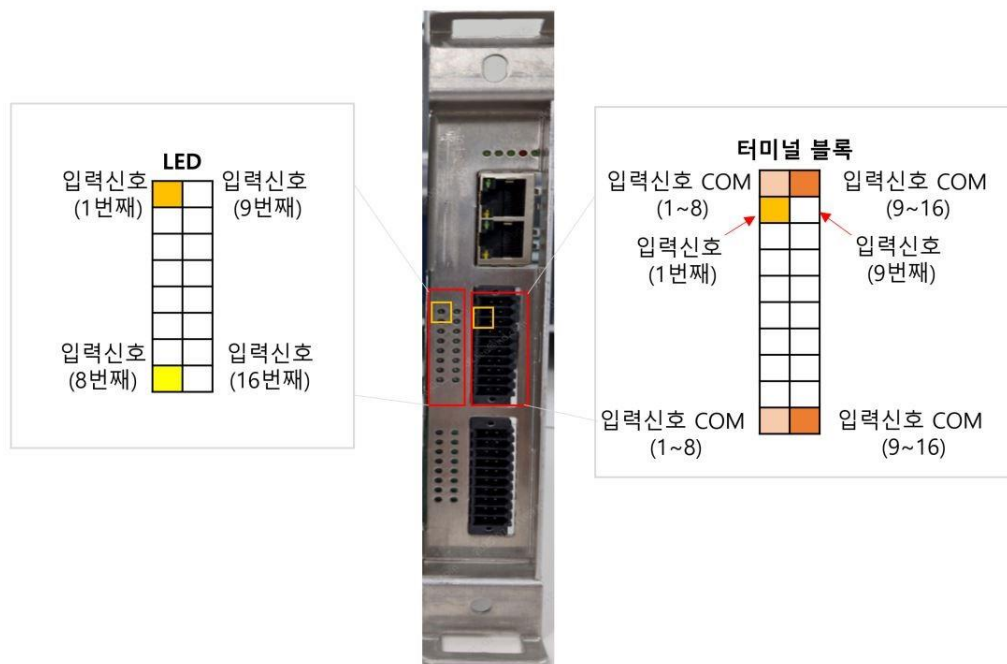
5.5.2.1 Digital Input

The following figure and table show the pin configuration of the terminal blocks for digital inputs. Each terminal block can receive 16 input signals, supporting NPN or PNP type inputs depending on the application.

When BD682 is additionally mounted, 16 digital inputs are added.

Refer to the function manual for how to configure NPN and PNP signals.

디지털 입력 커넥터 정보

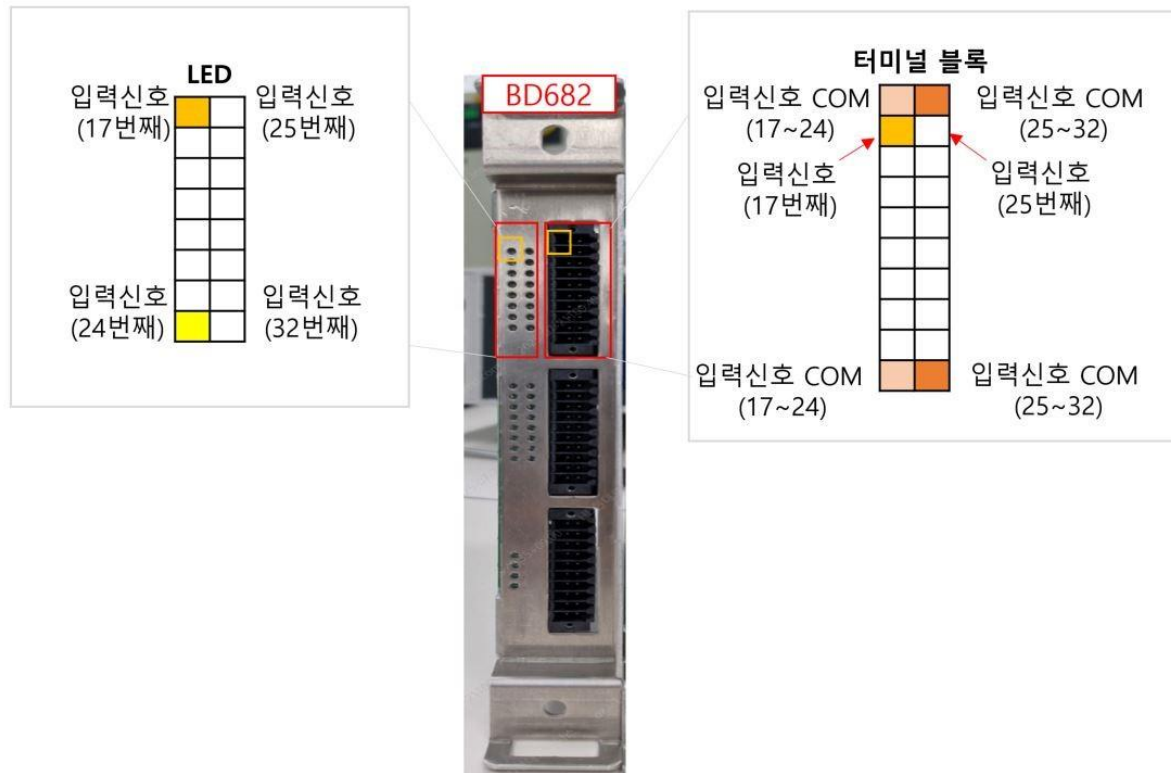


No.	Signal Name	Description	No.	Signal Name	Description
1	COM_IN_A	COM signal (1~8)	11	COM_IN_B	COM signal (9~16)
2	A1	Digital input 1	12	B1	Digital input 9
3	A2	Digital input 2	13	B2	Digital input 10
4	A3	Digital input 3	14	B3	Digital input 11
5	A4	Digital input 4	15	B4	Digital input 12
6	A5	Digital input 5	16	B5	Digital input 13
7	A6	Digital input 6	17	B6	Digital input 14

5. Optional Components of the Controller

8	A7	Digital input 7	18	B7	Digital input 15
9	A8	Digital input 8	19	B8	Digital input 16
10	COM_IN_A	COM signal (1~8)	20	COM_IN_B	COM signal (9~16)

When extended DIO board (BD682) is additionally mounted, the pin map is as follows:



No.	Signal Name	Description	No.	Signal Name	Description
1	COM_IN_A	COM signal (17~24)	11	COM_IN_B	COM signal (25~32)
2	A9	Digital input 17	12	B9	Digital input 25
3	A10	Digital input 18	13	B10	Digital input 26
4	A11	Digital input 19	14	B11	Digital input 27
5	A12	Digital input 20	15	B12	Digital input 28
6	A13	Digital input 21	16	B13	Digital input 29

7	A14	Digital input 22	17	B14	Digital input 30
8	A15	Digital input 23	18	B15	Digital input 31
9	A16	Digital input 24	19	B16	Digital input 32
10	COM_IN_A	COM signal (17~24)	20	COM_IN_B	COM signal (25~32)

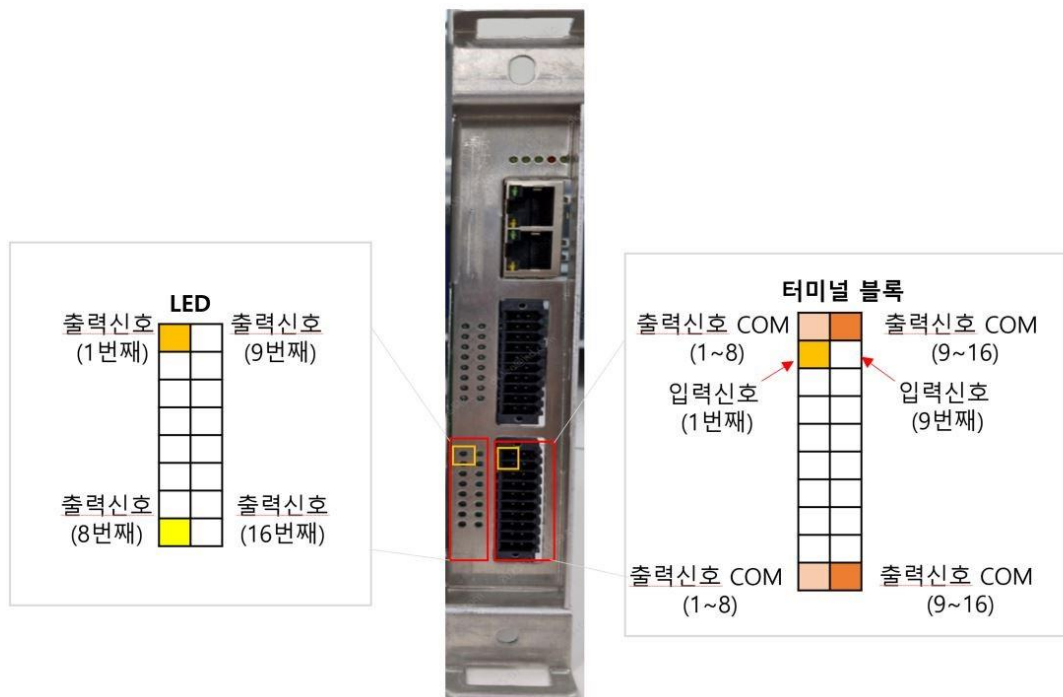
5. Optional Components of the Controller

5.5.2.2 Digital Output

The following figure and table show the pin configuration of the terminal blocks for digital outputs. Each terminal block can transmit 16 output signals, supporting NPN or PNP type outputs depending on the application.

When BD682 is additionally mounted, 16 digital outputs are added.

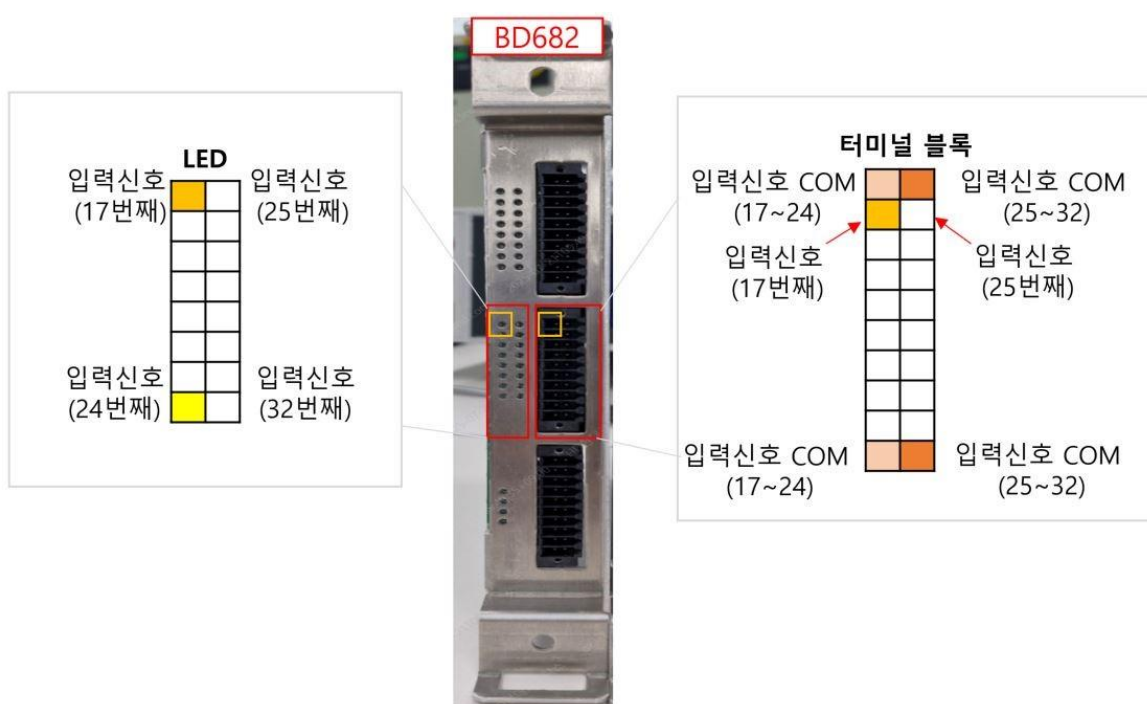
디지털 출력 커넥터 정보



No.	Signal Name	Description	No.	Signal Name	Description
1	COM_OUT_A	COM signal (1~8)	11	COM_OUT_B	COM signal (9~16)
2	A1	Digital Output 1	12	B1	Digital Output 9
3	A2	Digital Output 2	13	B2	Digital Output 10
4	A3	Digital Output 3	14	B3	Digital Output 11
5	A4	Digital Output 4	15	B4	Digital Output 12
6	A5	Digital Output 5	16	B5	Digital Output 13
7	A6	Digital Output 6	17	B6	Digital Output 14

8	A7	Digital Output 7	18	B7	Digital Output 15
9	A8	Digital Output 8	19	B8	Digital Output 16
10	COM_OUT_A	COM signal (1~8)	20	COM_OUT_B	COM signal (9~16)

When extended DIO board (BD682) is additionally mounted, the pin map is as follows:



No.	Signal Name	Description	No.	Signal Name	Description
1	COM_OUT_A	COM signal (17~24)	11	COM_OUT_B	COM signal (25~32)
2	A9	Digital Output 17	12	B9	Digital Output 25
3	A10	Digital Output 18	13	B10	Digital Output 26
4	A11	Digital Output 19	14	B11	Digital Output 27
5	A12	Digital Output 20	15	B12	Digital Output 28
6	A13	Digital Output 21	16	B13	Digital Output 29
7	A14	Digital Output 22	17	B14	Digital Output 30

5. Optional Components of the Controller

8	A15	Digital Output 23	18	B15	Digital Output 31
9	A16	Digital Output 24	19	B16	Digital Output 32
10	COM_OUT_A	COM signal (17~24)	20	COM_OUT_B	COM signal (25~32)

5.5.2.3 Conveyor

The following figure shows the configuration for conveyor synchronization with encoder inputs and limit switches.

It consists of a total of two input channels. Each channel supports two types of encoders (open collector/line driver).



No.	Signal Name	Description	No.	Signal Name	Description
1	PA2_P	Channel 2 line driver type encoder A signal input positive	11	PA1_P	Channel 1 line driver type encoder A signal input positive
2	PA2_N	Channel 2 line driver type encoder A signal input negative	12	PA1_N	Channel 1 line driver type encoder A signal input negative
3	PB2_P	Channel 2 line driver type encoder B signal input positive	13	PB1_P	Channel 1 line driver type encoder B signal input positive
4	PB2_N	Channel 2 line driver type encoder B signal input negative	14	PB1_N	Channel 2 line driver type encoder B signal input negative
5	LDLS2	Channel 2 line driver type encoder Limit switch	15	LDLS1	Digital Output 28
6	GND	Ground	16	GND	Ground
7	P2+	Channel 2 open collector encoder power	17	P1+	Channel 1 open collector encoder power

5. Optional Components of the Controller

8	A2	Channel 2 open collector type encoder A signal input	18	A1	Channel 1 open collector type encoder A signal input
9	B2	Channel 2 open collector type encoder B signal input	19	B1	Channel 1 open collector type encode B signal input
10	OCLS2	Channel 2 open collector type encoder Limit switch	20	OCLS1	Channel 1 open collector type encoder Limit switch

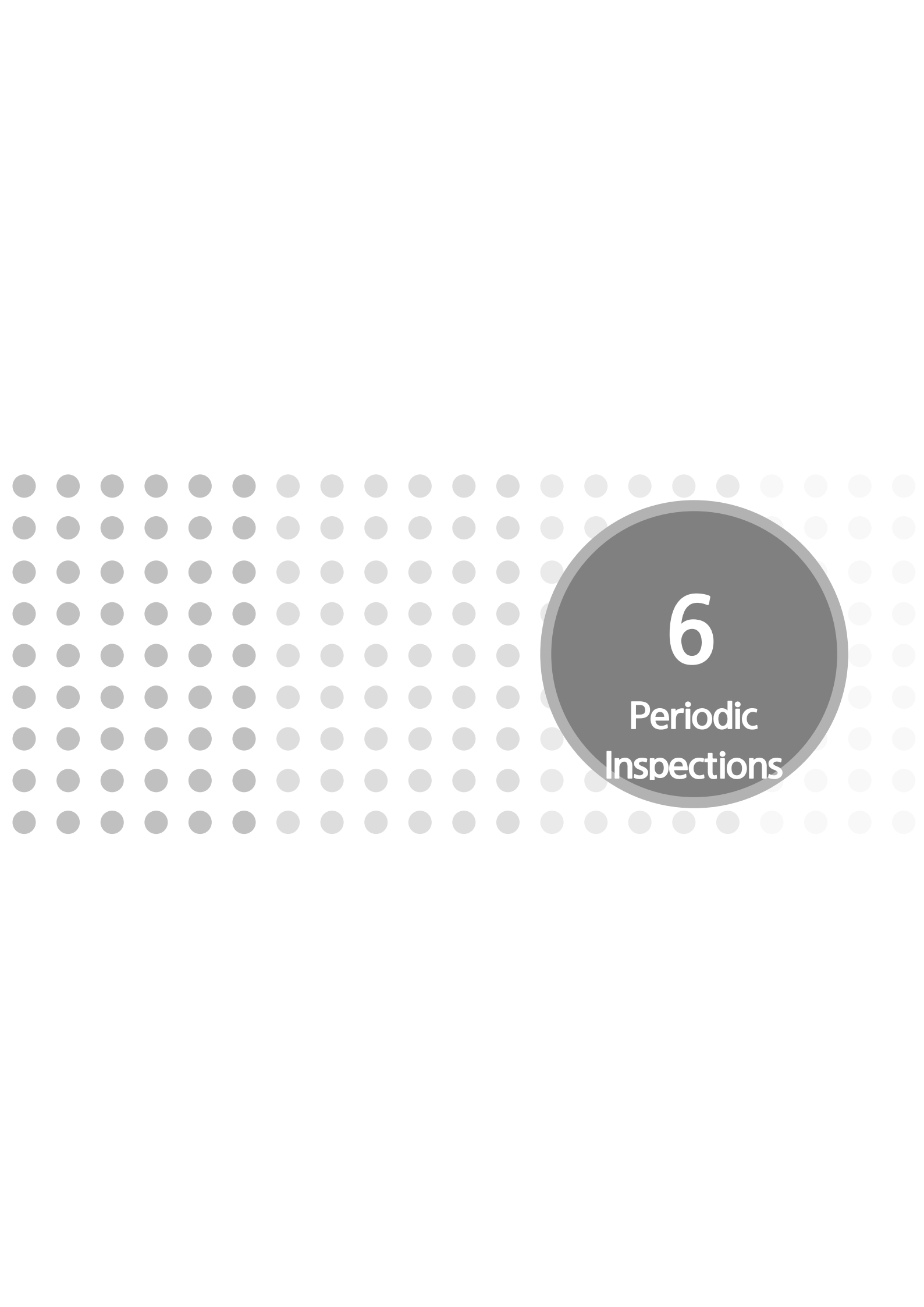
5.6. Safety communication board(BD671)

5.6.1. Overview

In Hi7, using the **PROFIsafe board (BD671)** allows processing of safety-related signals. This board is based on the PROFIsafe communication protocol and supports transmission of safety signals over standard fieldbus networks (e.g., PROFINET).

PROFIsafe is an international safety communication standard that enables the transmission of safety-related data together with standard data on PROFIBUS/PROFINET networks.

5.6.2. Information of hardware



6

Periodic
Inspections



6. Periodic Inspections

The periodic inspection of the controller is designed to minimize the occurrence of failures and to maintain the performance continuously. This section describes the cautions and contents of works during the periodic inspection.

6.1. Inspection Schedule

Inspections should be carried out according to the schedule shown in the figure below. Periodic inspections are designed to prevent failures in advance and to secure safety and maintain accuracy even when the controller and manipulator are used for a long time. Periodic inspections are absolutely necessary and must be conducted even during normal operation.

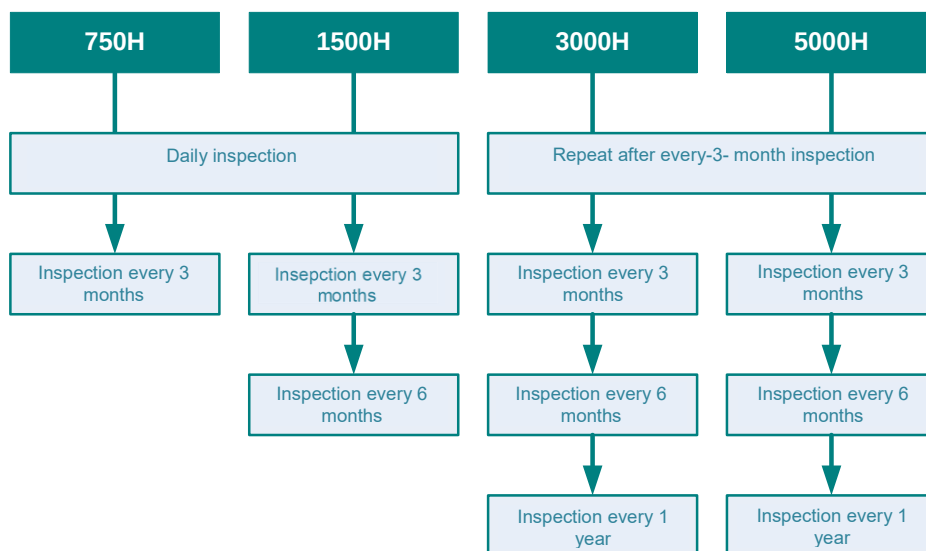


Figure 6.1 Inspection Schedule

6.2. General Cautions for Periodic Inspections

- The inspection work should be conducted by a person who has completed the courses implemented at the robot training center by our company.
- Before carrying out inspection works, check the parts, tools, drawings, etc. necessary for the work.
- Use the specialized replacement parts designated by our company.
- Turn off the power when inspecting the manipulator.
- When conducting works while keeping the door open, turn off the primary power and take precautions to prohibit dust, etc. to fly into the surrounding area.
- When required to touch the parts of the controller, you should take special care not to allow the IC to be destroyed by static electricity (Be careful also when contacting the connector).
- When performing periodic inspections while operating the manipulator, take precautions to prohibit anyone from entering inside the operation range.
- You should measure the voltage at a designated place and take precautions for electric shock and short-circuiting.
- Do not inspect the robot and the controller at the same time.
- After inspection, you must perform a trial-operation to check the operation of the robot before performing the normal operation.

6.3. General Inspections

Table 6-1 Daily Inspection

No.	Elements to inspect	Items to inspect	Remarks
1	Controller	Are the display lamps normal?	Check with your own eyes.
		Is the door closed properly?	Check with your own eyes.
		Is there any problem with the screen of the teach pendant?	Check with your own eyes.
2	Manipulator	Is there any noise during the operation?	Listen with your own ears.
		Is there any loosening of the screw at the front-end connection part?	Fasten.
		Is there any scratch, contamination, or damage to the wiring and wire harness of the manipulator?	Check with your own eyes.
		Is there any dirt or obstacle that causes damage to the manipulator?	Check with your own eyes, then clean.
3	Others	Is there any interference with the surrounding area of the controller and manipulator?	Check with your own eyes.

6.4. First Inspection (Inspection When Reaching 750 Hours)

Table 6-2 First Inspection

No.	Elements to Inspect	Items to Inspect	Remarks
1	External and main screws	Loosening of screws	Fasten.
2	Electric wiring connectors and wire harnesses of the manipulator	Loosening of connectors	Fasten.
3	Screws for fastening the dogs and limit switches	Loosening of screws	Fasten.

6.5. Cyclic Inspections

Table 6-3 Cyclic Inspections

No.	Cycle (Months)			Elements to Inspect	Items to Inspect	Remarks
	3	6	12			
1		○	○	Packing of the door	·Checking for deformation and detachment	
2	○	○	○	Back	·Checking dust on and rotation of the cooling fan blade part	
					·Checking the regenerative discharge resistor for damage and dust	
					·Checking the transformer room for any heat using the sense of touch, and then cleaning the room	
					·Checking the transformer terminal block for any loosening and damage	
3	○	○	○	Wire harness	·Checking the connectors for any loosening and damage	
4		○	○	Motor drive	·Checking the connectors and terminals for any loosening and damage	
5		○	○	Connectors of each board	·Checking for any loosening using the sense of touch	
6	○	○	○	Operation panel	· Checking the status of buttons and switches	
7		○	○	The whole controller	·Cleaning dust	
8	○	○	○	Nameplate	·Inspecting various nameplates	
9		○	○	Voltage measurement	·Voltage of the primary power	
					·CNFN1 B2-C2	
					CNPB1 PB-MB	
					·CN24VB1 P24B-24GND	
10		○	○	Grounding	·Checking the terminals for any loosening and detachment	
11		○	○	Battery	·Checking the voltage and replacing the battery Periodically	Main board LEDs
12	○	○	○	Teach pendant	Inspecting the exterior and checking the connector connection part	
					LCD Display Checking the status of the LCD display	
					·Checking the display of LEDs	
					·Checking the status of the buttons,	

6. Periodic Inspections

					switches, and LEDs	
13	⊙	⊙	⊙	Safety-related parts	·Checking the emergency stop switch (on the controller and teach pendant)	
	⊙	⊙	⊙		·Checking the main power circuit breaker switch (NFB1)	
	⊙	⊙	⊙		·Checking the Enable device of the teach pendant	
	⊙	⊙	⊙		·Checking the circuit protector (CP1)	
	⊙	⊙	⊙		·Checking the magnetic contacts (MC1, MC2)	
14	⊙	⊙	⊙	Safety-related board	·Checking the BD630 (Connectors, LEDs)	

6.6. Inspection When Going on a Long Vacation

When going on a long vacation, you should check the following items first before turning off the controller of the robot.

- (1) Check whether the yellow LED (BATLOW) for battery discharge detection on the main board is turned on. The yellow LED will be turned on if there is a problem with the battery. In this case, replace the battery with a battery of rated capacity. If the controller input power is turned off while there is a problem with the battery, various programs and integer data in the board will be removed after about seven days. Therefore, you must back them up using HRView or a USB memory.
- (2) Check if the door of the controller is secured.

6.7. Maintenance Parts

The characteristics of individual parts are described below.

Maintenance Parts of the Category A

 Caution	These are the important parts to be prepared for daily maintenance and inspection.
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To maintain normal operation, the parts of the category A-2 and parts of the category A-3 are the minimum necessary parts, and at least one set of each group should be prepared.

Table 6-4 Inspection of Maintenance Parts of the Category A

Type	Contents	Remarks (Reference)
Maintenance parts of the category A-1	Spare parts for standard parts	
Maintenance parts of the category A-2	Important backup parts	
Maintenance parts of the category A-3	Periodic replacement parts	

Table 6-5 Maintenance Parts of the Category A-1 (Spare Parts for Standard Parts)

No.	Part name	Type	Maker	Quantity(EA)	Remarks
1	Fuse (F1,F2)	GP75(250V, 7.5A)	Daito	2	BD6C3
2	Fuse (F3,F4)	GP75(250V, 7.5A)	Daito	2	BD6C3
3	Fuse (F5,F6)	GP75(250V, 7.5A)	Daito	2	BD6C3
4	Fuse(F1)	0458007.DR(63V, 7A)	Littelfuse	1	BD604

Table 6-6 Maintenance Parts of the Category A-2 (Important Backup Parts)

No.	Part name	Type	Maker	Quantity (EA)	Remarks
1	Motor drive	H6D6X H6D6A	Hyundai Robotics	1	Medium-sized robot Small-sized robot
2	Main control module	H6COM-T	Hyundai Robotics	1	
3	Teach pendant	TP630	Hyundai Robotics	1	
4	Power supply module	H7PSM30 H7PSM15	Hyundai Robotics	1	Large-sized robot Medium-sized robot Small-sized robot
5	Board	BD642	Hyundai Robotics	1	Servo board
		BD604	Hyundai Robotics	1	Backplane board

Table 6-7 Maintenance Parts of the Category A-3 (Periodic Replacement Parts)

No.	Part name	Type	Maker	Quantity (EA)	Remarks
1	Battery (3.6V AA size)	ER6V-T1	TOSHIBA (JAPAN)	1	Replace it every two years

Maintenance Parts of the Category B

 Caution	These are the maintenance parts to be prepared when multiple units are purchased.
---	---

Table 6-8 Maintenance Parts of the Category B

Type	Contents	Remarks (Reference)
Maintenance parts of the category B-1	Parts that should be purchased from Hyundai Robotics	
Maintenance parts of the category B-2	Parts that can be purchased directly from the parts maker	

Table 6-9 Maintenance Parts of the Category B-1 (Parts that should be purchased from Hyundai Robotics)

No.	Part Name	Type	Maker	Quantity (EA)	Remarks
1	Wire harness	CMC1	Hyundai Robotics	1	Large-sized/Medium-sized/Small-sized
		CMC2	Hyundai Robotics	1	Large-sized/Medium-sized
		CEC1	Hyundai Robotics	1	Large-sized/Medium-sized/Small-sized

Table 6-10 Table 6 10 Maintenance Parts of the Category B-2 (Parts that can be purchased directly from the parts maker)

No.	Part Name	Type	Maker	Quantity (EA)	Remarks
1	No fuse breaker (NFB)	-	-	1	
2	Magnetic contact (MC1, MC2)	-	-	2	
3	Circuit protector (CP1)	-	-	1	



Caution:

Considering that the boards are equipped with high-performance parts, take precautions about the following items for carrying out maintenance.

Preservation temperature 0° C-40° C / 32° F-104° F

For long-term preservation and maintaining high reliability, you should keep the temperature at $25 \pm 10^{\circ} \text{ C}$ / $77 \pm 50^{\circ} \text{ C}$ and avoid rapid temperature variation ($\pm 10^{\circ} \text{ C/hour}$), ($\pm 50^{\circ} \text{ F/hour}$).

Preservation humidity 20%– 80%

For long-term preservation and maintaining high reliability, you should keep the humidity at 45%– 65% and take special care not to allow condensation to take place.

Prevention of static electricity

If the tip of the electrode is preserved in a dry state, static electricity will be easily charged, and when the charged static electricity is discharged, the semiconductor will be easily destroyed. Therefore, when storing the substrate separately, please use antistatic wrapping paper.

Others

It is required to store the boards in a place where there is no toxic gas, no dust, or in a place where no load will be applied.

Appendices

Rules on Occupational Safety and Health Standards, and Notice for Safety Inspection

The industrial robot should be installed in consideration of the inspection standards, both of the Rules on Occupational Safety and Health Standards and of the Notice for Safety Inspection (if subject to inspection).

■ Rules on Occupational Safety and Health Standards

Article 222 (Teaching, etc.): When it is required to perform works such as teaching (this refers to setting or changing the operation sequence, position and speed of the manipulator, or checking the result, and will be the same in the following) on the robot within the operation range of an industrial robot (hereinafter “robot”), the business owner is required to take the measures described under the following items to prevent the danger that could be caused by the unexpected operation of or mistakes in the operation of the robot. However, if the works are performed with the robot’s driving source cut off, measures 1– 2 below may not be taken <Revised on April 7, 2016>.

1. Establish guidelines for the following items and make sure that works are performed according to the guidelines.
 - A. Robot operation method and sequence
 - B. Manipulator speed during a work
 - C. Signaling method when having two or more workers to carry out works
 - D. Measures to take when finding an abnormality
 - E. Measures to take when finding an abnormality and stopping the operation of the robot and restarting the operation
 - F. Other measures necessary to prevent the danger that could be caused by the unexpected operation of or mistakes in the operation of the robot
2. Workers engaged in the works or those who monitor the workers should take measures to stop the operation of the robot immediately if an abnormality is found.
3. In the process of performing works, it is required to take measures such as putting up a marking on the start switch, for example, to indicate that works are underway, to prevent workers other than the ones currently engaged in the works from manipulating the start switch.

Article 223 (Prevention of Dangers During Operation): The business owner is required to install a fence of 1.8 m or more in height (when considering the operation range of the robot, the height can be adjusted lower if there is no danger that could occur because of the height) to prevent the risk of injuries that may occur because of the operation of the robot (excluding the operation of the robot for the teaching according to Article 222 and also the operation of the robot in compliance with conditions of Article 224). In some sections where a fence cannot be installed because of the installation of a conveyor system, etc.,

a responsive, protective device such as a safety mat or optoelectronic protective device should be installed. However, if the Minister of Employment and Labor acknowledges that the safety standard for the robot meets the safety standards set by Korean industrial standards under Article 12 of the Industrial Standardization Act or internationally accepted safety standards, it is possible to forego measures according to the text of this document.

Article 224 (Including the measures to take when performing works such as repairing): When the work such as repairing, inspecting, adjusting (excluding teaching), cleaning, refueling the robot, or confirming relevant results within the operation range of the robot is underway, the business owner should take necessary measures to ensure that the robot can be stopped. These measures include locking up the start switch of the robot using a key and managing the key separately, or putting up a signboard on the start switch, indicating that work is underway. These measures should also be taken to ensure that anyone other than the workers engaged in the work will be prevented from manipulating the start switch. However, this is not the case if work must be performed even during the operation of the robot, and, in such a situation, measures of the individual items under Article 222 are taken to prevent the danger that could be caused by an unexpected operation or mistakes in operation

【Appendix 14】 Inspection Standards for Industrial Robots (Related to Article 30)

Automatic operation mode	A. In the automatic operation mode, safety devices such as fences should maintain the normal function, and if a person enters the protected area, the protective stop function should be activated. B. Protective stop or emergency stop should not be reset or disabled when the automatic operation mode is selected. C. Automatic operation mode should be deactivated when a stop signal is given. D. When the automatic mode is selected, the starting of the automatic mode should be possible only by a separate start operation outside the protected area. E. Changing from automatic operation mode to other operation modes should be possible only when driving is stopped.
Control of the pendant	If the pendant has a string, its length should be sufficient for the teaching operator to perform the work safely.
Electric connection devices	Electric connection devices such as electric outlets that are to be connected to the robot should not be separated arbitrarily.
Control of the pendant	The layout of the robotic system for the operation area, access, and free space should be appropriate in compliance with the following items.
Electric connection devices	A. The maximum area of the robot should be checked first. Then, the restricted area and operation area should be set, and there should be a free space between the robot and obstacles such as pillars of the building.
Robot system layout design	B. A passage such as a pedestrian passage should be secured for safe movement.
	C. Access to and from the path to the control system should be safe.

【Appendix 14】 Inspection Standards for Industrial Robots (Related to Article 30)

	D A safety passage for approaching for inspection, cleaning, repair, maintenance, etc. should be secured. E. There should be no risk of slipping, tripping, or falling over because of wiring or other dangerous sources. F. There should be no danger caused by the cable tray, etc. G. Operation controllers and auxiliary equipment (welding controllers, pneumatic valves, etc.) that workers need to access during automatic operation should be located outside the protected area.
Robot system stop function	All robot systems should have a protective stop function and also a separate emergency stop function. A. Emergency stop function 1) The robot system should have a single emergency stop function that operates for all relevant parts of the system. 2) In the case of multi-robots or multi-cells, the control range can be divided, and the control range should be clearly indicated using characters or symbols near the emergency stop device. 3) When there are more than two control positions, the emergency stop device installed in each control position should always maintain its function. 4) This function should take priority over all other robot controls and, in case of an emergency stop, should remove the driving power from the robot drive unit and maintain the stop state until initialization. 5) Initialization should be performed manually only, and restarting should not be performed immediately after initialization but should be carried out through a separate starting operation. 6) The performance of the emergency stop circuit should satisfy the safety-related control system performance requirements of item no. 4. B. Protective stop function 1) The robot system should be equipped with one or more protection stop circuits for connection with external protection devices. 2) When activated, the protective stop circuit should be capable of stopping the robot operation, removing the actuator power, and controlling all dangerous sources controlled by the robot. 3) Protective stop should be started manually or by the control logic. 4) At least one protection stop function should utilize the 0-stop method or 1-stop method. Note 1) 0-stop method: Stopping will be performed through immediate cut-off of the actuator power. Note 2) 1-stop method: The power will be cut off after the machine stops while the actuator power is supplied.

【Appendix 14】 Inspection Standards for Industrial Robots (Related to Article 30)

	5) The performance of the protective stop circuit should satisfy the safety-related control system performance requirements of item no. 4.
Manual resetting, starting, and restarting	The robot system should be in compliance with the following items to prevent unexpected starting. A. Starting and restarting should be possible only when all safety functions and protective measures are operating normally. B. If the power supply is interrupted and then resumed, a starting interlocking device should be installed to prevent the automatic starting of dangerous operations and should be reset only by manual operation. C. Start and restart control should be manually operated outside the protected area and should be impossible to activate within the protected area. D. The start and restart control unit should be located where the protected area can be visibly seen. However, if it is difficult to secure a view of the protected area, it is required to comply with one of the following items. 1) Installation of a detection device that can detect the presence of workers in the protected area 2) Installation of means to keep the door open 3) Installation of an additional device for time limit reset inside the protected area 4) Generation of audible and visual warning signals that can be sufficiently recognized inside the protected area, and provision of sufficient evacuation time E. When it comes to complying with item no. 4 of D above, it is required to install a sufficient number of emergency stop devices in the protected area at a location that the operator can access easily.
Protected area, fence, etc.	Protected areas and fences should be in compliance with the following items. However, in the case of cooperative robots, some applications may be excluded, depending on the operation method. A. The restricted area should be located within the protected area. B. The protected area should be established using a fence. However, when it comes to an opening for taking in and out materials or parts, the protected area should be established by a responsive, protective device, etc. C. The fence for the robot cell should be appropriate in compliance with the following items. 1) The fence should be firmly installed not to be easily damaged by external force and should be of a structure that can be removed only with tools. 2) If the fence has an opening for taking in and out materials or parts, it is required to physically prevent workers from accessing dangerous points under, next to, or through the opening, and if such measures are not possible, it is required to take additional measures such as using responsive, protective devices. 3) The height of the fence should be over 1,800 mm. However, if there is no danger that could occur because of the height, in consideration of both the operation range of the

【Appendix 14】 Inspection Standards for Industrial Robots (Related to Article 30)

	robot and the safety distance according to KS B ISO 13857, the height of the fence may be above 1,400 mm. 4) The movable fence (entry door) should be opened to the side or in the direction of moving away from a source of danger and should be structured in a way that it does not open toward the protected area. 5) An interlocking device should be installed to the movable fence (entry door), and it should be able to turn the source of danger into a safe state before the workers approach it. However, if there is a possibility of accessing the dangerous area before the source of danger is removed, there should be a locking device for the entry door, in addition to the interlocking device. 6) The movable fence (entry door) locking device should allow the start of the robot only when the movable fence (entry door) is closed and locked.
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Quality Assurance

To protect the interests of customers who have purchased products manufactured by Hyundai Robotics (hereinafter the “Company”) and sold by the Company or authorized distributors, the Company provides quality assurance against defects in the raw materials and in the manufacturing of products in accordance with the contents stipulated in the quality assurance. This quality assurance is provided only to the end users (hereinafter the “Customers”) of the products.

■ Coverage of Quality Assurance

Robots and relevant parts are covered by the Company’s quality assurance against defects in materials and manufacturing.

The Company’s sole responsibility and the sole remedy for the Customer with respect to all our products are limited to the repair or replacement of our products that are directly defective at our discretion. The Company will not compensate for any indirect or incidental, special, or consequential damages, such as loss of income, loss of use, loss of production, or damage to other products or equipment resulting from defects in the product.

■ Period of Quality Assurance

The Company guarantees the quality of the product for one year from the date that the Customer purchased the product and received it on-site, or the Customer issued the acceptance certificate after completing the trial-operation. However, if the contract date is different from the delivery date (installation and trial-operation completion date), the period of quality assurance will be based on the delivery date, and when the product is replaced with a new finished product, the period of assurance will be calculated from the date of replacement.

■ Limitation of and Exception for Assurance

To maintain quality assurance, you must follow the maintenance procedures and keep records as specified by the Company. When we decide that the Customer does not comply with the terms and conditions as follows, the quality assurance will be made invalid.

- Damage to or breakdown of the product because of the Customer’s carelessness, inexperience in operation, negligence, arbitrary modification, disassembly, or repair
- Failure of or damage to the product because of the use and installation of parts, consumables, software, etc. that are not certified by the company
- Failure of or damage to the product because of noncompliance with the instructions and precautions in the manual
- Failure of or damage to the product because of the use for purposes other than the original purpose
- Failure of or damage to the product caused by using it in inappropriate environments, dropping the product, or applying a strong impact on the product

- Failure of or damage to the product because of the installation, repair, and maintenance by a person (customer, uncertified or unauthorized mechanic, etc.) other than an installation expert
- When the service life of a consumable part is due
- If a service request for a defect is received after the warranty period has expired

We do not provide quality assurance against damage to the product caused by external circumstances beyond our reasonable control, such as theft, willful destruction, fire, natural disasters, wars, or acts of terror. Moreover, we are not responsible for the defects in the product, operation, and performance not pertaining to the range specified in the warranty.



Customer Support

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Hours of Operation: Weekdays (Monday– Friday) 09:00 – 18:00 | Closed on weekends and holidays

For detailed inquiries about products or services, please contact our customer support team.



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